

1 Mechanistic Explainability in Reinforcement
2 Learning

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4 Monday 28th September, 2026

Acknowledgements

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41

Contents

42	Acknowledgements	i
43	List of Figures	v
44	List of Tables	vii
45	Notation	ix
46	The Golem Myth	1
47	1 Introduction	3
48	1.1 Research Context	3
49	1.2 Research Questions	4
50	1.3 Contributions	5
51	1.4 Thesis Structure	6
52	2 Deep Learning	8
53	2.1 Historical Developments	8
54	2.2 Objective	10
55	2.3 Neural Network Structure	11
56	2.3.1 Neurons	12
57	2.3.2 Layers	13
58	2.3.3 Network	16
59	2.4 Learning	16
60	2.4.1 Computing the Gradient	17
61	2.4.2 Update rule	19
62	2.4.3 Convergence Properties	19
63	2.5 Evaluation	20
64	2.6 Conclusion	22
65	3 Latent Space Analysis	23
66	3.1 Concept Hypothesis	24
67	3.2 Probing	27
68	3.3 Directions	30
69	3.4 Superposition	32
70	3.5 Clustering	36
71	3.6 Conclusion	39

72	4 Reinforcement Learning	41
73	4.1 Historical Developments	41
74	4.1.1 Behavioral foundations (1900–1950)	41
75	4.1.2 Sequential decision-making (1950–1970)	42
76	4.1.3 Emergence of reinforcement learning (1970–2000)	42
77	4.1.4 Consolidation of reinforcement learning (2000–2012)	43
78	4.1.5 Deep reinforcement learning (2013–2026)	43
79	4.2 Agent Paradigm	44
80	4.3 Problem Formulation	46
81	4.3.1 Markov Decision Processes	46
82	4.3.2 Policy	47
83	4.3.3 Trajectory	48
84	4.3.4 Value Functions	49
85	4.3.5 Optimal Policy	50
86	4.4 Learning	51
87	4.4.1 Critic-Only Methods	51
88	4.4.2 Actor-Critic Methods	54
89	4.5 Conclusion	55
90	5 Explainability in Reinforcement Learning	57
91	5.1 Explanation in Artificial Intelligence	58
92	5.1.1 Explanation	58
93	5.1.2 Explainable Artificial Intelligence	60
94	5.1.3 Explainability Method in XAI	61
95	5.2 Taxonomy of XRL Methods	62
96	5.2.1 Subject	62
97	5.2.2 Source	63
98	5.2.3 Form	64
99	5.3 State-of-the-Art Review of XRL Methods	64
100	5.3.1 Policy Model	64
101	5.3.2 Saliency Map	66
102	5.3.3 Counterfactual	68
103	5.3.4 Agent Trajectory	69
104	5.3.5 Trajectory Prediction	70
105	5.3.6 Interaction Model	71
106	5.4 Conclusion	72
107	5.4.1 Absence of Explanation of the Learning Process	72
108	5.4.2 Lack of Unified Metrics for Explainability	74
109	5.4.3 Toward Intrinsic Explainability in XRL	74
110	6 Clustering Agent Representations	75
111	6.1 Representation Convergence Hypothesis	76
112	6.2 Latent Clustering of Surrogate Policies	77
113	6.2.1 Multiple surrogate policies	78
114	6.2.2 Training surrogate policies	78
115	6.2.3 Cluster extraction	79
116	6.3 Evaluation Metrics	83
117	6.3.1 Behavior Reconstruction	83
118	6.3.2 Adjusted Rand Index	84
119	6.3.3 Clustering Universality	85

120	6.3.4	Abstraction Score	86
121	6.3.5	Adjusted Clustering Universality	87
122	6.4	Conclusion	88
123	7	Experiments	89
124	7.1	Software Implementation	89
125	7.1.1	Initial policy training	90
126	7.1.2	Trajectory dataset generation	91
127	7.1.3	Surrogate policy training	91
128	7.1.4	Surrogate policy evaluation	92
129	7.1.5	Latent space analysis	92
130	7.2	Experimental Configuration	92
131	7.2.1	Environments	92
132	7.2.2	Neural architecture	94
133	7.2.3	Other CAR parameters	95
134	7.3	Policy Performance and Surrogate Fidelity	95
135	7.3.1	Initial policy training	95
136	7.3.2	Surrogate policy training	96
137	7.4	Clustering Quality of the Latent Space	97
138	7.4.1	Analysis across layers	98
139	7.4.2	Analysis of the selected layers	99
140	7.5	Semantic Interpretation of the Clusters	101
141	7.5.1	Breakout	102
142	7.5.2	Mario Bros	104
143	7.5.3	Ms. Pac-Man	106
144	7.5.4	Pong	108
145	7.5.5	Space Invaders	110
146	7.5.6	Surround	112
147	7.6	Conclusion	114
148	8	Conclusion	115
149	8.1	Summary of Contributions	115
150	8.2	Limitations	116
151	8.3	Perspectives	117
152	1	Additional Results	118
153	1.1	Breakout	119
154	1.2	Mario Bros	122
155	1.3	Ms. Pac-Man	125
156	1.4	Pong	128
157	1.5	Space Invaders	131
158	1.6	Surround	134
159		Acronyms	137
160		Bibliography	140
161		LLM Use Policy	151
162		Institutional Acknowledgements	152

List of Figures

164	1	The creation of a Golem, by Mikoláš Aleš.	2
165	1.1	Structure of the thesis.	6
166	2.1	Evolution of deep learning.	9
167	2.2	Artificial neuron.	13
168	2.3	Multilayer Perceptron Architecture.	14
169	2.4	Interaction between layers.	15
170	2.5	<i>Figure skipped in draft mode (set to render it).</i>	19
171	2.6	<i>Figure skipped in draft mode (set to render it).</i>	21
172	3.1	The same latent space activations without and under the Concept Hypothesis.	25
173	3.2	Decision boundaries of the linear probes for concepts A, B, and C.	28
174	3.3	Probing analysis of a chess-playing language model.	29
175	3.4	Interpretable directions in the latent space.	31
176	3.5	Interpretable directions in the latent space of a generative model.	32
177	3.6	Entanglement in the latent space of a generative face model.	33
178	3.7	Transformation from a dense to a sparse latent space.	34
179	3.8	The Golden Gate Bridge concept direction of a sparse autoencoder.	35
180	3.9	Visualization of the latent space structure.	37
181	3.10	Clusters discovered by deep clustering on natural images.	39
182	3.11	Overview of the four latent space analysis methods studied in this chapter, applied to the same point cloud.	40
183			
184			
185	4.1	Evolution of reinforcement learning prior to its fusion with deep learning.	43
186	4.2	Evolution of deep reinforcement learning.	44
187	4.3	The agent paradigm.	45
188			
189	5.1	Yearly growth of Explainable Reinforcement Learning research with at least one citation on arXiv.	58
190	5.2	Explanation process.	60
191	5.3	Policy approximation via an interpretable model.	66
192	5.4	Saliency heatmaps for Breakout	67
193	5.5	Attention-based heatmaps for Breakout and Ms. Pac-Man	68
194	5.6	Counterfactual explanation for Space Invaders	69
195	5.7	Prediction of sub-rewards as a function of the selected action.	71
196	5.8	Agent–environment interaction model for StarCraft II.	72
197			

198	5.9	t -SNE projection of a neural network's latent space.	73
199	6.1	Illustration of the Clustering Agent Representation method.	77
200	6.2	Example of a surrogate policy architecture π'	79
201	6.3	Two-dimensional projection of a latent space for Space Invaders	82
202	6.4	Illustration of <i>Clustering Universality</i> in the ideal case.	86
203	7.1	The CAR pipeline.	90
204	7.2	The six Atari environments used to evaluate CAR.	93
205	7.3	Architecture of the surrogate policies.	95
206	7.4	Training of the initial and surrogate policies.	97
207	7.5	CU, AS, and ACU averaged over cluster numbers as a function of layer depth.	99
208	7.6	CU, AS, and ACU with respect to the number of clusters.	100
209	7.7	Clusters A–B identified by CAR in the Breakout environment (layer 4, $K = 2$).	103
210	7.8	Clusters C–G identified by CAR in the Breakout environment (layer 6b, $K = 5$).	103
211	7.9	Clusters A–C identified by CAR in the Mario Bros environment (layer 3, $K = 3$).	105
212	7.10	Clusters D–H identified by CAR in the Mario Bros environment (layer 3, $K = 5$).	105
213	7.11	Clusters A–C identified by CAR in the Ms. Pac-Man environ- ment (layer 6b, $K = 3$).	107
214	7.12	Clusters D–H identified by CAR in the Ms. Pac-Man environ- ment (layer 4, $K = 5$).	107
215	7.13	Clusters A–B identified by CAR in the Pong environment (layer 2, $K = 2$).	109
216	7.14	Clusters C–E identified by CAR in the Pong environment (layer 3, $K = 3$).	109
217	7.15	Clusters A–D identified by CAR in the Space Invaders environ- ment (layer 4, $K = 4$).	111
218	7.16	Clusters E–I identified by CAR in the Space Invaders environ- ment (layer 2, $K = 5$).	111
219	7.17	Clusters A–D identified by CAR in the Surround environment (layer 8b, $K = 4$).	113
220	7.18	Clusters E–J identified by CAR in the Surround environment (layer 7b, $K = 6$).	113
221			
222			
223			
224			
225			
226			
227			
228			
229			
230			
231			
232			
233			

List of Tables

235	5.1	Taxonomy of explainability methods in reinforcement learning. .	65
236	7.1	Main software libraries used at each stage of the CAR pipeline. .	91
237	7.2	Main experimental parameters.	96
238	7.3	Evaluation of <i>Behavior Reconstruction</i> across the six Atari environments.	97
239	7.4	Layer maximizing the average ACU in each environment.	101
240	7.5	CU, AS, and ACU of the two clusterings analyzed in the Breakout environment.	102
241	7.6	CU, AS, and ACU of the two clusterings analyzed in the Mario Bros environment.	104
242	7.7	CU, AS, and ACU of the two clusterings analyzed in the Ms. Pac-Man environment.	106
243	7.8	CU, AS, and ACU of the two clusterings analyzed in the Pong environment.	108
244	7.9	CU, AS, and ACU of the two clusterings analyzed in the Space Invaders environment.	110
245	7.10	CU, AS, and ACU of the two clusterings analyzed in the Surround environment.	112
246	1.1	CU — Breakout	119
247	1.2	AS — Breakout	119
248	1.3	ACU — Breakout	120
249	1.4	Observation baseline (ρ^{obs}) — Breakout	120
250	1.5	Untrained-network (latent) baseline (ρ^{untr}) — Breakout	120
251	1.6	Action baseline (ρ^{act}) — Breakout	121
252	1.7	CU — Mario Bros	122
253	1.8	AS — Mario Bros	122
254	1.9	ACU — Mario Bros	123
255	1.10	Observation baseline (ρ^{obs}) — Mario Bros	123
256	1.11	Untrained-network (latent) baseline (ρ^{untr}) — Mario Bros	123
257	1.12	Action baseline (ρ^{act}) — Mario Bros	124
258	1.13	CU — Ms. Pac-Man	125
259	1.14	AS — Ms. Pac-Man	125
260	1.15	ACU — Ms. Pac-Man	126
261	1.16	Observation baseline (ρ^{obs}) — Ms. Pac-Man	126
262	1.17	Untrained-network (latent) baseline (ρ^{untr}) — Ms. Pac-Man	126
263	1.18	Action baseline (ρ^{act}) — Ms. Pac-Man	127

271	1.19	CU — Pong .	128
272	1.20	AS — Pong .	128
273	1.21	ACU — Pong .	129
274	1.22	Observation baseline (ρ^{obs}) — Pong .	129
275	1.23	Untrained-network (latent) baseline (ρ^{untr}) — Pong .	129
276	1.24	Action baseline (ρ^{act}) — Pong .	130
277	1.25	CU — Space Invaders .	131
278	1.26	AS — Space Invaders .	131
279	1.27	ACU — Space Invaders .	132
280	1.28	Observation baseline (ρ^{obs}) — Space Invaders .	132
281	1.29	Untrained-network (latent) baseline (ρ^{untr}) — Space Invaders .	132
282	1.30	Action baseline (ρ^{act}) — Space Invaders .	133
283	1.31	CU — Surround .	134
284	1.32	AS — Surround .	134
285	1.33	ACU — Surround .	135
286	1.34	Observation baseline (ρ^{obs}) — Surround .	135
287	1.35	Untrained-network (latent) baseline (ρ^{untr}) — Surround .	135
288	1.36	Action baseline (ρ^{act}) — Surround .	136

289

Notation

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293

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This section summarizes the notation used throughout this thesis. It follows the conventions of Goodfellow et al. [47] for deep learning and of Sutton and Barto [65] for reinforcement learning. Vectors and matrices are written in bold upright letters and sets in calligraphic letters. Unlike Sutton and Barto, random variables are written in lower case, like their values, and the reward r_t is indexed by the time step of the action a_t that produced it. Unlike Goodfellow et al., σ denotes any activation function and η the learning rate.

297

Numbers and Arrays

a	A scalar
$\mathbf{a}$	A vector
$\mathbf{A}$	A matrix
a_i	Element i of vector $\mathbf{a}$, with indexing starting at 1
$A_{i,j}$	Element i, j of matrix $\mathbf{A}$
$\mathbf{e}^{(i)}$	Standard basis vector, with a 1 at position i

298

Sets

$\mathcal{A}$	A set
$\mathbb{R}$	The set of real numbers
$[a, b]$	The real interval including a and b
$ \mathcal{A} $	Number of elements of the set $\mathcal{A}$
$\mathcal{P}(\mathcal{A})$	Set of probability distributions over $\mathcal{A}$

299

Calculus, Probability and Functions

$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
$\nabla_{\boldsymbol{\theta}} y$	Gradient of y with respect to $\boldsymbol{\theta}$

$\mathbf{J}_f$	Jacobian matrix of the function f
$x \sim P$	x is sampled from the distribution P
$\mathbb{E}[\cdot]$	Expectation
$f : \mathcal{A} \rightarrow \mathcal{B}$	The function f with domain $\mathcal{A}$ and range $\mathcal{B}$
$f \circ g$	Composition of the functions f and g
$\hat{f}$	An approximation of f ; the hat denotes an approximation
$\ \mathbf{x}\ _p$	L^p norm of $\mathbf{x}$
$\arg \max_a f(a)$	A value of a at which $f(a)$ is maximal
σ	Activation function, applied element-wise

Deep Learning

$\mathcal{X}, \mathcal{Y}$	Input and output spaces
$\mathbf{x}, \mathbf{y}$	An input and its target output
$\hat{\mathbf{y}}$	Output predicted by the network
$\mathcal{D}$	A dataset
N	Number of examples in a dataset
$(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$	The i -th example of a dataset
$\mathcal{L}$	Loss function
$\mathcal{L}^{(i)}(\boldsymbol{\theta})$	Loss of the i -th example
$\boldsymbol{\theta}$	Parameters of a model, a vector of $\mathbb{R}^d$
θ_j	The j -th parameter
$\hat{f}_{\boldsymbol{\theta}}$	Network with parameters $\boldsymbol{\theta}$
$\boldsymbol{\theta}_k$	Parameters after $k - 1$ steps of gradient descent
η	Learning rate
$\mathbf{g}$	Gradient estimate used by stochastic gradient descent
L	Index of the output layer of a network, the input layer being indexed 0
n_l	Number of neurons of layer l
$\mathbf{W}^{(l)}, \mathbf{b}^{(l)}$	Weight matrix and bias vector of layer l
$\mathbf{h}^{(l)}$	Latent representation (activations) of layer l
$h_i^{(l)}$	Activation of neuron i of layer l
$\phi^{(l)}$	Function computed by layer l
$f_{\text{enc}}, f_{\text{dec}}$	Encoder and decoder of an autoencoder
$\mathbf{z} \in \mathcal{Z}$	Code produced by an encoder, and the space of codes

Latent Space Analysis

c_j	Presence of concept j
$\mathbf{v}_i$	Direction associated with concept i
α	Strength of a perturbation along a direction
$\psi_{\theta}^{(l,j)}$	Probe predicting concept j from layer l
$\lambda, \lambda_1, \lambda_2, \lambda_3$	Weighting coefficients of a loss

Reinforcement Learning

$(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$	A Markov decision process
$\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}$	Sets of states, observations, actions and rewards
$\mathcal{S}^{\text{init}}, \mathcal{S}^{\text{ter}}$	Sets of initial and terminal states
s, o, a, r	A state, an observation, an action and a reward
s', o'	Next state and next observation
$s^{\text{init}}, s^{\text{ter}}$	An initial state and a terminal state
$p(s', o, r \mid s, a)$	Probability of the transition to (s', o, r) from state s and action a
$\beta(s \mid o)$	Probability of state s given observation o
$\pi(a \mid o)$	Probability of taking action a given observation o under policy π
π_*	Optimal policy
ε	Probability of taking a random action in an ε -greedy policy
t	Discrete time step
T	Final time step of an episode
s_t, o_t, a_t, r_t	State, observation, action and reward at time step t
τ	A trajectory
τ_t	Part of the trajectory τ starting from time step t
$\tau^{\text{ter}}, \tau^{\text{ep}}$	A terminal trajectory and an episode
$\tau \sim \pi$	Trajectory generated by following the policy π
$\Pr_{\pi}(\tau \mid s, o)$	Probability of τ under π , starting from state s and observation o
γ	Discount factor
G_t	Return following time step t
$v_{\pi}(s, o)$	Value of state s with observation o under policy π
$q_{\pi}(s, a)$	Value of taking action a in state s under policy π
q_*	State-action value function of the optimal policy
$\hat{v}(s, o, \mathbf{w})$	Approximate state value given the weight vector $\mathbf{w}$ (critic)

$\hat{q}(o, a, \mathbf{w})$	Approximate state-action value given the weight vector $\mathbf{w}$
π_{θ}	Policy with parameters θ (actor)
$\pi_k, \hat{q}_k$	Policy and value approximation at iteration k
$A(s, a)$	Advantage of action a in state s

Clustering Agent Representation

π	Initial policy to be explained
π'_i	The i -th surrogate policy, with parameters θ_i
n	Number of surrogate policies
$f_{\text{enc},i}, f_{\text{dec},i}$	Encoder and decoder of the i -th surrogate policy
ℓ	Parameters (logits) of an action distribution
$\mathcal{O}_{\text{data}}$	Set of observations used for training and clustering
$\mathcal{H}_i$	Latent representations of $\mathcal{O}_{\text{data}}$ given by $f_{\text{enc},i}$
f_{clust}	Clustering function
$\mathcal{C}_i$	Clustering of $\mathcal{H}_i$
K	Number of clusters
$\mathcal{C}_{\text{obs}}, \mathcal{C}_{\text{act}}$	Clusterings of the observations and of the action distributions
$\mathcal{C}_{\text{untr},u}$	Clustering given by the u -th of U untrained networks
$\rho_i^{\text{obs}}, \rho_i^{\text{act}}, \rho_i^{\text{untr}}$	Similarity of $\mathcal{C}_i$ to the three baseline clusterings
$M, G_{\pi}^{(e)}$	Number of evaluation episodes, and return of π in episode e
RI, ARI	Rand Index and Adjusted Rand Index
BR, CU, AS, ACU	<i>Behavior Reconstruction, Clustering Universality, Abstraction Score, Adjusted Clustering Universality</i>

The Golem Myth

305 A creation that surpasses its creator.

306 Long before modern technology, humanity had already questioned the possi-
 307 bility of losing control over its creations. One of the earliest expressions of this
 308 idea can be found in the story of the Golem. In Hebrew tradition, the term
 309 Golem refers to an artificial creature created by humans [29]. These creatures are
 310 made of clay and brought to life using instructions inscribed on their bodies, as
 311 illustrated in Figure 1. The earliest accounts describe the Golem as a protective
 312 and helpful figure.

313 This vision becomes more ambiguous in the seventeenth century, as illustrated
 314 by the legend of the Golem of Prague. In this story, the initially small Golem
 315 gradually grows in size and strength until it can no longer be safely controlled
 316 [29]. The creation intended to protect others therefore becomes a potential
 317 threat to them. The only remaining solution is to destroy the creature in order
 318 to protect the community.

319 The Golem has inspired similar narratives over time, including the modern
 320 notion of the robot. The concept of the robot as an artificial servant was
 321 introduced in the play *Rossum's Universal Robots*, written in 1920 [4]. In the
 322 play, a scientist creates artificial beings designed to perform labour. In this
 323 first version, robots are organic creatures forged through advances in chemistry.
 324 Conceived as servants, these artificial beings turn against humanity and destroy
 325 civilisation.

326 The connection between the Golem and the robot was explicitly made by the
 327 author, as illustrated by his statement that “the Robot is the Golem made flesh
 328 through mass production” [24]. Despite their destructive role in the play, the
 329 word robot is now used to designate artificial beings created to serve humans.
 330 This shows that, from its origin, the word robot has embodied the hopes and
 331 fears surrounding technological progress.

332 Today, this tension is particularly evident in the field of artificial intelligence.
 333 With the rise of computer technology, organic robots were gradually replaced in
 334 the collective imagination by mechanical ones, but the underlying fears remained.
 335 These concerns are even stronger as progress in the field has continued to
 336 accelerate. In these times of technological revolution, no one can predict how
 337 this development will unfold. What is certain, however, is that this field offers
 338 an almost literal adaptation of the Golem myth.

339 Among the subfields of artificial intelligence, reinforcement learning is most
 340 closely aligned with the goal of creating autonomous agents while minimizing
 341 reliance on human guidance. In this learning paradigm, an agent interacts with an
 342 environment, selects actions, and receives feedback according to the outcomes of



Figure 1: The creation of a Golem, by Mikoláš Aleš [2].

its behaviour [65]. Through repeated interaction with its environment, the agent learns a strategy to achieve the objective. As a result, the agent's behavior is not explicitly programmed but learned through interactions with the environment [44]. This minimal guidance enables reinforcement learning systems to achieve superhuman capabilities [64, 75].

However, maintaining control over such agents presents a major challenge, as the autonomy that makes this learning approach powerful also makes their behavior difficult to control. Despite the difficulty, understanding the behaviour of such agents is the objective of this thesis. This work therefore places within a long-standing line of thought that can be summarized by the following question: how can we ensure that our creations remain aligned with our intentions?

Chapter 1

Introduction

This initial chapter provides a general introduction to this work. It begins by presenting the research context that motivates the study. It then introduces the research questions addressed throughout the thesis. To address these questions, it then presents the main contributions. Finally, it concludes with an overview of the structure of the thesis.

1.1 Research Context

As is evident in recent years, Artificial Intelligence (AI) has advanced rapidly.

This progress is driven largely, by major development in Deep Learning (DL). This type of approach relies on artificial Neural Networks (NNs), which offer an unprecedented level of performance. Inspired by their biological counterparts, these networks are known as connectionist models. In such models, information is represented and transformed through patterns of interactions among interconnected elementary computational units [18].

One of the fields significantly transformed by DL is Reinforcement Learning (RL), which is the main focus of this thesis. The purpose of RL is to enable an agent to learn how to make decisions through interactions with an environment [65]. At each step, the agent observes the current state of the environment and selects an action. This action changes the state of the environment and results in a reward. The objective of the agent is then to learn a behavior that maximizes the cumulative rewards obtained over time. Due to their flexibility, NNs provide a powerful means of representing the agent's information-processing capabilities. Their use has consequently made it possible to apply RL to problems that were previously intractable [40, 64, 75].

The ability of RL to operate in complex environments introduces a new challenge. When an agent learns to act in such environments, it can develop complex strategies. Although such agents can achieve strong performance, these strategies can be difficult to understand. From an external perspective, we observe sequences of interactions between the agent and its environment, but these observations are not enough to infer the agent's strategy.

An alternative approach is to analyze the agent itself. However, due to their connectionist nature, NNs that model the agent's policy are not directly interpretable. The information behind a decision is distributed across many

units and connections making it impossible to trace. NNs are therefore often regarded as “black box” models [59].

To address this black-box problem, the field of Explainable Artificial Intelligence (XAI) has emerged [69, 82]. In the specific context of RL, this research area is referred to as Explainable Reinforcement Learning (XRL) [89, 116]. XRL develops methods for producing explanations that improve our understanding of agent behavior. Such explanations can serve several purposes, including increasing human trust, identifying biases and verifying that previously identified problems have been effectively corrected [82, 116].

This thesis presents XRL methods from different approaches to provide a broader view of the field. We focus on the mechanistic approach to explainability, which we refer to as Mechanistic Explainability (ME) (Definition 1.1.1). Although still relatively underexplored, this approach seems particularly promising to us. In ME, the goal is to understand how information is represented within the NN. From this perspective, explainability is not achieved despite the use of NNs, but through them.

Definition 1.1.1: Mechanistic Explainability

ME is the approach that seeks to explain the behavior of an NN by analyzing its internal mechanisms.

To understand how NNs work internally, ME studies the intermediate representations learned by these networks. These representations are called Latent Representations (LRs) [38]. Together, they form the network’s Latent Space (LS). Analyzing the structure of this LS provides insights into how information relevant to decision-making is organized.

To develop an XRL method based on ME, we need to address several research questions, which are detailed in the following section.

1.2 Research Questions

The first question concerns the analysis of LRs themselves. Numerous methods have been proposed to analyze the LSs. These methods originate from different research communities are rarely compared within a common framework. This observation leads to our first research question:

Research Question 1

What approaches have been proposed in the literature to analyze the internal workings of NNs?

The second question concerns the organization of existing research in XRL. Several taxonomies have been proposed to organize existing XRL methods [89, 116]. However, these taxonomies often rely on approaches initially developed for explainability in supervised settings. As a result, the organization of existing XRL methods remains ambiguous, making it difficult to structure the field. This observation leads to our second research question:

Research Question 2

How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of RL?

424

425 The analysis of existing XRL approaches also reveals a third limitation.
 426 Among existing methods for explaining RL agents, few adopt a mechanistic
 427 perspective. This observation leads to our third research question:

Research Question 3

How can the internal representations of NNs be exploited to identify the mechanisms underlying the decision-making process of RL agents?

428

429 The analysis of existing XRL methods highlights a fourth question concerning
 430 the evaluation of explanations. Although numerous approaches have been
 431 proposed to explain the behavior of RL agents, there is no widely accepted set of
 432 objective criteria for evaluating the quality and relevance of the explanations they
 433 produce. Existing evaluations often rely on user studies or qualitative analyses,
 434 making it difficult to systematically assess the validity of an explanation approach
 435 [51, 116].

436 This issue is particularly important for approaches that seek to explain agent
 437 behavior by analyzing LRs. Multiple hypotheses can be proposed about how
 438 information is structured within a LS. It is therefore important to define objective
 439 criteria for comparing these competing hypotheses. This leads to our fourth
 440 research question:

Research Question 4

How can the quality and relevance of explanations produced for RL agents be objectively quantified?

441

1.3 Contributions

442

443 To address these four research questions, this thesis is based on three main
 444 contributions.

445 The first contribution has not yet been published and addresses Research
 446 Question 1. In this contribution, we make explicit the working hypothesis that
 447 underlies much of the literature on LS analysis. We then organize existing LS
 448 analysis methods into families to provide an overview of the different approaches.

449 The second contribution addresses Research Question 2. This contribution
 450 takes the form of a first paper [112] and an extended English version [125],
 451 both of which were submitted for review and accepted for publication. In this
 452 contribution, we first clarify what explainability means in RL by providing a clear
 453 frame of reference. Building on these frame, we then conduct a comprehensive
 454 analysis of the state of the art in XRL.

455 The third contribution addresses Research Questions 3 and 4. In Alaarabiou
 456 et al. [126], which has been reviewed and accepted for publication, we introduce
 457 Clustering Agent Representation (CAR), an XRL method based on the principles
 458 of ME. By analyzing the LRs, CAR produces explanations in the form of clusters

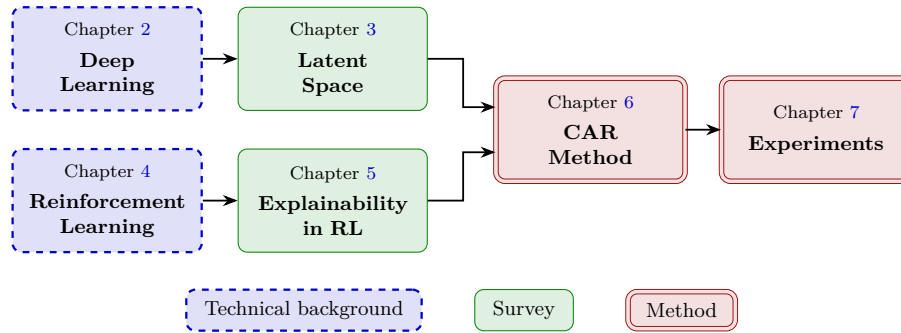


Figure 1.1: Structure of the thesis.

of states that are similar from the agent’s own perspective. We further show that independently trained networks develop similarly structured representations, indicating that these representations reflect properties of the task rather than of a particular network.

This thesis can be viewed as a synthesis of these three contributions.

1.4 Thesis Structure

This thesis brings together the contributions developed during this doctoral work. It is therefore written with the reviewers in mind while remaining as accessible as possible to readers. To this end, the background chapters introduce the concepts on which the subsequent chapters rely. Allowing readers to follow the thesis without having to consult external material.

The thesis is organized into a sequence of chapters, with each chapter falling into one of these three categories:

Technical background chapters introduce fundamental concepts required to understand the subsequent contributions.

Survey chapters review the existing literature around a specific research question. The organization adopted in each of these surveys are original and therefore constitute contributions of this thesis.

Method chapters present CAR, our ME method for RL.

Following these categories, the chapters are organized as shown in Figure 1.1 and detailed below:

Deep Learning (Chapter 2, technical background)

This chapter introduces the fundamental concepts of DL.

Latent Space Analysis (Chapter 3, survey)

This chapter addresses Research Question 1 by providing a survey of existing approaches for analyzing the LSs.

Reinforcement Learning (Chapter 4, technical background)

This chapter introduces the fundamental concepts of RL.

487 **Explainability in Reinforcement Learning (Chapter 5, survey)**

488 This chapter addresses Research Question 2 by reviewing existing approaches
489 to explainability in RL.

490 **The CAR Method (Chapter 6, method)**

491 This chapter addresses Research Questions 3 and 4 by proposing a new
492 method for explainability in RL based on the analysis of LRs.

493 **Evaluation of the CAR Method (Chapter 7, method)**

494 This chapter presents the implementation and experimental evaluation of the
495 proposed method.

496 **Conclusion**

497 This chapter summarizes the main contributions of the thesis, discusses the
498 limitations and presents possible directions for future research.

499 The mathematical notation used throughout this thesis is summarized on
500 page ix, and the acronyms are listed on page 137. The use of Large Language
501 Models (LLMs) in the preparation of this thesis is described in the LLM Use
502 Policy, on page 151.

503 Now that the research context and objectives have been presented, we can
504 turn to the first chapter, which introduces the fundamentals of DL.

Chapter 2

Deep Learning

This chapter introduces the fundamental concepts of Deep Learning (DL), the training paradigm used to optimize deep Neural Networks (NNs). The objective is to provide the necessary foundation in DL for the ideas developed throughout the rest of this thesis. Its structure is as follows:

Section 2.1: An overview of the historical developments that led to the emergence of the field.

Section 2.2: A formalization of the objective underlying DL algorithms.

Section 2.3: An examination of the structure of artificial NNs.

Section 2.4: An explanation of how the learning process shapes NNs.

Section 2.5: A discussion of how such models are evaluated.

Section 2.6: A summary of the concepts introduced in the chapter and a discussion of the theoretical limitations of DL.

2.1 Historical Developments

NNs and DL emerged from an ambition to study the expressive capabilities of organic brain. The first major milestone came in 1943 with the proposal of a mathematical model of an artificial neuron, with a formulation of networked interactions [6]. This work established the key idea that networks of simple units can give rise to complex computations. The focus was on representational power, not on the ability of such systems to learn. As a result, this work remained largely theoretical and had limited practical applications.

A major step toward operationalizing the idea of NNs was achieved with the perceptron algorithm in 1958 [9]. It was designed to adjust its parameters from input–output pairs in order to approximate a target function. The perceptron fundamental limitations were exposed in 1969, showing that it could not learn functions as simple as the exclusive-or [12]. At the same time, it was demonstrated that stacking multiple perceptrons could, in principle, represent such functions. However, no effective training procedure for these multi-layer architectures was available at the time. It was only in 1986 that researchers introduced gradient

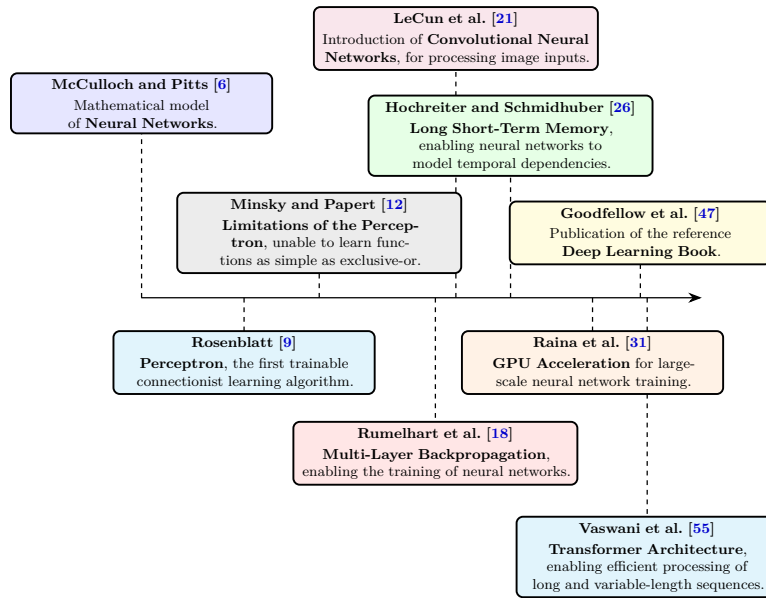


Figure 2.1: Evolution of deep learning.

descent as an effective method for training deep NNs [18]. From this point onward, the emergence of DL is retrospectively situated, even though the term itself appeared later. DL is defined as the training of deep NNs using gradient-based optimization methods.

It took several decades before DL reached the level of prominence it has today. One major obstacle was the absence of theoretical guarantees regarding the convergence of learning algorithms. Although the universal approximation theorem shows that NNs can represent a wide class of functions [20], it did not ensure that such representations could be effectively learned through optimization methods like gradient descent. This lack of rigorous guarantees initially reduced interest within the academic community. As a result, early progress relied largely on empirical breakthroughs.

These empirical advances were made possible by two key developments: the availability of large-scale datasets and significant improvements in computational infrastructure. DL methods require vast amounts of data to perform well, and the rise of the internet alongside the digitization of information enabled the creation of such datasets [34]. The use of graphics processing units for large-scale training, demonstrated in 2009, greatly increased computational capacity and made it feasible to train large-scale models [31, 36].

At the same time, researchers developed architectures specifically designed for different types of data, allowing DL to address an increasingly wide range of problems. Convolutional Neural Networks (CNNs) were introduced in 1989 to process images, using local, weight-shared filters that exploit the spatial structure of pixel grids [21]. Long Short-Term Memory (LSTM) networks were later proposed in 1997 to process temporal sequences, introducing gating mechanisms able to retain relevant information over long time horizons [26]. More recently, the Transformer architecture, relying entirely on the attention

mechanism, was introduced in 2017 to process long, variable-length sequences of strongly interconnected elements, allowing every element to attend directly to every other element regardless of their distance within the sequence [55]. This growing body of knowledge was consolidated in 2016 in a reference textbook that formalized DL as a coherent field of study [47].

These changes paved the way for the rise of DL as it is known today. DL has achieved remarkable success across a wide range of domains, including computer vision [36], complex games [64], content recommendation [46], bio-chemistry [95], as well as the now essential digital assistants based on Large Language Models (LLMs) [83]. The diversity of these applications explains the enthusiasm for this technology, which continues to redefine the boundaries of what was once considered automatable.

Figure 2.1 provides a timeline of the key milestones that shaped deep learning.

2.2 Objective

Now that the main developments that led to the rise of DL have been outlined, it is necessary to formalize what is actually being achieved when performing DL. To ensure clarity, the supervised learning setting will be considered, specifically applied to a regression task. This framework serves as a standard reference in the field, as all building blocks of DL applications are built upon this fundamental formalism.

In this setting, learning consists of progressively shaping a NN that serves as an approximation of a target function. At initialization, the network does not yet provide a meaningful representation of this function. Since a NN is a parametric function, its parameters are progressively updated during training so as to make it a better approximation of the target function. It gradually becomes capable of capturing the relationship between inputs and outputs through successive parameter updates during training. For this reason, DL is naturally situated within the framework of machine learning. A standard definition of machine learning is:

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E . [27]

When this formulation is adapted to DL, the following correspondences hold:

- The computer program corresponds to the learning algorithm, which is responsible for adjusting the network's parameters.
- The task T consists of learning a NN that provides a good approximation of the target function.
- The performance measure P corresponds to a function used to evaluate the quality of this approximation.
- The experience E takes the form of a dataset composed of input-output pairs derived from the function to be approximated.

The function to be approximated is denoted by f . As with any function, it defines a mapping from an input to an output: $\mathbf{x} \in \mathcal{X}$ denotes an input and

606 $\mathbf{y} \in \mathcal{Y}$ the corresponding output. The function f is therefore characterized
 607 by an input space $\mathcal{X}$ and an output space $\mathcal{Y}$, and can be formally written as
 608 $f : \mathcal{X} \rightarrow \mathcal{Y}$. For simplicity of exposition, both $\mathcal{X}$ and $\mathcal{Y}$ are assumed to be vector
 609 spaces throughout this thesis. This assumption is adopted solely to facilitate the
 610 presentation, as the framework can be readily extended to more general settings.
 611 The notation $\hat{\cdot}$ indicates that a given object is an approximation of another.
 612 Since the goal of the trained NN is to approximate f , this approximation is
 613 denoted by $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$. Given an input $\mathbf{x} \in \mathcal{X}$, the output of the approximating
 614 function $\hat{f}(\mathbf{x})$ is denoted by $\hat{\mathbf{y}}$.

615 It is important to emphasize that the target function f is generally not known
 616 explicitly. If it were, there would be no need to learn an approximation, since
 617 the true function would already provide the exact mapping. In practice, only a
 618 finite set of observations generated by f is available, organized in what is called
 619 a dataset. To distinguish between the different samples, an index i is introduced,
 620 allowing input-output pairs of the form $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$ to be defined. This leads
 621 to the following formulation for a dataset $\mathcal{D}$ of size N , where N denotes the
 622 number of available examples:

$$\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^N. \quad (2.1)$$

623 Now that the representation of the target function f has been established,
 624 the next question is what makes a function $\hat{f}$ a good approximation of it. For
 625 a given input $\mathbf{x}^{(i)}$, this means that the NN output $\hat{\mathbf{y}}^{(i)}$ should be close to the
 626 true output $\mathbf{y}^{(i)}$. To quantify the quality of this approximation, a function $\mathcal{L}$ is
 627 introduced that measures the discrepancy between predictions and target values.
 628 This function called the loss function plays a central role as it guides the learning
 629 process. In regression settings, a common choice is the Mean Squared Error
 630 (MSE), defined as:

$$\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) = \|\hat{\mathbf{y}} - \mathbf{y}\|_2^2. \quad (2.2)$$

631 Thanks to this loss, it is now possible to rigorously define the performance
 632 measure P . This performance measure corresponds to the average loss over the
 633 dataset and can therefore be written as:

$$P = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\hat{f}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)}). \quad (2.3)$$

634 This formulation captures the mathematical core of the learning objective,
 635 namely finding a NN $\hat{f}$ that minimizes the prediction error over the dataset $\mathcal{D}$.
 636 Since the target function f may take many different forms, $\hat{f}$ must be expressive
 637 enough to approximate a broad range of such functions. NN architectures are
 638 well suited to this requirement, as they can represent a wide range of functional
 639 mappings, whose composition is detailed in the following section.

640 2.3 Neural Network Structure

641 NNs are called connectionist models. Rather than modeling information process-
 642 ing through a set of explicit rules, they rely on a flow of signals distributed within
 643 a computational structure. Their architecture is composed of a large number
 644 of simple computational units, which operate on partial representations and

exchange signals across the network. A complex pattern of weighted connections between these units enables the transformation and propagation of information.

To simplify the presentation, the focus is placed on the Multilayer Perceptron (MLP) architecture. Although more specialized architectures exist, the MLP forms a fundamental building block of DL. All the essential mechanisms are present in it, making it a particularly suitable framework for introducing key concepts.

This section therefore begins by examining the artificial neuron, the fundamental computational unit underlying the MLP. It then explains how these neurons are organized into layers. Finally, it concludes with a general description of the NN as a whole.

2.3.1 Neurons

The artificial neuron is the basic building block of NNs. It is through these units that all computations within the network are carried out. A neuron receives a set of input signals from other neurons, where these inputs are represented as real-valued numbers. It produces an output, also a real-valued number, which is transmitted to subsequent neural units. Its internal state, called the activation, reflects its level of response and encodes the information processed for a given input. This activation, or lack thereof, is then used as input information by other neurons to determine their own activations, thereby propagating information through the network.

A neuron is characterized by:

- a set of incoming connections from other neurons,
- a weight associated with each of these connections,
- a bias term,
- a differentiable nonlinear activation function,
- a scalar activation state.

The activation of a neuron is determined by a weighted combination of its input signals, to which a bias is added, followed by the application of a nonlinear activation function. Mathematically, for a neuron receiving inputs $(x_1, \dots, x_n)$, its activation h is given by:

$$h = \sigma \left(\sum_{i=1}^n w_i x_i + b \right), \quad (2.4)$$

where:

- w_i is the weight associated with the i -th incoming connection,
- b is the bias,
- σ is the activation function.

Figure 2.2 shows a schematic view of this basic unit. The set of weights $w_1, \dots, w_i$ with the bias b constitute the parameters of the neuron, and they

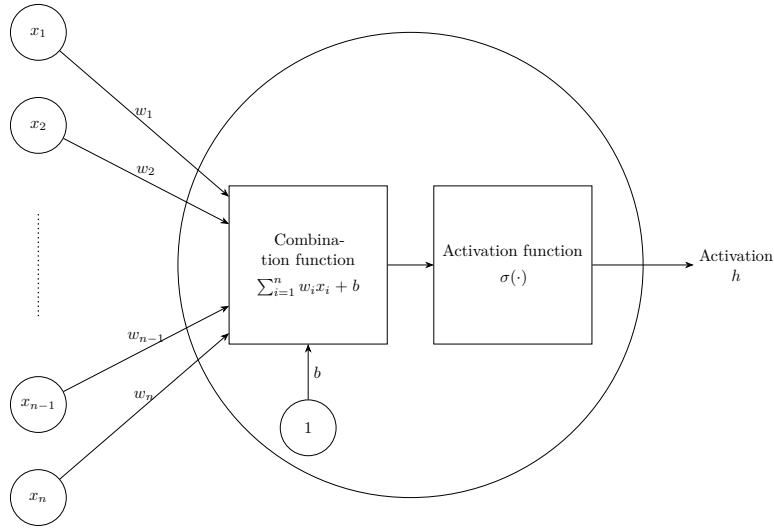


Figure 2.2: Artificial neuron.

are the elements that evolve during training. The magnitude of a weight w , relative to the others, indicates how strongly a neuron depends on the activation of the neurons to which it is connected. A large positive weight means that a high activation in the connected neuron tends to increase the activation of the receiving neuron, whereas a negative weight implies the opposite effect. In this way, the weights modulate the strength and nature of the connections between neurons, thereby defining the overall pattern of interaction within the network.

The activation function σ models how a neuron transforms incoming signals into its output activation. Originally, this function was inspired by biological neurons, with the goal of mimicking the firing behavior of organic cells. However, modern DL no longer relies on this biological interpretation. In practice, any function that is nonlinear, differentiable, and computationally efficient can be used. Nonlinearity is a crucial property, since without it a NN would reduce to a composition of linear transformations, and would therefore be unable to approximate complex functions.

2.3.2 Layers

In MLP architectures, neurons are organized into successive layers, as illustrated in Figure 2.3. This layered structure determines how neurons are connected. In this setting, each neuron receives inputs from all neurons in the preceding layer and sends its output to all neurons in the following layer. As a result, neurons within the same layer do not interact with each other.

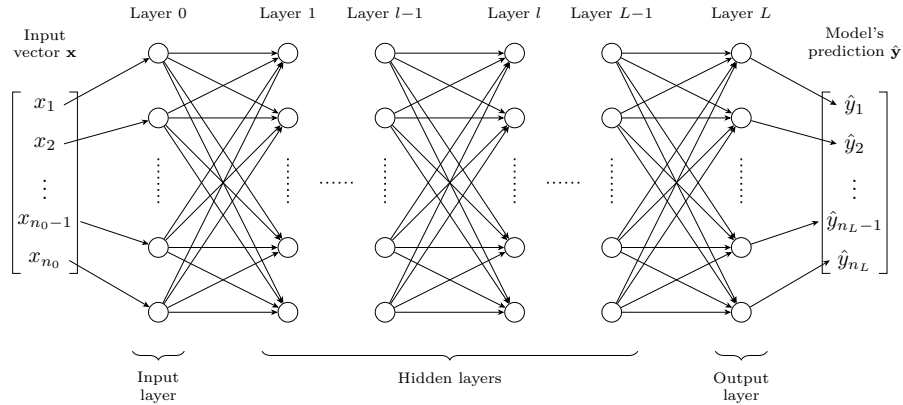


Figure 2.3: Multilayer Perceptron Architecture.

Three types of layers are distinguished:

- an *input layer*, which directly receives input,
- one or more *hidden layers* (also referred to as intermediate layers), which perform intermediate transformations,
- an *output layer*, which produces the final prediction.

The input and output layers play a specific role. The input layer has no preceding layer, as its neurons directly encode the components of the input vector $\mathbf{x}$. Conversely, the output layer has no subsequent layer, and its neurons produce the model's prediction $\hat{\mathbf{y}}$. This structure imposes a direct correspondence between the dimensionality of the input space and the number of neurons in the input layer, and similarly between the output space and the number of neurons in the output layer.

When describing NNs, especially in MLPs, it is more common to reason at the level of layers rather than individual neurons. This perspective allows interactions between layers to be modeled in a matrix form. Such a representation enables efficient parallel computation and forms the basis of modern DL implementations [36]. Within this framework, it is useful to introduce the notion of layer activations. The activation of a layer corresponds to the activations of all its neurons, organized into a vector. For a layer l composed of n_l neurons, the activations are given by:

$$\mathbf{h}^{(l)} = [h_1^{(l)}, \dots, h_{n_l}^{(l)}]. \quad (2.5)$$

Using this formulation, the interaction between two consecutive layers can be written compactly as:

$$\mathbf{h}^{(l)} = \sigma^{(l)} \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad (2.6)$$

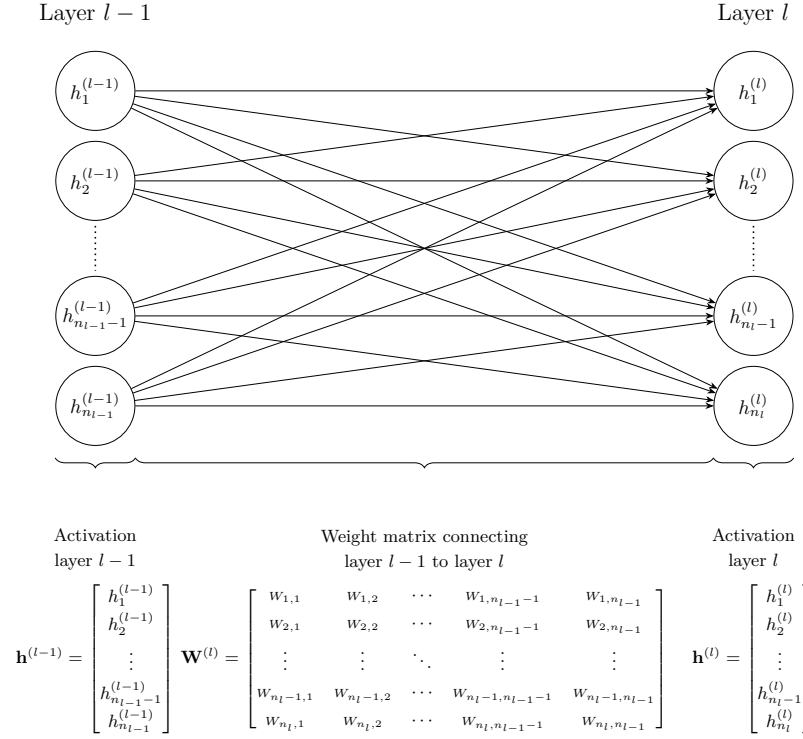


Figure 2.4: Interaction between layers.

where:

- $\mathbf{h}^{(l-1)}$ represents the activations vector of the previous layer,
- $\mathbf{W}^{(l)} \in \mathbb{R}^{n_l \times n_{l-1}}$ is the weight matrix connecting layer $l - 1$ to layer l ,
- $\mathbf{b}^{(l)} \in \mathbb{R}^{n_l}$ is the bias vector associated with layer l ,
- $\sigma^{(l)}$ is the activation function applied element-wise to the neurons of layer l .

This matrix-vector representation is visualised in Figure 2.4, which highlights how each neuron in layer l receives weighted signals from all neurons in the previous layer.

It is important to examine the role played by the intermediate layers of a NN. At least one hidden layer is required to represent sufficiently complex functions, and in practice, modern architectures often include many more. Theoretical results have shown that, for a fixed number of parameters, increasing the number of layers enhances the expressive capacity of NNs [48]. Each layer can be viewed as an additional stage in the computation performed by the network. This suggests that deeper networks can represent increasingly complex functions by composing a larger number of transformations. The precise relationship between depth and function complexity remains primarily supported by empirical observations rather than formal theoretical justification [47].

2.3.3 Network

With a layer-wise representation, the propagation of information can be described as a sequence of transformations applied from the input layer (indexed 0) to the output layer (indexed L):

$$\hat{f}(\mathbf{x}) = \mathbf{h}^{(L)} = \sigma^{(L)} \left(\mathbf{W}^{(L)} \sigma^{(L-1)} \left(\dots \sigma^{(1)} \left(\mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right) \dots \right) + \mathbf{b}^{(L)} \right) = \hat{\mathbf{y}}. \quad (2.7)$$

For a more compact description of information propagation, the layer activations are defined recursively as:

$$\mathbf{h}^{(l)} = \begin{cases} \mathbf{x} & \text{if } l = 0, \\ \sigma^{(l)} \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right) & \text{if } 1 \leq l \leq L. \end{cases} \quad (2.8)$$

To simplify notation, it is convenient to group all parameters of the network into a single set. The model parameters are thus denoted by:

$$\boldsymbol{\theta} = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^L. \quad (2.9)$$

In this thesis, $\boldsymbol{\theta}$ is a vector in $\mathbb{R}^d$, where d denotes the total number of parameters, such that $\boldsymbol{\theta} = [\theta_1, \dots, \theta_d]$. The notation $\hat{f}_{\boldsymbol{\theta}}$ expresses the dependence of the NN on its parameters. Increasing the number of parameters, either by adding more neurons per layer or by increasing the number of layers, can enhance the expressive power of the model, allowing it to approximate more complex functions. However, this also increases computational cost and training time.

It is important to distinguish between the architecture and the parameters of a NN. When referring to the architecture of $\hat{f}$, the number of layers, the number of neurons per layer, and the choice of activation functions are considered. The parameters correspond to $\boldsymbol{\theta}$, which controls the strength of the connections between neurons. The following sections examine how these parameters can be adjusted to obtain a better approximation of the target function.

2.4 Learning

Now that the structure of a NN has been detailed, the next step is to examine how its parameters are adjusted to accomplish the assigned task. This brings the discussion to the core of DL, namely the learning process itself. As a reminder, the objective of training a NN is to adjust its parameters so that the network provides a good approximation of the target function. Mathematically, this objective is expressed as:

$$\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^N \mathcal{L} \left(\hat{f}_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)} \right). \quad (2.10)$$

As shown in Equation (2.10), the loss for example i depends on the input-target pair $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})$, the loss function $\mathcal{L}$, the network architecture $\hat{f}$, and the parameters $\boldsymbol{\theta}$. Since the dataset, loss function, and network architecture are fixed during training, only the parameters $\boldsymbol{\theta}$ are optimized. To simplify the

notation, only the dependence on $\boldsymbol{\theta}$ is kept, and the loss for example i is written as

$$\mathcal{L}^{(i)}(\boldsymbol{\theta}) = \mathcal{L}\left(\hat{f}_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)}\right). \quad (2.11)$$

Now that the learning objective has been defined, the next question is how to modify the parameters $\boldsymbol{\theta}$ to minimize the loss. The NN architecture $\hat{f}$ used to approximate the target function f is highly complex. It involves a large number of parameters as well as many nonlinear components. As a result, the optimum cannot be determined analytically by directly studying the properties of the function.

Instead, approximate optimization methods are required to find suitable parameter values for NNs. The class of algorithms used for this optimization is known as gradient descent methods. Numerous variants of gradient descent have been developed for NN training. For simplicity, the focus here is on Stochastic Gradient Descent (SGD) with a batch size of 1. As in previous cases, this choice is made because this setting captures the essential intuition behind the training method.

This section presents the learning process in three steps. The first derives the gradient of the loss using the chain rule. The second shows how this gradient is used to update the network parameters. The third discusses the convergence properties of the resulting algorithm.

2.4.1 Computing the Gradient

The first step of SGD is to randomly initialize the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$ for a given network architecture $\hat{f}$. This initial state is denoted by $\boldsymbol{\theta}_1$. The goal is to iteratively update this vector so that the network gets better at approximating the target function. To do so, it is necessary to quantify the impact of each parameter θ_j on the loss $\mathcal{L}^{(i)}(\boldsymbol{\theta})$. This is done using the partial derivative of the loss with respect to each parameter, denoted by $\frac{\partial \mathcal{L}^{(i)}(\boldsymbol{\theta})}{\partial \theta_j}$.

Computing this derivative is not straightforward because a NN is composed of a sequence of functions, and the parameter θ_j can belong to any layer of the network. Its influence on the loss must therefore be propagated through all subsequent layers. To describe these dependencies, each layer can be viewed as a function, denoted by $\phi^{(l)}$ for layer l . This function takes the activation vector $\mathbf{h}^{(l-1)} \in \mathbb{R}^{n_{l-1}}$ produced by the previous layer as input. It produces the activation vector $\mathbf{h}^{(l)} \in \mathbb{R}^{n_l}$ of the current layer. This can be expressed as

$$\phi^{(l)}: \mathbb{R}^{n_{l-1}} \rightarrow \mathbb{R}^{n_l}, \quad \mathbf{h}^{(l)} = \phi^{(l)}(\mathbf{h}^{(l-1)}). \quad (2.12)$$

Each layer function can have multiple inputs and multiple outputs, so the ordinary derivative is not sufficient to describe how its outputs change with respect to its inputs. The Jacobian matrix extends the derivative to functions with multiple inputs and outputs, quantifying how each output component changes with respect to each input component. Consider the function $\phi^{(l)}$ and its Jacobian $\mathbf{J}_{\phi^{(l)}}$ evaluated at $\mathbf{h}^{(l-1)}$. The Jacobian collects the partial derivatives

of each output component with respect to each input component:

$$\mathbf{J}_{\phi^{(l)}}(\mathbf{h}^{(l-1)}) = \begin{bmatrix} \frac{\partial h_1^{(l)}}{\partial h_1^{(l-1)}} & \cdots & \frac{\partial h_1^{(l)}}{\partial h_{n_{l-1}}^{(l-1)}} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_{n_l}^{(l)}}{\partial h_1^{(l-1)}} & \cdots & \frac{\partial h_{n_l}^{(l)}}{\partial h_{n_{l-1}}^{(l-1)}} \end{bmatrix}. \quad (2.13)$$

Since a NN is composed of several layers, a change in an early layer can affect the outputs of all subsequent layers. It is therefore necessary to propagate this change through the sequence of layer functions until it reaches the network output. The chain rule provides the mathematical tool required to describe this propagation. To illustrate this, consider two consecutive layers, $\phi^{(l)}$ and $\phi^{(l+1)}$. The chain rule states that the Jacobian of their composition is given by the product of their Jacobians. Each Jacobian is evaluated at the input received by the corresponding layer:

$$\mathbf{J}_{\phi^{(l+1)} \circ \phi^{(l)}}(\mathbf{h}^{(l-1)}) = \mathbf{J}_{\phi^{(l+1)}}(\mathbf{h}^{(l)}) \mathbf{J}_{\phi^{(l)}}(\mathbf{h}^{(l-1)}). \quad (2.14)$$

The network is composed of a sequence of layer functions $\phi^{(1)}, \dots, \phi^{(L)}$. Each layer depends only on the output of the previous layer. Consider a parameter θ_j belonging to layer l . A change in θ_j first affects the output $\mathbf{h}^{(l)}$ of its own layer. This change then propagates through the subsequent layers, affecting $\mathbf{h}^{(l+1)}$, and so on. The effect reaches the output $\mathbf{h}^{(L)}$, from which the loss $\mathcal{L}^{(i)}(\boldsymbol{\theta})$ is computed. This layer-by-layer propagation is referred to as backpropagation:

$$\frac{\partial \mathcal{L}^{(i)}(\boldsymbol{\theta})}{\partial \theta_j} = \mathbf{J}_{\mathcal{L}}(\mathbf{h}^{(L)}, \mathbf{y}^{(i)}) \mathbf{J}_{\phi^{(L)}}(\mathbf{h}^{(L-1)}) \cdots \mathbf{J}_{\phi^{(l+1)}}(\mathbf{h}^{(l)}) \mathbf{J}_{\phi^{(l)}}(\theta_j), \quad (2.15)$$

where:

- $\mathbf{J}_{\mathcal{L}}(\mathbf{h}^{(L)}, \mathbf{y}^{(i)})$ denotes the Jacobian of $\mathcal{L}$ with respect to its first argument, evaluated at $(\mathbf{h}^{(L)}, \mathbf{y}^{(i)})$,
- each subsequent $\mathbf{J}_{\phi^{(m+1)}}(\mathbf{h}^{(m)})$ denotes the Jacobian of the function $\phi^{(m+1)}$ with respect to its input $\mathbf{h}^{(m)}$,
- $\mathbf{J}_{\phi^{(l)}}(\theta_j)$ denotes the Jacobian of the function $\phi^{(l)}$ with respect to the parameter θ_j .

Once $\frac{\partial \mathcal{L}^{(i)}(\boldsymbol{\theta})}{\partial \theta_j}$ can be computed for every parameter, all of these values are stacked into one vector, called the gradient:

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}^{(i)}(\boldsymbol{\theta}) = \left[\frac{\partial \mathcal{L}^{(i)}(\boldsymbol{\theta})}{\partial \theta_1}, \dots, \frac{\partial \mathcal{L}^{(i)}(\boldsymbol{\theta})}{\partial \theta_d} \right]. \quad (2.16)$$

Each component answers the same question for a different parameter, indicating whether a small increase in that parameter increases or decreases the loss, and by how much. A positive value means increasing the parameter increases the loss, whereas a negative value means the opposite. The gradient thus indicates, for every parameter, which direction to move in order to reduce the loss.

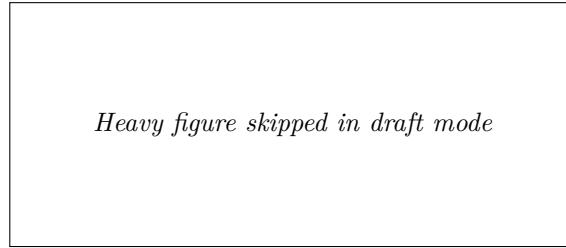


Figure 2.5: *Figure skipped in draft mode (set `\draftfiguresfalse` to render it).*

843 2.4.2 Update rule

844 Now that information is available on how changes in the parameters affect the
 845 loss, the objective is to minimize it. To do so, the parameters are updated in
 846 the direction opposite to the gradient. Indeed, the gradient is followed in the
 847 opposite direction because it indicates the direction in which the loss increases,
 848 whereas the objective is to reduce it. This leads to the gradient descent update
 849 rule:

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}^{(i)}(\boldsymbol{\theta}_k), \quad (2.17)$$

850 where $\eta > 0$ is the learning rate, which controls the step size, and $\nabla_{\boldsymbol{\theta}} \mathcal{L}^{(i)}(\boldsymbol{\theta}_k)$
 851 denotes the gradient of the loss with respect to the parameters, evaluated at the
 852 current values $\boldsymbol{\theta}_k$.

853 The parameter η must remain small, as the gradient only provides reliable
 854 information in a local neighborhood of the current parameters. As a result,
 855 a single step is not sufficient to significantly reduce the loss, so the process is
 856 iterated many times. Producing a sequence of parameter vectors $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots$ that
 857 progressively improve the model. In practice, training is stopped when successive
 858 updates no longer produce a significant decrease in the loss.

859 Algorithm 1 summarizes this whole training procedure.

860 To develop an intuition for this update rule, Figure 2.5 illustrates it applied
 861 to the simple convex loss function $\mathcal{L}(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2$, where the red path shows
 862 the parameter updates converging to the minimum. The point $\boldsymbol{\theta}_1$ represents the
 863 initial parameter values, with θ_1 and θ_2 randomly initialized, and the subsequent
 864 points, from $\boldsymbol{\theta}_2$ to $\boldsymbol{\theta}_6$, represent the successive updates produced by applying
 865 Equation (2.17), progressively moving the trajectory toward the minimum of the
 866 loss function.

867 2.4.3 Convergence Properties

868 Having established how SGD updates the parameters, a natural question is
 869 whether this procedure is guaranteed to converge. It has been mathematically
 870 shown that gradient descent applied to a convex loss with respect to the param-
 871 eters converges to a global optimum, independently of the initialization. However,
 872 NN training involves highly non-convex loss surfaces, so the optimization trajec-
 873 tory becomes strongly dependent on the initial conditions, and no such guarantee
 874 holds. Figure 2.6 illustrates the effect of non-convex functions, different starting
 875 points lead to distinct local minima.

Algorithm 1: Stochastic Gradient Descent**Input:**Dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^N$ Architecture $\hat{f}$ Loss function $\mathcal{L}$ Learning rate $\eta > 0$ Initial parameters θ (optional)**Output:**Trained parameters θ

```

1 if  $\theta$  is given then
2   |  $\theta_1 \leftarrow \theta$ ;
3 else
4   | Initialize  $\theta_1$  randomly;
5  $k \leftarrow 1$ ;
6 while stopping criterion not satisfied do
7   | Select a random example  $(\mathbf{x}^{(i)}, \mathbf{y}^{(i)}) \in \mathcal{D}$ ;
8   | Model prediction for the selected example
   |  $\hat{\mathbf{y}}^{(i)} \leftarrow \hat{f}_{\theta_k}(\mathbf{x}^{(i)})$ ;
9   | Per-example loss between the prediction and the ground truth
   |  $\mathcal{L}^{(i)}(\theta_k) \leftarrow \mathcal{L}(\hat{\mathbf{y}}^{(i)}, \mathbf{y}^{(i)})$ ;
10  | Gradient of the per-example loss, since it is differentiable
   |  $\mathbf{g} \leftarrow \nabla_{\theta} \mathcal{L}^{(i)}(\theta_k)$ ;
11  | Error minimization via a gradient descent step
   |  $\theta_{k+1} \leftarrow \theta_k - \eta \mathbf{g}$ ;
12  |  $k \leftarrow k + 1$ ;
13 Return  $\theta_k$ 

```

876 Despite the lack of theoretical guarantees of convergence to a global optimum,
877 the simple procedure of random initialization followed by iterative gradient-based
878 updates is sufficient in practice to train complex NNs.

879 2.5 Evaluation

880 Now that the process of training a model has been described, the question of
881 how to evaluate it must be addressed. In the supervised learning setting, this
882 evaluation relies on two aspects:

- 883 • its performance on the data it was trained on,
- 884 • its performance on data it was not trained on.

885 The first aspect measures how well the loss minimization process has suc-
886 ceeded. The second measures how well the model generalizes. Indeed, the dataset
887 $\mathcal{D}$ represents a small subset of all possible inputs. To estimate how the model

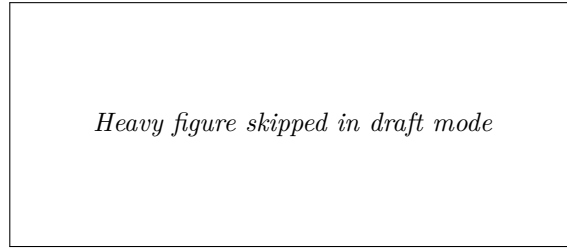


Figure 2.6: *Figure skipped in draft mode (set `\draftfiguresfalse` to render it).*

will perform in general, it is important to evaluate how it performs on data it has not seen during training.

In order to quantify these two aspects, the dataset is split into two subsets:

- $\mathcal{D}_{\text{train}}$ is the set on which the model minimizes its loss during the training.
- $\mathcal{D}_{\text{test}}$ is kept aside during training in order to evaluate the model's ability to generalize.

When training is completed, the first step is to verify that the model performs well on the data it was trained on. To do so, the average loss on the training set, denoted $\bar{\mathcal{L}}_{\text{train}}$, is studied. This is an instance of the performance measure P defined in Equation (2.3), restricted to $\mathcal{D}_{\text{train}}$:

$$\bar{\mathcal{L}}_{\text{train}} = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)}) \in \mathcal{D}_{\text{train}}} \mathcal{L}(\hat{f}_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)}). \quad (2.18)$$

If $\bar{\mathcal{L}}_{\text{train}}$ is not sufficiently low, this means that the model has not been able to learn the task. It is not necessary to go further in the analysis of the model's quality. Before restarting the training process, various design choices are then adjusted. Based on what has been discussed so far in this chapter, these include modifications to the number of layers, the number of neurons per layer, activation functions, the learning rate. However, in practice, there are many other possible variations. There is no universal rule for choosing the appropriate value for each of them. They are therefore chosen empirically, based on models that perform well on similar tasks, combined with intuition. Their selection is often costly and remains a challenging problem in DL.

Assuming now that the model achieves satisfactory performance on the training dataset $\mathcal{D}_{\text{train}}$. Its ability to generalize must now be studied. To do so, the corresponding average loss on $\mathcal{D}_{\text{test}}$ is computed:

$$\bar{\mathcal{L}}_{\text{test}} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)}) \in \mathcal{D}_{\text{test}}} \mathcal{L}(\hat{f}_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)}). \quad (2.19)$$

If $\bar{\mathcal{L}}_{\text{test}} \gg \bar{\mathcal{L}}_{\text{train}}$, the model fails to generalize. This situation most often arises due to a phenomenon known as overfitting. In such cases, the model does not learn general patterns but instead memorizes the training examples. This typically occurs when the number of parameters is too large relative to

the amount of training data. The simplest remedy is therefore to reduce the number of parameters in the model. However, it is still necessary to ensure that the model remains sufficiently expressive to perform well on the data.

If $\bar{\mathcal{L}}_{\text{test}} \approx \bar{\mathcal{L}}_{\text{train}}$, this indicates that the training procedure has successfully produced a NN that generalizes well beyond the training data [105]. The resulting model can therefore be considered suitable for the task for which it was trained, marking the end of the NN training process.

2.6 Conclusion

This chapter has introduced the fundamental concepts underlying DL, including the general objective of NN models, their structure, the training procedure, and their evaluation. For clarity of presentation, several simplifying assumptions have been made throughout this chapter. The problem has been considered in a regression setting, using a MLP trained with SGD and a batch size of one. These choices were made deliberately to expose the core mechanisms of DL as clearly as possible.

Despite these simplifications, the framework introduced here captures mechanisms that are shared across a broad range of DL methods: A sequence of parameter updates, each based on information about the loss around the current parameters. The apparent simplicity of this learning procedure therefore contrasts with the complexity of the functions that NNs can represent. Understanding how such complex behavior emerges from a relatively simple optimization process remains an important challenge in the study of DL.

Mathematical tools allow us to describe the functions represented by NNs and the procedures used to train them. However, they do not yet fully explain their remarkable empirical performance. Understanding the origin of this success therefore requires looking beyond the optimization procedure itself.

A common feature across NN architectures is the presence of intermediate representations. Whether considering a MLP, a CNN, an LSTM, or an architecture based on attention, the input is progressively transformed into intermediate representations before producing the final output. Their presence provides a common perspective for understanding how NNs transform information. The collection of these intermediate representations is referred to as the Latent Space (LS). Studying this LS can provide insights into how NNs learn and generalize, which motivates the following chapter.

Chapter 3

Latent Space Analysis

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep Learning (DL) introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, Neural Networks (NNs) gradually construct representations that become increasingly relevant to the task under consideration [18].

These representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training. Since these representations are learned internally by the network and are not directly observable from the input or output spaces, they are commonly referred to as Latent Representations (LRs). The collection of these LRs learned throughout the network is referred to as a Latent Space (LS).

These LSs are now at the core of a growing number of practical applications. Neural video compression exploits the compactness of LSs to encode information efficiently [72]. In Reinforcement Learning (RL), agent internal representations are used for efficient exploration in extremely open environments [56]. Recommendation systems model users and content within a shared LS to compute meaningful similarities [53]. Retrieval-augmented generation systems index and query knowledge bases through LRs [85]. Finally, Large Language Models (LLMs) and multimodal architectures rely on the quality of learned representations to align heterogeneous modalities such as text, images, and audio [97]. These examples represent only some of the most widespread applications of LRs.

Given their extensive use, the study of LRs has become a central topic in DL research. A major objective is to understand how information is organized within these spaces, which concepts are encoded, and how these LRs influence model behavior.

The remainder of this chapter is organized as follows:

- Section 3.1 introduces the hypothesis that underlies much of the research on LS analysis, which we refer to as the *Concept Hypothesis* (Hypothesis 1).
- Sections 3.2, 3.3, 3.4, and 3.5 examine four families of LS analysis methods, respectively based on probing, interpretable directions, feature superposition, and clustering.
- Section 3.6 concludes with a comparative synthesis of these methods and a discussion of the open questions that remain.

3.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on LS analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, LS, concept, and semantics. Throughout this thesis, we refer to this hypothesis as *Concept Hypothesis* (Hypothesis 1). The goal of this section is to make this conceptual framework explicit by defining its main components.

The *Concept Hypothesis* (Hypothesis 1) originates from the need to explain the generalization ability of deep NNs observed after training, as discussed in Section 2.5. A model is said to generalize when, for an input $\mathbf{x} \in \mathcal{X}$ not seen during training, it still produces an output $\hat{\mathbf{y}}$ close to the true output $\mathbf{y}$. Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 3.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task [38].

Definition 3.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction.

The ability of NNs to perform abstraction arises from their intermediate activations. A NN does not directly map $\mathbf{x}$ to $\hat{\mathbf{y}}$, it computes a sequence of intermediate activations across hidden layers. These activations are numerical encodings of the input referred to as embeddings. Once training is complete, these embeddings are structured to support the performance of a specific task. To mark this distinction, we will refer to these trained embeddings as LRs, as defined in Definition 3.1.2.

Definition 3.1.2: Latent Representation

A LR is the internal encoding of an input given by an intermediate layer of a trained NN.

Each hidden layer refines the LR of the previous layer. Each transformation progressively reshapes the LR into a form more directly aligned with the output

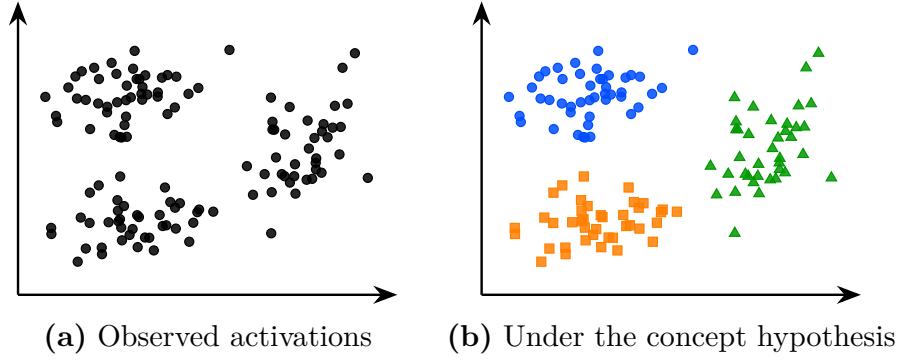


Figure 3.1: The same latent space activations seen without (a) and under (b) the *Concept Hypothesis* (Hypothesis 1).

space [18]. This process can be described as a sequence of transformations where each $\mathbf{h}^{(l)}$ is a LR:

$$\mathbf{x} \rightarrow \mathbf{h}^{(1)} \rightarrow \mathbf{h}^{(2)} \rightarrow \dots \rightarrow \mathbf{h}^{(L-1)} \rightarrow \hat{\mathbf{y}}$$

The set of all such LRs produced by a layer over a dataset defines its LS, as defined in Definition 3.1.3.

Definition 3.1.3: Latent Space

The LS is the set of all LRs produced by a layer over a dataset. It is shaped by learning and organizes LRs according to task-relevant properties.

Under the *Concept Hypothesis* (Hypothesis 1), the analysis of the LS is assumed to provide access to abstract structures learned by the model, as illustrated in Figure 3.1. The same set of LRs can be viewed either as an unstructured point cloud or, under this hypothesis, as a collection of points organized by concept membership, in which case the structure of the LS becomes meaningful and amenable to analysis. In Figure 3.1, the colors represent hypothetical concept memberships assigned by the model. In this thesis, a concept is defined as an abstract representation that organizes information and supports reasoning (see Definition 3.1.4).

Definition 3.1.4: Concept

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

Our definition of concepts is consistent with both the pragmatic tradition in philosophy and the information-processing view of cognition. Within these frameworks, concepts arise from the need to simplify an otherwise intractable reality. Because the environment contains more information than any cognitive system can process, intelligent behavior relies on abstractions, called concepts, that discard irrelevant details while preserving the information required to achieve a given objective [1, 39]. These concepts are realized as representations that reduce complexity and enable reasoning under limited computational resources [15, 25].

1039 Although these theories were originally developed to explain human cognition,
 1040 the *Concept Hypothesis* (Hypothesis 1) assumes that the same principles can be
 1041 extended to deep NNs.

1042 In order to influence the model’s computations, concepts must necessarily
 1043 admit a concrete realization within the LS. We refer to this realization as the
 1044 Concept Structure (CS). The definition of the CS is provided in Definition 3.1.5.

Definition 3.1.5: Concept Structure

A CS is the specific form for how a concept is instantiated within the LS.

1045

1046 The methods reviewed in this chapter all build on the *Concept Hypothe-*
 1047 *sis* (Hypothesis 1) formalized in Hypothesis 1, but differ in how they characterize
 1048 the CS of a concept. This choice directly determines how concepts can be
 1049 identified and extracted from the LS. As a result, several families of LS analysis
 1050 methods have emerged in the literature, each adopting a different view of how
 1051 concepts are structured within the LS.

Hypothesis 1: Concept Hypothesis

NNs generalize by performing abstraction. Through training, their LS be-
 comes organized around concepts. Concepts are task-dependent abstractions
 that capture regularities relevant to the task. Analyzing the LS therefore
 provides access to the CSs through which these concepts are instantiated in
 the model.

1052

1053 Now that the definition of a *Concept Hypothesis* (Hypothesis 1) has been
 1054 established, we can consider its relationship to human. According to our def-
 1055 inition, we can imagine concepts that are useful for the machine but do not
 1056 make sense to humans. This relationship is captured by the notion of semantics
 1057 [110], defined in Definition 3.1.6. Some concepts developed in LS can be readily
 1058 associated with human concepts and are said to have high semantic value. Others
 1059 may contribute to task performance while remaining difficult for humans to
 1060 understand, and are said to have low semantic value.

Definition 3.1.6: Semantics

Semantics refers to the alignment between concepts in the model LS and
 concepts used by a human observer. A concept is semantic if it can be
 associated with a human interpretable concept.

1061

1062 It is important to note that this conceptual framework is not supported
 1063 by a complete theoretical understanding of deep NNs. Current mathematical
 1064 tools do not provide a rigorous characterization of their internal representations
 1065 and do not formally prove that concepts emerge as identifiable CSs within LS.
 1066 Nevertheless, this perspective constitutes a widely adopted working hypothesis
 1067 in the field of representation and LS analysis.

1068 The remainder of this chapter is devoted to reviewing methods for analyzing
 1069 LSs and extracting the concepts they encode. We begin with probing, one of
 1070 the most direct approaches for assessing whether concepts are represented in the
 1071 LS.

3.2 Probing

Probing methods aim to determine whether a given semantic concept is encoded within the LS of a NN [109, 115]. The idea behind these methods is that if a concept can be reliably decoded from LRs, it is likely relevant to the problem being solved. Studying such concepts can therefore provide insight into how the network internally represents the problem.

A probing analysis begins by identifying a set of semantic concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concepts. We denote by c_j the presence or absence of concept j in example $\mathbf{x}$. The dataset can therefore be written as follows:

$$\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)}, c_1^{(i)}, \dots, c_n^{(i)})\}_{i=1}^N.$$

The data are subsequently projected into the LS of the pretrained model. For an input $\mathbf{x}$, we write $\mathbf{h}^{(l)}(\mathbf{x}) \in \mathbb{R}^{n_l}$, or simply $\mathbf{h}^{(l)}$, the LR obtained at layer l . A classifier, referred to as a probe, is then trained to predict the presence of a given concept from this LR. It is denoted by $\psi_{\boldsymbol{\theta}}^{(l,j)}$, where $\boldsymbol{\theta}$ represents its parameters, l indicates the layer from which the LR is extracted, and j identifies the concept being predicted. The probe maps the LR to a scalar value in $[0, 1]$, which can be interpreted as the probability that the concept is present. The parameters $\boldsymbol{\theta}$ are learned by minimizing a loss function over the training examples using Stochastic Gradient Descent (SGD). This gives the following mathematical formulation:

$$\psi_{\boldsymbol{\theta}}^{(l,j)} : \mathbb{R}^{n_l} \rightarrow [0, 1], \quad \min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^N \mathcal{L} \left(\psi_{\boldsymbol{\theta}}^{(l,j)} \left(\mathbf{h}^{(l)}(\mathbf{x}^{(i)}) \right), c_j^{(i)} \right). \quad (3.1)$$

The architecture of the probe must be kept intentionally simple. A probe with excessive expressive capacity may be able to predict the concept by exploiting complex patterns within the representation. These patterns do not necessarily indicate that the concept is explicitly encoded in the LS. Instead, they may result from the ability of the probe itself to combine and transform the available information. Consequently, a highly expressive probe makes it difficult to distinguish between information that is directly accessible in the representation and information that is reconstructed by the classifier. Restricting the capacity of the probe therefore helps ensure that its predictions primarily reflect the information already present in the LS.

For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the NN and potentially usable during decision-making. If the classifier achieves performance significantly above chance level, this suggests that the LS contains exploitable information related to the considered concept.

Figure 3.2 illustrates the decision boundaries of such linear probes for three concepts, where each colored region indicates the area classified positively by the corresponding probe.

An application of this technique can be found in studies investigating whether LLMs develop internal representations of their environment [109, 115]. In these works, researchers train LLM architectures to play games such as chess or Othello in a purely textual setting. The model is provided only with sequences of moves

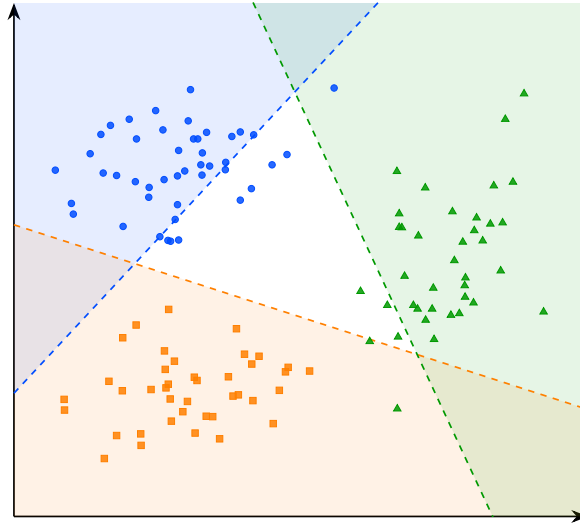


Figure 3.2: Decision boundaries of the linear probes for concepts A, B, and C.

1114 and information about the actions it can take, all expressed in textual form. The
 1115 central question is whether the LLM develops an internal representation of the
 1116 game state. To address this question, the probed concepts correspond to the
 1117 presence or absence of different game pieces on specific board positions.

1118 As illustrated in Figure 3.3, probes are able to reconstruct the full board
 1119 state from the model’s LR, even though the only available information to the
 1120 model is a sequence of text. In this figure, each row corresponds to a piece type,
 1121 comparing the ground truth with the probe’s predictions and confidence. The
 1122 close agreement between the ground truth and predicted positions suggests that
 1123 the model develops a structured representation of the underlying environment
 1124 from which the data are generated.

1125 However, the fact that a concept can be decoded from LR does not necessarily
 1126 imply that the model actually uses this information to perform its task. To assess
 1127 whether LR plays a causal role in the model’s output, it is possible to perform
 1128 interventions on them. An intervention consists of modifying a LR in a controlled
 1129 manner by activating or deactivating a given concept, and then studying its
 1130 impact on the model’s output. If the resulting change in the model’s output is
 1131 consistent with the applied perturbation, this suggests that the representation
 1132 had a causal role.

1133 To this end, the LR is minimally modified such that the probe prediction for
 1134 the target concept c_j changes state, while the predictions associated with other
 1135 concepts remain unchanged. This goal can be formulated as an optimization
 1136 problem over the LR itself. The optimization is performed directly on the LR,

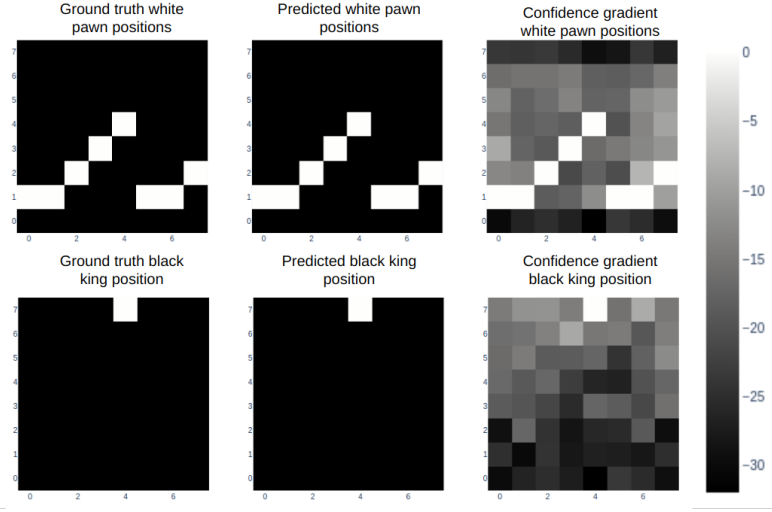


Figure 3.3: Probing analysis of a large language model trained to play chess [115].

1137 using SGD. The solution to this problem is the optimal modified LR, defined as:

$$\begin{aligned}
 \mathbf{h}^{(l)*} = \arg \min_{\mathbf{h}^{(l)'}} & \underbrace{\lambda_1 \mathcal{L}_{\text{flip}}(\psi_{\boldsymbol{\theta}}^{(l,j)}(\mathbf{h}^{(l)'}), c_j)}_{\text{flip target concept } c_j} \\
 & + \underbrace{\lambda_2 \sum_{k \neq j} \mathcal{L}_{\text{preserve}}(\psi_{\boldsymbol{\theta}}^{(l,k)}(\mathbf{h}^{(l)'}), c_k)}_{\text{preserve other concepts } c_{k \neq j}} \\
 & + \underbrace{\lambda_3 \|\mathbf{h}^{(l)'} - \mathbf{h}^{(l)}\|_2^2}_{\text{minimality}}
 \end{aligned}$$

1138 where:

- 1139 • $\mathbf{h}^{(l)}$ is the original LR of the input at layer l ,
- 1140 • $\mathbf{h}^{(l)'}$ is the modified LR being optimized,
- 1141 • $\mathbf{h}^{(l)*}$ is the optimal modified LR,
- 1142 • $\mathcal{L}_{\text{flip}}$: loss encouraging the prediction of the target concept c_j to change
1143 state (i.e., flipping the predicted label),
- 1144 • $\mathcal{L}_{\text{preserve}}$: loss enforcing invariance of all non-target concepts c_k for $k \neq j$,
- 1145 • $\|\mathbf{h}^{(l)'} - \mathbf{h}^{(l)}\|_2^2$: regularization term ensuring the modified LR remains close
1146 to the original one,
- 1147 • $\lambda_1, \lambda_2, \lambda_3$: weighting coefficients controlling the trade-off between the in-
1148 tervention strength, the preservation of other concepts, and the minimality
1149 of the perturbation.

1150 In the context of LLMs studied previously, such interventions have been used
1151 to remove specific pieces from the model’s internal representation of the game

state [109, 115]. The main observation is that, in the vast majority of cases, the model behaves as if the piece had effectively disappeared. The LLM continues to generate legal moves while correctly accounting for the absence of the removed piece. This provides evidence for the causal role of these internal representations in the model’s behavior.

A main limitation of the probing approach is that the concepts under investigation must be predefined before any analysis can be performed. This introduces a strong bias in what can be studied. Moreover, annotating a dataset for each concept is extremely time-consuming and requires significant human effort. Interventions in LS are also difficult to interpret, as it is not always clear what constitutes a meaningful change in the network’s output under a given perturbation.

Moreover, probing methods provide a binary view of concepts, focusing on whether a given concept is present or absent in a LR. This binary perspective can be overly restrictive, as concepts may be represented in a more continuous manner. To obtain a more detailed characterization of these representations, the next section turns to methods that describe a concept as a direction in LS along which it varies continuously.

3.3 Directions

Probing describes a concept as a binary attribute, either present or absent in a LR. The methods in this section instead associate each concept with a direction in the LS. Moving a LR along this direction is expected to strengthen or weaken the concept smoothly. This provides information that a binary probe cannot express.

Empirically, it has been observed that shifting a LR along such a direction can induce coherent and semantically meaningful changes in the model’s output [91, 113]. Such transformations can be mathematically express as:

$$\mathbf{h}' = \mathbf{h} + \alpha \mathbf{v}_i,$$

where:

- $\mathbf{h}$ denotes the original LR,
- $\mathbf{v}_i$ is a unit vector associated with concept i ,
- α controls the strength of the intervention,
- $\mathbf{h}'$ is the resulting perturbed LR.

In many cases, the resulting change in the model’s output varies smoothly with α , suggesting an approximately linear CS with respect to certain semantic factors. This observation has led to the hypothesis that neural representations may organize concepts in a partially linear manner within the LS.

Vectors of this form are referred to as interpretable directions, as they correspond to transformations that can be associated with specific semantic attributes. The objective of this line of work is therefore to identify directions in LS that align with meaningful variations in the encoded concepts. This is illustrated schematically in Figure 3.4, where arrows represent semantic shifts

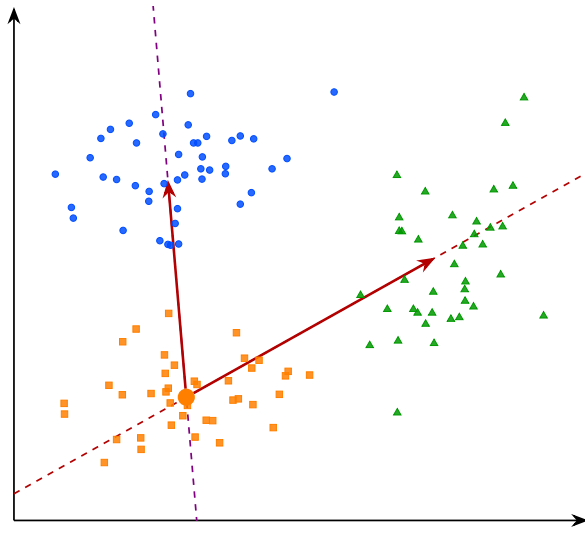


Figure 3.4: Interpretable directions in the latent space.

from one concept toward others, and moving along these directions induces a coherent change in the model’s behavior, suggesting that the transition between concepts is continuously encoded along these axes.

This phenomenon has been extensively studied in generative image models. As illustrated in Figure 3.5, where each row corresponds to a distinct direction $\mathbf{v}_i$ and each column to an increasing value of α , controlled perturbations of LRs along interpretable directions can induce specific semantic transformations, such as stroke thickness modification in digit images, head rotation in facial images, or texture and background changes in natural images [91]. These observations have led to the development of the field of LS editing, whose objective is to manipulate generated content directly through modifications in LS rather than through changes in the input prompt.

A central challenge remains identifying such interpretable directions. Most existing approaches rely on auxiliary models that encode human defined semantics, such as CLIP [113], to relate latent variations to textual concepts. Consequently, these methods inherit a limitation similar to that of probing techniques, since concept discovery depends on a pre-existing semantic framework defined by humans.

To overcome this limitation, a research direction known as unsupervised semantic discovery has emerged. Its goal is to uncover interpretable directions in LS without explicit semantic supervision. In these approaches, one model applies perturbations along random directions in LS, while a second model attempts to recognize the applied transformation. If a stable correspondence emerges between the perturbation and its prediction, this suggests that the considered direction corresponds to a coherent CS [91]. In this framework, human intervention occurs only a posteriori, to verify whether the discovered directions indeed correspond to interpretable semantic variations.

Despite these advances, several challenges remain. It appears that not all concepts encoded by NNs are linearly identifiable in LS. In practice, interpretable direction methods typically discover only a few dozen directions per dataset,



Figure 3.5: Interpretable directions in the latent space of a generative model [91].

1223 which intuitively seems insufficient to account for the full diversity of semantic
 1224 variations that a generative model is capable of producing.

1225 In addition, LRs often exhibit a phenomenon known as entanglement, illus-
 1226 trated in Figure 3.6, where multiple semantic attributes are mixed within the
 1227 same directions of the LS. For instance, in a generative face model, moving along
 1228 the “glasses” direction may simultaneously modify age and gender, whereas a
 1229 disentangled LR would let each axis control a single semantic attribute while
 1230 leaving the others unchanged. This lack of separability suggests that semantic
 1231 information is not cleanly organized in the LS. This observation motivates the
 1232 *Superposition Hypothesis* (Hypothesis 2), which proposes a different view of how
 1233 NNs organize information internally. The next section introduces this hypothesis.

1234 3.4 Superposition

1235 The previous section highlighted a limitation of interpretable directions. Only a
 1236 limited number of concepts can be represented by a linear direction, and these di-
 1237 rections are often entangled with multiple semantic concepts. To account for this,
 1238 an extension of the *Concept Hypothesis* (Hypothesis 1) has been proposed under
 1239 the name *Superposition Hypothesis* (Hypothesis 2), formalized in Hypothesis 2
 1240 [104].

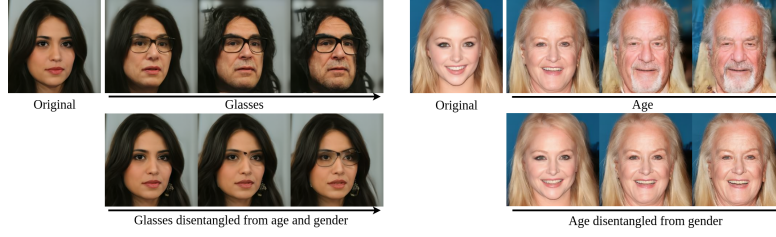


Figure 3.6: Illustration of entanglement in the latent space of a generative face model [113].

Hypothesis 2: Superposition Hypothesis

Concepts are preferentially represented by distinct linear directions in LS. When the number of concepts exceeds the available dimensions, multiple concepts are encoded in superposition within the same directions.

1241

1242 According to this hypothesis, in an unconstrained LS, each concept is repre-
 1243 sented in its own independent linear direction. The idealized non-superposed
 1244 LR can be mathematically expressed as follows:

$$\mathbf{h} = \sum_{i=1}^n \alpha_i \mathbf{v}_i, \quad \forall i \neq j, \quad \mathbf{v}_i^\top \mathbf{v}_j \approx 0, \quad \|\alpha\|_0 \ll n,$$

1245

where:

1246

- $\mathbf{h}$ denotes a LR,

1247

- n is the total number of concepts,

1248

- $\mathbf{v}_i$ represents the direction associated with concept c_i ,

1249

- α_i corresponds to the activation strength of concept c_i in the LR $\mathbf{h}$,

1250

- $\mathbf{v}_i^\top \mathbf{v}_j \approx 0$ indicates that concept directions are approximately orthogonal, meaning each concept independently controls a single factor of variation,

1251

1252

- $\|\alpha\|_0 \ll n$ indicates that only a small fraction of concepts are active for a given LR.

1253

1254

1255 However, NNs operate in finite-dimensional spaces, so when the number of
 1256 concepts exceeds the available dimensions, such a one-to-one correspondence
 1257 becomes impractical. The network must therefore reuse its representational
 1258 dimensions, allowing multiple concepts to be encoded using the same directions.
 1259 This shared encoding is referred to as superposition [108]. This phenomenon
 1260 becomes an obstacle for any method that attempts to analyze LRs directly in
 the original activation space.

1261

1262 Building on this observation, several works based on the *Superposition Hypoth-*
 1263 *esis* (Hypothesis 2) have developed methods that aim to recover less superposed
 1264 LRs. A common approach is to map LS into a higher-dimensional space while
 1265 imposing additional constraints that encourage individual concepts to be repre-
 sented separately. This transformation is illustrated in Figure 3.7, while LRs are

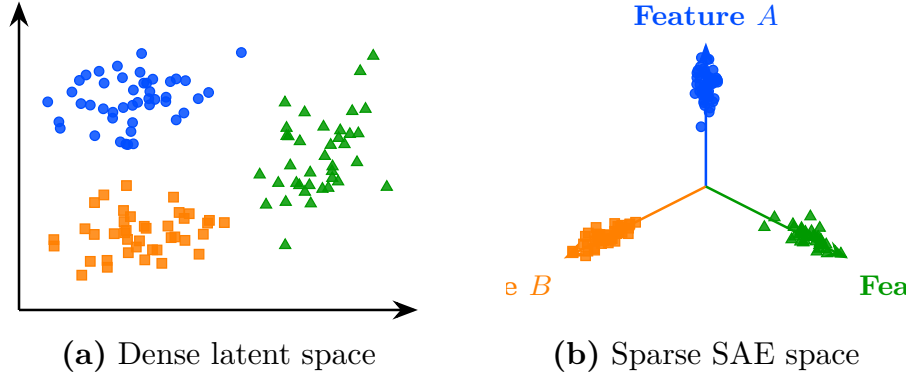


Figure 3.7: Transformation from a dense latent space (a) to a sparse SAE representation (b).

entangled in the initial LS, the sparse representation aligns each concept with a dedicated feature axis.

Sparse Autoencoders (SAEs) provide a practical implementation of this approach and can be defined as follows:

$$\mathbf{z} = f_{\text{enc}}(\mathbf{h}), \quad \hat{\mathbf{h}} = f_{\text{dec}}(\mathbf{z})$$

$$\mathcal{L}_{\text{auto}} = \|\mathbf{h} - \hat{\mathbf{h}}\|_2^2 + \lambda \|\mathbf{z}\|_1$$

where:

- $f_{\text{enc}}(\mathbf{h})$ is the encoding function that projects the original LR $\mathbf{h}$ into a higher-dimensional sparse space,
- $f_{\text{dec}}(\mathbf{z})$ is the decoding function that reconstructs the original LR from the sparse code,
- $\mathbf{z} \in \mathcal{Z}$ is the sparse representation of $\mathbf{h}$, with $\dim(\mathbf{z}) \gg \dim(\mathbf{h})$,
- $\hat{\mathbf{h}}$ is the reconstructed LR,
- $\|\mathbf{h} - \hat{\mathbf{h}}\|_2^2$ is the reconstruction loss ensuring that the original LR can be faithfully recovered,
- $\|\mathbf{z}\|_1$ is the sparsity regularization term that encourages most components of $\mathbf{z}$ to be zero,
- λ is a weighting coefficient controlling the trade-off between reconstruction fidelity and sparsity.

The parameters of these projection functions f_{enc} and f_{dec} are learned by minimizing $\mathcal{L}_{\text{auto}}$ using SGD. Under this formulation, each dimension of $\mathcal{Z}$ is interpreted as a dedicated concept direction.

In practice, both f_{enc} and f_{dec} are implemented as linear projections between the original LS and the high-dimensional space $\mathcal{Z}$. This choice is motivated by the same principle underlying probing methods. The transformation applied to the LR should remain as simple as possible. Using more expressive projection



Figure 3.8: A concept direction corresponding to the Golden Gate Bridge, discovered by a sparse autoencoder applied to a large language model [119].

functions could introduce additional structures into the transformed LR. In this case, it would become difficult to determine whether an observed concept was already encoded in the model’s LR, or whether it was introduced by the projection itself. Restricting f_{enc} and f_{dec} to linear projections therefore helps ensure that the concepts identified in $\mathcal{Z}$ can be attributed to the original LR, rather than to the transformation used to analyze it.

The method proves to be powerful when successfully applied to language models [119]. In these applications, SAEs are used to desuperpose the LR of LLMs. Once this LR is obtained, it has been shown that many concept vectors exhibit semantic properties. A example is the Golden Gate Bridge concept direction, illustrated in Figure 3.8, which activates systematically whenever the input text or image refers to the Golden Gate Bridge, regardless of the language or modality. This demonstrates that the learned concept is not tied to a specific surface form but captures a deeper semantic abstraction shared across modalities.

SAEs also provide a way to intervene directly on the LR learned by the model. A concept can be artificially activated by modifying its corresponding feature in the sparse representation $\mathbf{z}$, even when the input contains no reference to it. The modified LR is then projected back into the original activation space, after which the model proceeds with standard next-token generation from $\mathbf{h}^*$:

$$\mathbf{h}^* = f_{\text{dec}}(\mathbf{z}^*), \quad \text{where} \quad \mathbf{z}^* = \mathbf{z} + \alpha \mathbf{e}^{(i)},$$

where:

- $\mathbf{z}$ is the original sparse representation of the input,
- $\mathbf{z}^*$ is the modified sparse representation,
- $\mathbf{e}^{(i)}$ is the basis vector of $\mathcal{Z}$ associated with the target concept direction,
- α controls the strength of the artificial activation,
- $\mathbf{h}^*$ is the resulting edited LR in the original activation space.

With this method, a reference to the Golden Gate Bridge can be injected into the LLM’s internal representation. This causes the model’s output to shift toward discussing the Golden Gate Bridge, even when the prompt contains no reference to it. This provides evidence that the learned feature plays a causal

role in the model’s behavior. Such interventions enable control over specific model behaviors through LS editing.

Despite their empirical success, *Superposition Hypothesis* (Hypothesis 2) exhibit several important limitations that question their assumptions. These limitations can be summarized as follows:

- **Instability of the decomposition.** Repeating the SAE training process multiple times on the same model does not yield the same decomposition of the LS. Different runs produce different concept directions in $\mathcal{Z}$, even when achieving comparable reconstruction performance, which raises questions about the reliability of the discovered CSs.
- **Lack of semantic coverage.** In the high-dimensional space $\mathcal{Z}$, a large fraction of dimensions do not correspond to any interpretable semantic concept. This abundance of non-semantic directions raises questions about whether the decomposition truly reflects the model’s CSs.
- **Sparsity–semantics trade-off.** Increasing the sparsity coefficient λ beyond a certain threshold degrades the semantic quality of the learned features [114]. Suggesting that enforcing excessive sparsity comes at the cost of interpretability.
- **Failure of scaling to remove superposition.** Under the *Superposition Hypothesis* (Hypothesis 2), increasing the dimensionality of the LS should reduce superposition. However, this is not observed in practice. Even in very high-dimensional LSs, representations remain superposed, indicating that scaling alone does not resolve the entanglement.

These limitations suggest that *Superposition Hypothesis* (Hypothesis 2), while powerful, do not fully resolve the problem of CS. Convincing empirical results do not necessarily validate the underlying theoretical assumptions.

So far, every method described, whether based on probing, interpretable directions, or superposition, relies on a common assumption. The CS can be recovered from LR through a linear decomposition. This assumption has enabled substantial progress in understanding neural representations, but it also imposes a strong constraint on how concepts are expected to be represented. The following section examines clustering as an alternative approach that does not rely on this assumption.

3.5 Clustering

As the previous section noted, every LS analysis method presented so far rests on a linear view of the CS. For probing methods, concepts must be at least partially linearly separable for simple probes to detect them. Interpretable direction methods posit that variations associated with a concept can be captured by moving along a linear axis of the LS. The *Superposition Hypothesis* (Hypothesis 2) proposes that concepts are represented through linear directions that become entangled because of representational capacity constraints. Although these assumptions have led to empirical successes, several observations suggest that the linearity hypothesis may not fully capture the CS.

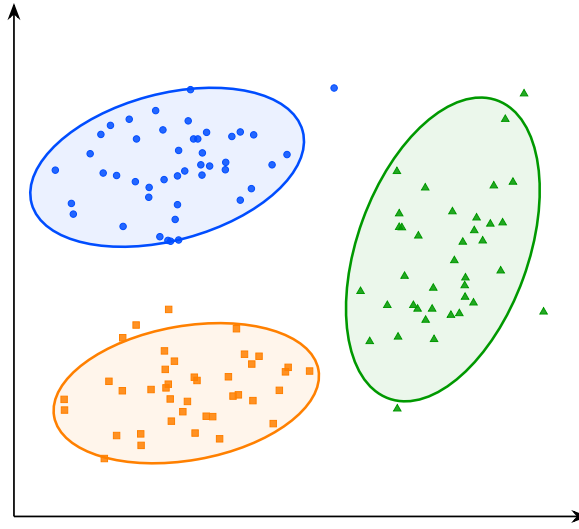


Figure 3.9: Visualization of the latent space structure.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches, enforcing excessive sparsity degrades interpretability. This suggests that the underlying representation may not be perfectly described by a sparse linear decomposition. Taken together, these observations indicate that semantic concepts may not always exhibit a linear CS. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of LSs.

To the best of our knowledge, clustering-based analysis is the only family of LS analysis methods that does not rely on any assumption of a linear CS [120, 123]. Rather than assuming that concepts correspond to directions, clustering methods only assume that conceptual similar inputs produce nearby LRs. Conversely, inputs that differ significantly from a concept perspective are expected to be located farther apart in the LS. This is a considerably weaker assumption, in the sense that it is empirically verified in a wide range of settings.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of LRs that occupy the same region of the LS. This is shown schematically in Figure 3.9, where the points represent LRs associated with concepts, and the ellipses indicate the approximate extent of each group and highlight their geometric separation.

This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is projected into a LS using a NN. The resulting LRs form a point cloud on which standard clustering algorithms can be applied. The obtained clusters are then interpreted as representing distinct concepts.

Deep clustering has been particularly successful in unsupervised image classification. For this type of task, a specific NN architecture called an autoencoder is commonly used. An autoencoder is composed of an encoder f_{enc} and a decoder f_{dec} . The encoder maps an input $\mathbf{x} \in \mathcal{X}$ to a low-dimensional LR $\mathbf{z} \in \mathbb{R}^d$, the LR produced by the encoder. The decoder then maps this LR back to the input

space to produce a reconstruction $\hat{\mathbf{x}}$. The model is trained by SGD to minimize the reconstruction error between the original input and its reconstruction using a Mean Squared Error (MSE) loss. This objective encourages the LS to preserve the most informative features of the input while enforcing a compact representation. This can be expressed mathematically as follows:

$$\mathbf{x} \xrightarrow{f_{\text{enc}}} \mathbf{z} \xrightarrow{f_{\text{dec}}} \hat{\mathbf{x}}, \quad d \ll \dim(\mathcal{X}), \quad \mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \left\| \mathbf{x}^{(i)} - \hat{\mathbf{x}}^{(i)} \right\|_2^2.$$

Once the autoencoder has been trained sufficiently well, the reconstruction loss $\mathcal{L}_{\text{rec}}$ should be low enough for the learned representation to be useful. A clustering algorithm can then be applied to the compressed representations of the images. Let $\mathcal{D}$ denote a dataset of inputs, through the encoder f_{enc} , this dataset is mapped to a point cloud in the LS:

$$\mathcal{Z} = \{\mathbf{z}^{(i)} \in \mathbb{R}^d \mid \mathbf{z}^{(i)} = f_{\text{enc}}(\mathbf{x}^{(i)}), \mathbf{x}^{(i)} \in \mathcal{D}\}.$$

A classical clustering algorithm is applied to the set $\mathcal{Z}$ of LR to identify groups of similar LR. Each resulting cluster can therefore be interpreted as a group of images that the model considers similar.

An example of this methodology can be observed in Figure 3.10, where the approach is applied to the STL-10 dataset [33] using k -means clustering [11] with a number of clusters arbitrarily fixed to $K = 10$ [49]. For each cluster, the corresponding line shows the 10 samples closest to the associated centroid in the LS.

We observe that the method captures coherent semantic categories such as animals, vehicles, and flying entities. This result is particularly striking given that the only available information to the system consists of raw images. It suggests that the compression process induces a LS in which semantically meaningful CSs emerge.

More advanced approaches combine representation learning and clustering within a single optimization process. In this setting, additional loss terms encourage LR to organize themselves around cluster centroids during training. This often produces LSs that are easier to analyze and improves the quality of the discovered semantic CSs [52, 122].

Despite its effectiveness, clustering-based analysis presents several important limitations.

First, as with most unsupervised semantic discovery techniques, interpreting the resulting clusters remains partly subjective. Benchmark datasets often provide class labels for evaluation, but the discovered clusters may not match the categories that humans consider relevant.

Second, clustering methods are highly sensitive to design choices, as the number of clusters, the distance metric, and the clustering algorithm itself can all significantly influence the results, making parameter selection challenging and often requiring substantial empirical tuning.

Finally, to the best of our knowledge, no intervention technique has yet been proposed in the context of clustering-based analysis. Unlike probing or SAE analysis, which allow targeted modifications of LR to study their causal role, clustering methods currently offer no mechanism to alter the model’s behavior by acting directly on the discovered CSs.

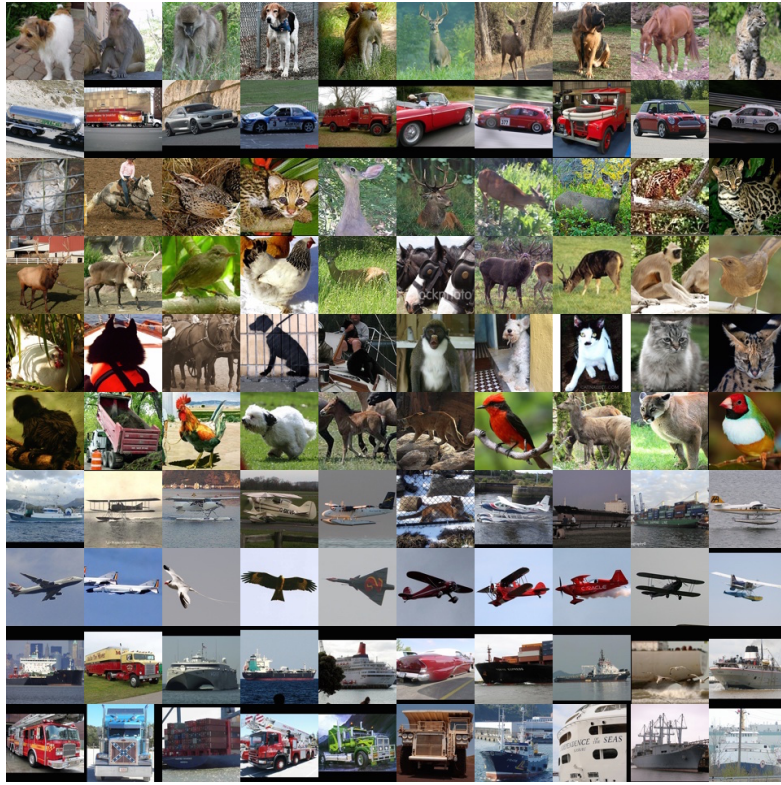


Figure 3.10: Clusters discovered by a deep clustering model applied to natural images [49].

3.6 Conclusion

The four families of methods reviewed in this chapter are summarized in Figure 3.11. They include probing, interpretable directions, sparse autoencoders, and clustering, all of which build upon the *Concept Hypothesis* (Hypothesis 1) introduced in Section 3.1. Each family approaches the analysis of LRs from a different angle. However, they share a common goal of understanding how LRs are structured and what information they contain.

However, a fundamental difficulty remains. There is currently no consensus on how to rigorously evaluate whether these methods truly capture the internal concepts of a model. Most existing metrics assess reconstruction quality or linear separability, but these do not directly measure whether the discovered CSs correspond to meaningful abstractions from the model’s perspective. This leaves open the question of what it means, formally, for a concept to be represented in a NN.

Addressing this question may require contributions from disciplines beyond computer science and mathematics. As the notion of semantics is inherently tied to human interpretation, a purely formal or computational account may be insufficient. Insights from philosophy, cognitive science, or psychology could prove valuable in grounding the notion of concept in a way that is both theoretically rigorous and empirically tractable. Despite these open questions, the methods

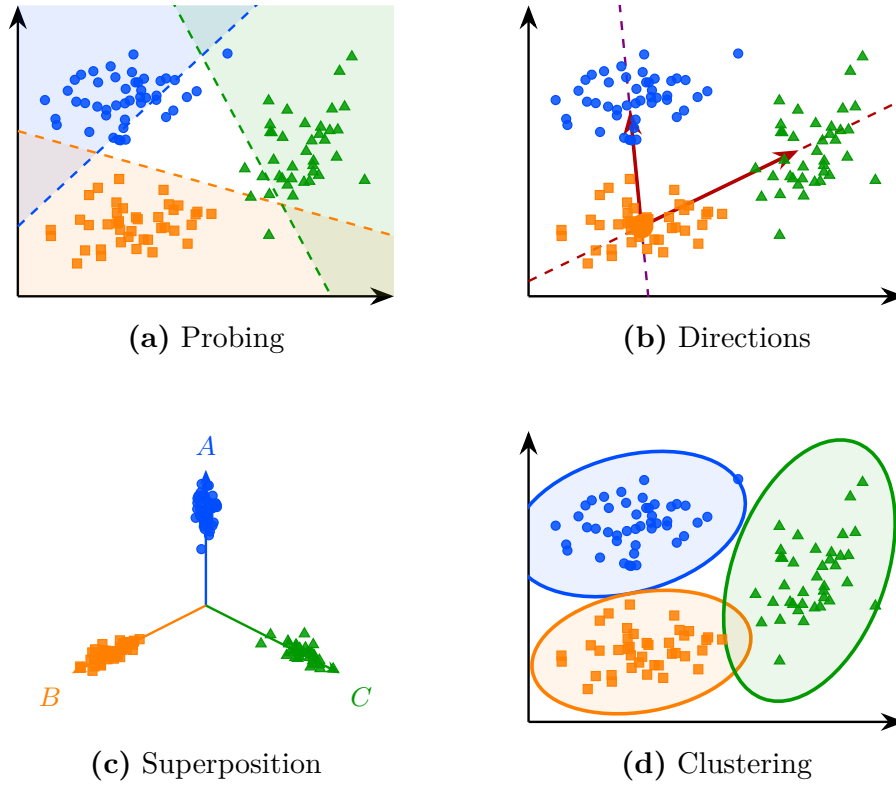


Figure 3.11: Overview of the four latent space analysis methods studied in this chapter, applied to the same point cloud.

1451 studied in this chapter provide a practical foundation for analyzing the internal
 1452 representations of NNs.

1453 Building on the *Concept Hypothesis* (Hypothesis 1) and the insights developed
 1454 throughout this chapter, this thesis investigates a representation-based approach
 1455 to explainability in RL. To establish the necessary foundations, the next chapter
 1456 introduces the principles of RL. The subsequent chapter then presents the
 1457 proposed method for analyzing these representations and studying their potential
 1458 for explainability in RL (Chapter 6).

Chapter 4

Reinforcement Learning

This chapter introduces the fundamental concepts of Reinforcement Learning (RL), providing the necessary foundation for the ideas developed throughout the rest of this thesis. Its structure is as follows:

Section 4.1: An overview of the milestones that led to the emergence of RL as a unified field.

Section 4.2: A clarification of what is meant by an agent and how it relates to the RL algorithm.

Section 4.3: A formalization of the problem that RL algorithms aim to solve.

Section 4.4: An explanation of how policies are progressively refined through interaction with the environment to approach the optimal policy.

Section 4.5: A summary of the key ideas and a discussion of the broader significance of RL.

4.1 Historical Developments

RL is today presented as a major field within artificial intelligence. However, it originates from several research traditions that progressively converged toward a common framework. In this section, we outline the main stages in the emergence of this field, highlighting the key ideas and milestones that shaped its evolution.

4.1.1 Behavioral foundations (1900–1950)

The origins of RL can be traced back to 20th-century animal psychology studies [3]. These studies aimed to understand how behaviors are formed and modified. The central idea emerging from this work is that learning is driven by the consequences of actions. Behaviors followed by positive outcomes are more likely to be repeated, whereas those associated with negative outcomes tend to gradually decrease in frequency.

This perspective was later developed within behaviorism. In this school of psychology, rewards and punishments are used to gradually shape behavior

1488 through repeated interaction with the environment [5]. The term “reinforcement”
 1489 in the context of learning comes from this school of thought, where it refers
 1490 precisely to the process of strengthening a behavior through its consequences.

1491 This idea led to the suggestion that learning principles observed in living
 1492 organisms could be transferred to artificial systems. Machines were therefore
 1493 envisioned as potentially learning through reward-like mechanisms, guided by
 1494 evaluative feedback signals resembling a “pleasure–pain system” [7]. Although
 1495 these studies did not involve formal mathematical models, they introduced the
 1496 central idea of RL, trial-and-error learning guided by environmental feedback.

1497 4.1.2 Sequential decision-making (1950–1970)

1498 A second foundation of RL comes from optimal control theory. This field studies
 1499 how to determine sequences of decisions that optimize a cumulative performance
 1500 criterion over time. In this context, the decision-making entity is called an agent.
 1501 The agent interacts with the environment by selecting actions. The function that
 1502 maps an observed state to the chosen action is called a policy. The objective is
 1503 to find an optimal policy, in the sense that it maximizes the expected cumulative
 1504 reward over time.

1505 This framework led to the formalization of Markov Decision Processes (MDPs),
 1506 which provide a mathematical model for sequential decision-making [10]. MDPs
 1507 describe an agent interacting with an environment through a set of states,
 1508 actions, rewards, and transition rules. The earliest method proposed to solve
 1509 MDPs is dynamic programming [8]. It proceeds by breaking down a complex
 1510 decision problem into smaller, more manageable subproblems, solving each
 1511 one, and combining the results to obtain an overall solution. However, dynamic
 1512 programming requires a full description of the environment, which is only feasible
 1513 for small problems.

1514 4.1.3 Emergence of reinforcement learning (1970–2000)

1515 In order to address this issue, temporal-difference learning methods were intro-
 1516 duced [19]. This mechanism enables incremental learning directly from experience
 1517 without requiring a complete model of the environment. It can be interpreted as
 1518 a sample-based approximation of the Bellman equations.

1519 Building on these ideas, the notion of value functions emerged as a way
 1520 to estimate the long-term benefit of being in a given state or taking a given
 1521 action. These functions assign a numerical score to states or state-action pairs,
 1522 reflecting the expected cumulative reward an agent can obtain from that point
 1523 onward. They thus provide a compact way to evaluate and compare different
 1524 courses of action. This line of work culminated in Q-learning, which showed that
 1525 optimal action-value functions can be learned directly from interaction with the
 1526 environment, without requiring a model of its dynamics [22, 23].

1527 The publication of the foundational reference work [30] contributed to the
 1528 formalization of RL as a coherent field. From that point onward, the field can be
 1529 considered well established and widely recognized as a distinct area of research.

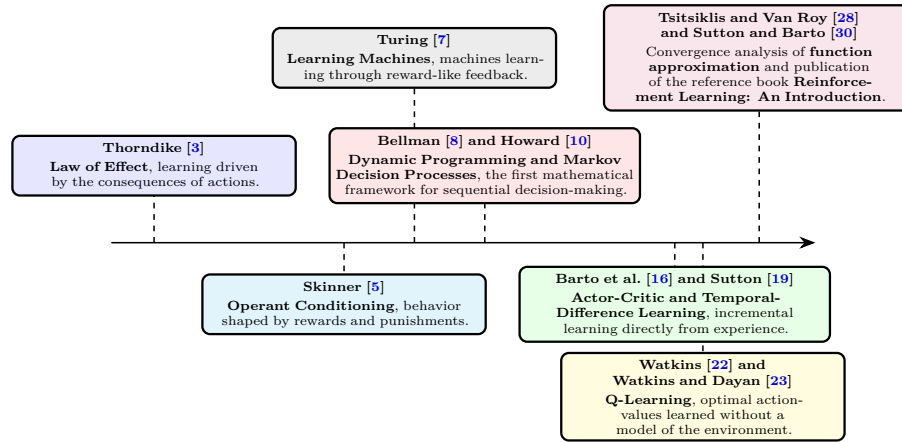


Figure 4.1: Evolution of reinforcement learning prior to its fusion with deep learning.

4.1.4 Consolidation of reinforcement learning (2000–2012)

A limitation of early algorithms such as Q-learning is their reliance on explicit state-action tables. Although these methods remove the need to know the full dynamics of the MDP, they still require storing a value for each possible state-action pair. As a result, the agent must effectively visit and maintain estimates for a potentially enormous number of states in the environment. This requirement becomes infeasible in large-scale MDPs, where the state space is too large to be exhaustively explored or stored in memory.

To address this issue, research shifted toward more general representations of value functions and policies. Instead of maintaining explicit tables, function approximation methods were introduced. These allowed value functions and policies to be represented using parametric models such as linear approximators or perceptron-like architectures [16, 28]. This change enabled generalization across states, reducing the dependence on exhaustive exploration of the state space. These parametric representations nonetheless remained limited in expressiveness. The next step consisted in replacing such approximators with Neural Networks (NNs), which would soon reshape the field.

Figure 4.1 provides a timeline of the key milestones that shaped the development of reinforcement learning prior to its fusion with deep learning.

4.1.5 Deep reinforcement learning (2013–2026)

The introduction of Deep Learning (DL) into RL marked a major turning point in the field. This shift was initiated by Deep Q-Network (DQN), which used a deep NN to approximate action-value functions at scale. This made it possible to train an agent to play Atari games at a human level directly from raw pixels [40]. The use of deep NNs provided RL with a powerful and flexible function approximator. This removed an important practical limitation by making it possible to efficiently represent the complex functions required by RL.

A series of subsequent algorithms further extended and improved this approach. Deep Deterministic Policy Gradient (DDPG) extended actor-critic

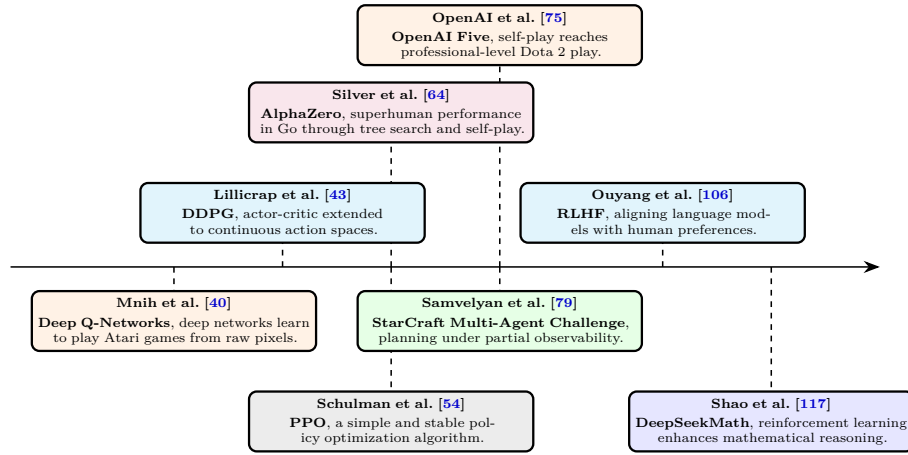


Figure 4.2: Evolution of deep reinforcement learning.

1559 methods to continuous action spaces [43]. Proximal Policy Optimization (PPO)
 1560 later became a widely adopted algorithm in RL due to its simplicity, stability,
 1561 and ability to handle both continuous and discrete action spaces [54].

1562 These advances established a foundation for large-scale policy optimization.
 1563 Notably, AlphaZero demonstrated that the combination of deep NNs, RL, and
 1564 tree search could reach superhuman performance in the game of Go [64]. This
 1565 success was later extended to even more complex real-time strategy environments
 1566 such as StarCraft II, where agents must handle long-term planning, partial
 1567 observability, and large action spaces [79]. Similarly, in Dota 2, large-scale multi-
 1568 agent RL systems such as OpenAI Five showed that self-play combined with
 1569 deep policy optimization can achieve professional-level performance in highly
 1570 dynamic and stochastic environments [75].

1571 More recently, RL has expanded beyond its traditional application domains.
 1572 With the emergence of increasingly agentic views of Large Language Models
 1573 (LLMs), it has become possible to train these models to perform specific tasks
 1574 using RL-based techniques. In particular, RL from human feedback has become
 1575 a method for aligning LLMs with human preferences. It does so by formalizing
 1576 the problem as an MDP and using RL to solve it [106]. Similarly, DeepSeekMath
 1577 shows how RL can enhance the mathematical reasoning abilities of LLMs [117].
 1578 It demonstrates that an artificial agent can learn to reason in natural language in
 1579 order to accomplish a given task. The skills acquired during this learning process
 1580 also transfer to other similar tasks, leading to improved overall performance. For
 1581 instance, training a LLM with RL on mathematical problem-solving can also
 1582 enhance its programming capabilities.

1583 Figure 4.2 provides a timeline of the key milestones that shaped the emergence
 1584 of deep reinforcement learning.

1585 4.2 Agent Paradigm

1586 The RL framework is built around the agent paradigm, in which the decision-
 1587 making entity is referred to as the agent. A standard definition is:

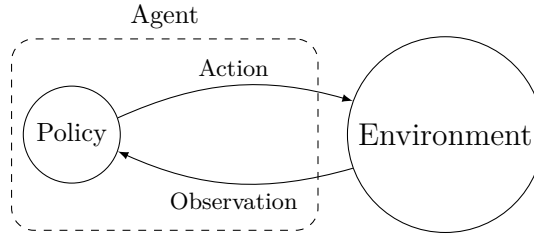


Figure 4.3: The agent paradigm.

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.
[98]

In the RL setting, this definition translates into an interaction between the agent and its environment. The agent perceives its environment through observations. These observations are then processed by a decision-making mechanism referred to as the policy. The policy determines how the agent modifies its environment through the actions it selects. Figure 4.3 illustrates this interaction.

The objective of an RL agent is to maximize the cumulative reward it receives through its interactions with the environment. To achieve this objective, the agent's policy must progressively improve through experience. The procedure responsible for this learning process is the RL algorithm, which progressively modifies the policy based on the experience collected through interaction.

In this thesis, the RL algorithm is considered as a process external to the agent. Other presentations of RL describe the agent as both the learner and the decision-making entity [30]. We chose to distinguish the agent from the RL algorithm because this provides a more general definition of an agent.

An agent can therefore exist without learning, for example when its behavior is entirely specified by a set of programmed rules. Such an agent still perceives, processes information, and acts upon its environment, but its behavior is not the result of RL.

This distinction also makes the definition consistent with modern implementations of RL [57, 61], in which the acting policy and the learning algorithm are separate software components. Several copies of the policy, each interacting with its own instance of the environment, collect experience in parallel, while a single learning algorithm uses this experience to update the shared policy parameters.

1615 4.3 Problem Formulation

1616 Having outlined the main developments shaping RL and introduced the concept
 1617 of the agent, the RL problem can now be stated formally. In essence, RL aims to
 1618 build a agent that makes sequential decisions in order to maximize a cumulative
 1619 reward signal over the long term. The purpose of this section is to translate
 1620 this intuitive goal into a mathematical framework. To this end, the presentation
 1621 proceeds step by step:

- 1622 1. Modeling the RL problem as a Markov Decision Process.
- 1623 2. Defining the notion of a policy, which formalizes the agent’s behavior.
- 1624 3. Describing the interaction between the agent and the environment through
 1625 trajectories.
- 1626 4. Introducing value functions to evaluate policies.
- 1627 5. Defining the notion of an optimal policy.

1628 4.3.1 Markov Decision Processes

1629 To specify the task and the agent’s capacity to perceive and act, the problem is
 1630 modeled as a MDP.

1631 Before providing a more detailed definition, it is useful to clarify the scope of
 1632 the MDP problems considered in this thesis. We therefore restrict the analysis
 1633 to the following configuration:

- 1634 • The single-agent learning setting is considered. If any other agents are
 1635 present in the environment, they are treated as part of the environment
 1636 dynamics.
- 1637 • Finite-horizon MDPs are considered. Each interaction sequence between
 1638 the MDP and the policy has a defined beginning and a defined end. Between
 1639 these two points, the number of interaction steps may vary depending on
 1640 the situation, but it always remains finite.
- 1641 • The partially observable framework is adopted, where the agent does
 1642 not have direct access to the true state of the system, but instead relies
 1643 on limited observations that provide only partial information about the
 1644 environment.

1645 The resulting framework can be formally described as a single-agent finite-
 1646 horizon partially observable Markov decision process. For simplicity, we will
 1647 refer to this setting as an “MDP” throughout this thesis.

1648 A MDP is defined by the tuple $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$:

- 1649 • $\mathcal{S}$: The set of states, representing all possible configurations of the en-
 1650 vironment. A state belonging to this set is denoted by $s \in \mathcal{S}$. Among
 1651 these, two special subsets are distinguished: initial states, from which an
 1652 interaction can begin, and terminal states, in which the interaction ends.
 1653 The set of initial states is denoted by $\mathcal{S}^{\text{init}}$, with an initial state written as
 1654 $s^{\text{init}} \in \mathcal{S}^{\text{init}}$. The set of terminal states is denoted by $\mathcal{S}^{\text{ter}}$, with a terminal
 1655 state written as $s^{\text{ter}} \in \mathcal{S}^{\text{ter}}$.
- 1656 • $\mathcal{O}$: The set of observations that the agent can perceive from the environ-
 1657 nment. A observation belonging to this set is denoted by $o \in \mathcal{O}$. Unlike
 1658 the state $s \in \mathcal{S}$, which is hidden from the agent, the observation $o \in \mathcal{O}$
 1659 provides partial information about the current state.
- 1660 • $\mathcal{A}$: The set of actions that the agent may execute. A action belonging to
 1661 this set is denoted by $a \in \mathcal{A}$. Actions influence the dynamics of the MDP,
 1662 thereby shaping its future course. They represent the agent's capacity to
 1663 act upon its environment.
- 1664 • $\mathcal{R}$: The set of rewards that the agent can receive from the environment.
 1665 A reward belonging to this set is denoted by $r \in \mathcal{R}$. $\mathcal{R}$ is a subset of $\mathbb{R}$.
 1666 Rewards can be either positive or negative, depending on whether the
 1667 corresponding outcome is considered beneficial or detrimental for the agent.
 1668 The agent's goal is to maximize the cumulative sum of these rewards over
 1669 time.
- 1670 • p : The transition function, which defines the dynamics of the environment.
 1671 Given a state s and an action a , this function defines a probability distribu-
 1672 tion over the joint outcome consisting of the next state s' , the observation
 1673 o , and the reward r . This function is denoted by p , and is defined as:
 1674 $p : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S} \times \mathcal{O} \times \mathcal{R})$. The notation $(s', o, r) \sim p(s, a)$ means that
 1675 the tuple (s', o, r) is sampled from the distribution induced by $p(\cdot \mid s, a)$.
 1676 The notation $p(s', o, r \mid s, a)$ denotes the probability of observing the tuple
 1677 (s', o, r) under the distribution $p(\cdot \mid s, a)$.

1678 The two quantities, observation o and state s , are not independent. They are
 1679 jointly generated via $(s', o, r) \sim p(s, a)$, which implicitly defines the relationship
 1680 between them. In our MDP formalization, this relationship is left implicit
 1681 because it is never manipulated directly during learning.

1682 However, to express mathematical equations, it is convenient to introduce
 1683 the conditional distribution over states given an observation. We denote this
 1684 distribution by $\beta : \mathcal{O} \rightarrow \mathcal{P}(\mathcal{S})$. The notation $\beta(s \mid o)$ denotes the probability of
 1685 state s given the observation o , while $s \sim \beta(\cdot \mid o)$ indicates that the state s is
 1686 sampled from this distribution.

1687 4.3.2 Policy

1688 The agent's decision-making capability is modeled by a policy function. The
 1689 policy defines the stochastic mapping between observations $o \in \mathcal{O}$ and actions
 1690 $a \in \mathcal{A}$. A policy function is denoted by π , and is defined as: $\pi : \mathcal{O} \rightarrow \mathcal{P}(\mathcal{A})$. The
 1691 notation $a \sim \pi(\cdot \mid o)$ means that the action a is sampled from the distribution

induced by $\pi(\cdot | o)$. The notation $\pi(a | o)$ denotes the probability of selecting action a under the distribution $\pi(\cdot | o)$.

It is important to note that this policy does not include any explicit mechanism for estimating the underlying state from the observations. Instead, the policy must learn from experience how to infer the information about the underlying state that is not directly available in the observations. This reflects the design of many modern RL implementations, where policies are often parameterized by architectures capable of summarizing past observations, such as recurrent or attention-based networks, without explicitly modeling the uncertainty over the underlying state.

4.3.3 Trajectory

Now that both the MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$ and the policy π have been defined, the interaction between them can be characterized. A sequence of consecutive interactions between an environment and a policy is called a trajectory. A trajectory is generated through the repeated interaction between the policy and the environment, following the sequence below:

1. The environment, being in state s , provides an observation o to the policy π . Based on this observation, the policy outputs a probability distribution over actions $\pi(\cdot | o)$, from which an action is sampled: $a \sim \pi(\cdot | o)$.
2. The sampled action is executed in the environment. The environment then generates a transition: $(s', o', r) \sim p(\cdot | s, a)$. It moves to a new state s' , and returns a new observation o' and a reward r .

Indexing each interaction by a discrete time step t , a trajectory τ of length T can be written as:

$$\tau = (s_1, o_1, a_1, r_1, s_2, o_2, a_2, r_2, \dots, s_T, o_T, a_T, r_T),$$

where r_t is the reward received after taking action a_t at time step t .

A trajectory is random: it is sampled from a distribution determined by the initial state s , the initial observation o , the environment dynamics p , and the policy π . Since the dynamics p are fixed, they are omitted from the notation. The notation $\tau \sim \pi$ means that the trajectory τ is generated by following the policy π , and $\Pr_\pi(\tau | s, o)$ denotes the probability of observing the trajectory τ when starting from state s with observation o .

An important property of the trajectory distribution is its recursive structure. Since both the policy π and the transition function p are known, the distribution over trajectories starting from s_1 with observation o_1 can be expressed in terms of the distribution over trajectories starting from the next state s_2 with observation o_2 . Specifically, the probability of a trajectory τ factorizes as:

$$\Pr_\pi(\tau | s_1, o_1) = \pi(a_1 | o_1) p(s_2, o_2, r_1 | s_1, a_1) \Pr_\pi(\tau_{2:} | s_2, o_2),$$

where $\tau_{t:} = (s_t, o_t, a_t, r_t, \dots, s_T, o_T, a_T, r_T)$ denotes the part of the trajectory starting from time step t .

There exist two special types of trajectories. First, a trajectory that terminates in a terminal state is called a terminal trajectory, and is denoted by τ^{ter} . Second, a terminal trajectory that starts from an initial state s^{init} and an initial observation is called an episode, and is denoted by τ^{ep} .

4.3.4 Value Functions

To evaluate the ability of a policy π to collect rewards, the notion of return is introduced. The return G_t is defined as the discounted sum of the rewards collected along a terminal trajectory from time step t onward:

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots + \gamma^{T-t} r_T = \sum_{k=t}^T \gamma^{k-t} r_k,$$

where $\gamma \in [0, 1]$ is the discount factor. The return of the whole trajectory is therefore G_1 . This parameter controls the time horizon of the agent's objective. When γ is close to 0, the agent focuses almost exclusively on immediate rewards, leading to short-sighted behavior. As γ increases toward 1, future rewards gain more weight, and the agent adopts a more far-sighted strategy. In theory, since the goal is to maximize long-term cumulative reward, the discount factor would ideally be set to $\gamma = 1$. In practice, γ is often chosen slightly below 1 (e.g., 0.99) to stabilize learning, and to reflect the fact that distant future rewards are inherently more uncertain.

The return also admits a recursive form, which is central to the dynamic programming and RL algorithms presented later:

$$G_t = r_t + \gamma G_{t+1}, \quad G_{T+1} = 0.$$

From the notion of return, functions that estimate the expected return of a policy from a given starting point can be defined. These functions are called value functions. The first one considered is the state-value function v . It tells us how desirable a particular state is under a given policy π , given an observation o , the environment dynamics p , and the discount factor γ . Strictly speaking, this function should be written as $v(s, o, p, \gamma, \pi)$ to reflect all its dependencies. However, since both the environment dynamics p and the discount factor γ are fixed throughout learning, they are omitted from the notation. To emphasize that the value is tied to a specific policy, π is placed as a subscript. This yields the compact notation $v_\pi : \mathcal{S} \times \mathcal{O} \rightarrow \mathbb{R}$.

If $v_\pi(s, o) > v_\pi(s', o')$, then, under policy π , the agent prefers being in state s with observation o over state s' with observation o' . Formally, the state-value function is defined as the expected return over the terminal trajectories $\tau^{\text{ter}} \sim \pi$:

$$v_\pi(s, o) = \mathbb{E}_\pi \left[G_t \mid s_t = s, o_t = o \right].$$

It is worth noting that the state-value function v_π is recursive, a property that follows directly from the recursive form of the return:

$$\begin{aligned} v_\pi(s, o) &= \mathbb{E}_\pi \left[G_t \mid s_t = s, o_t = o \right] \\ &= \mathbb{E}_\pi \left[r_t + \gamma G_{t+1} \mid s_t = s, o_t = o \right] \\ &= \mathbb{E}_\pi \left[r_t + \gamma v_\pi(s_{t+1}, o_{t+1}) \mid s_t = s, o_t = o \right] \end{aligned}$$

If the expectation is expressed explicitly in terms of the distributions p and π , the Bellman equation for v_π is obtained:

$$\begin{aligned}
v_\pi(s, o) &= \mathbb{E}_\pi \left[r_t + \gamma v_\pi(s_{t+1}, o_{t+1}) \mid s_t = s, o_t = o \right] \\
&= \sum_{a \in \mathcal{A}} \pi(a \mid o) \sum_{\substack{s' \in \mathcal{S} \\ o' \in \mathcal{O} \\ r \in \mathcal{R}}} p(s', o', r \mid s, a) \left[r + \gamma v_\pi(s', o') \right].
\end{aligned}$$

Now that the value of a state has been expressed, the next step is to consider the value of taking a specific action in a given state. Indeed, it is useful to estimate the expected future gain of an action a in state s , in order to compare different actions and determine which one is the most promising under a given policy π . This function is called the state-action value function $q_\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. If $q_\pi(s, a) > q_\pi(s, a')$, then, under policy π , choosing action a in state s yields a higher expected return than choosing action a' . It is defined as:

$$q_\pi(s, a) = \mathbb{E}_\pi \left[r_t + \gamma G_{t+1} \mid s_t = s, a_t = a \right].$$

The function q_π can be related to v_π by recognizing that, after taking action a in state s , the remaining return from the next state s' with observation o' is precisely $v_\pi(s', o')$. This yields:

$$\begin{aligned}
q_\pi(s, a) &= \mathbb{E}_\pi \left[r_t + \gamma G_{t+1} \mid s_t = s, a_t = a \right] \\
&= \mathbb{E}_\pi \left[r_t + \gamma v_\pi(s_{t+1}, o_{t+1}) \mid s_t = s, a_t = a \right] \\
&= \sum_{\substack{s' \in \mathcal{S} \\ o' \in \mathcal{O} \\ r \in \mathcal{R}}} p(s', o', r \mid s, a) \left[r + \gamma v_\pi(s', o') \right]
\end{aligned}$$

4.3.5 Optimal Policy

The goal of RL is to find a policy that maximizes the expected cumulative reward from any observation until the end of the trajectory. This policy is called the optimal policy and is denoted by π_* , and its state-action value function by $q_* = q_{\pi_*}$. In a partially observable setting, the optimal action is the one that maximizes the expected state-action value over the distribution of possible states given the observation:

$$\pi_*(a \mid o) = \begin{cases} 1 & \text{if } a = \arg \max_{a' \in \mathcal{A}} \sum_{s \in \mathcal{S}} \beta(s \mid o) q_*(s, a'), \\ 0 & \text{otherwise} \end{cases}$$

As shown, both v_π and q_π can be expressed explicitly in terms of the environment dynamics p and the policy π . If p , β and π are fully known, one can theoretically compute the optimal policy by unfolding the graph of all possible trajectories starting from a given state s and observation o . This is possible precisely because v_π and q_π are recursive functions. Solving an MDP in this way is known as dynamic programming.

In practice, most environments do have too many states to allow an explicit representation of the dynamics p . Storing and computing over the full transition

function becomes intractable. To overcome this limitation, instead of directly constructing the optimal policy, an iterative approach is adopted: starting from an initial policy, it is progressively improved so that it converges toward π_* . This iterative improvement through interaction with the environment is precisely what constitutes learning in RL, and it is the subject of the next section.

4.4 Learning

Now that the optimal policy π_* has been defined, the question is how to obtain it. The recursive Bellman equations for v_π and q_π suggest dynamic programming approaches, but these scale poorly to large state spaces. RL offers an alternative by solving the MDP iteratively and generating a sequence of policies that progressively approach π_* through direct interaction with the environment. This section introduces two major families of RL algorithms, critic-only methods and actor-critic methods.

At a high level, every algorithm presented below repeats the same loop: the current policy interacts with the environment, the resulting trajectories are sent to the RL algorithm, and the algorithm uses them to produce a new policy.

4.4.1 Critic-Only Methods

The central idea behind all critic-only approaches is to iteratively construct an approximation of the state-action value function q_π , denoted by $\hat{q}$. Although $\hat{q}$ is only an approximation of the true q_π , the policy always acts greedily with respect to this approximation.

Since $\hat{q}$ serves as the basis for the agent's decisions, it cannot rely on more information than the agent itself has access to. It must therefore be defined solely over observations, as a function $\hat{q} : \mathcal{O} \times \mathcal{A} \rightarrow \mathbb{R}$. Its objective is to approximate the true q_π , which depends on the underlying states s , using only the partial information available through o . In practice, this does not necessarily pose a problem, as observations often contain sufficient information to make accurate predictions about the underlying state. Moreover, the strong representational capacity of NNs allows the critic to capture these relationships without explicitly representing the mapping between observations and states.

This leads to the following mathematical formulation:

$$\pi(a \mid o) = \begin{cases} 1 & \text{if } a = \arg \max_{a' \in \mathcal{A}} \hat{q}(o, a'), \\ 0 & \text{otherwise.} \end{cases} \quad (4.1)$$

It is precisely this way of modeling the policy that gives this family of methods its name. The policy is not represented explicitly as a standalone function. Rather, it simply emerges from searching over the set of actions to maximize $\hat{q}$.

To give a clear example of critic-only methods, they are presented in their simplest form: model-free, without temporal-difference learning, without replay buffer, without bootstrapping, and using only ε -greedy exploration. Naturally, many variants and extensions of this basic scheme exist, but the core idea remains the same.

When implementing a RL algorithm, one of the first questions is how to represent the value function. In critic-only methods, this means choosing a model for $\hat{q}$. Any machine learning model capable of regression task can be used for this purpose. The parameters of the model are denoted by the weight vector $\mathbf{w}$, and the approximation is written $\hat{q}(o, a, \mathbf{w})$ when this dependence matters. In deep RL, a NN is used to model this function, in which case $\mathbf{w}$ represents the weights and biases of the network.

In our simple setting, exploration is handled via ε -greedy selection. With probability $1 - \varepsilon$, the policy selects the greedy action that maximizes $\hat{q}$, and with probability ε it selects an action uniformly at random. Formally, this ε -greedy policy is obtained by modifying the greedy policy defined above as follows:

$$\pi(a \mid o) = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{|\mathcal{A}|} & \text{if } a = \arg \max_{a' \in \mathcal{A}} \hat{q}(o, a'), \\ \frac{\varepsilon}{|\mathcal{A}|} & \text{otherwise.} \end{cases} \quad (4.2)$$

The trajectories generated by the resulting policy now depend on both π and ε , which is written $\tau \sim (\pi, \varepsilon)$. The parameter ε controls the degree of exploration. If the policy always chooses the action it currently believes to be optimal, it may miss better alternatives that $\hat{q}$ has not yet learned to value correctly. Exploring suboptimal actions from time to time is therefore essential to verify that $\hat{q}$ accurately reflects the true values, while still focusing most of the time on the most promising actions. In more advanced implementations, ε can be adapted over the course of learning to control exploration dynamically, for instance by decreasing it gradually over time.

The learning process consists of two steps that repeat until convergence:

1. **Data collection.** The current policy π interacts with the MDP. For a given initial state s^{init} and observation o_1 , the interaction produces an episode $\tau^{\text{ep}} \sim (\pi, \varepsilon)$. For each time step t along the episode, the return from that step onward, G_t , serves as the target value for the observation-action pair (o_t, a_t) . This procedure is detailed in Algorithm 2.
2. **Model update.** The tuples (o_t, a_t, G_t) obtained in the previous step serve as supervised training data: the observation-action pair (o_t, a_t) is the input, and the return G_t is the associated target. These examples are used to train the current approximation $\hat{q}_k$, associated with the current policy π_k . The improved approximation is denoted by $\hat{q}_{k+1}$, which now predicts returns more accurately under π_k . Since $\hat{q}_{k+1}$ has changed, the policy derived from it also changes. This new policy is denoted by π_{k+1} .

Because the policy is now π_{k+1} , the approximation $\hat{q}_{k+1}$ must be trained to predict returns under π_{k+1} . This requires returning to step 1 to collect fresh interaction data generated by π_{k+1} . The two steps thus alternate until the policy and the value approximation stabilize. Algorithm 3 summarizes this whole critic-only learning loop.

Even this simple algorithm already illustrates a difficulty inherent to all forms of RL: the instability of the learning process. Each update from $\hat{q}_k$ to $\hat{q}_{k+1}$ causes the policy to shift from π_k to π_{k+1} . For the learning to remain stable, the change from $\hat{q}_k$ to $\hat{q}_{k+1}$ should be as small as possible. This way, $\hat{q}_{k+1}$, trained to predict returns under π_k , remains reasonably accurate under π_{k+1} . However,

Algorithm 2: Data Collection**Input:**MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$ Policy π Discount factor $\gamma \in [0, 1]$ **Output:**Dataset $\mathcal{D}$ *Sample an initial state.*1 $t \leftarrow 1$;2 Sample $s_1 = s^{\text{init}} \in \mathcal{S}^{\text{init}}$;3 Observe o_1 ;*Episode generation by interaction with the MDP.*4 **repeat***Action selection.*5 Sample $a_t \sim \pi(\cdot \mid o_t)$;*Environment transition.*6 Sample $(s_{t+1}, o_{t+1}, r_t) \sim p(\cdot \mid s_t, a_t)$;7 $t \leftarrow t + 1$;8 **until** $s_t \in \mathcal{S}^{\text{ter}}$;*Return computation for each step of the trajectory.*9 $T \leftarrow t - 1$;10 **for** $t = 1, \dots, T$ **do**11 $G_t \leftarrow \sum_{k=t}^T \gamma^{k-t} r_k$;*Dataset construction*12 $\mathcal{D} \leftarrow \{(s_t, o_t, a_t, G_t)\}_{t=1}^T$;13 **Return** $\mathcal{D}$

1874 the smaller the change, the slower the learning progresses. A trade-off must
 1875 therefore be struck between stability and learning speed. Many improvements
 1876 introduced later in the literature aim at stabilizing this learning loop [60].

1877 Convergence guarantees for such algorithms have been proved under a different
 1878 set of assumptions than the setting presented here. For example, in the classical
 1879 convergence results, the MDP is fully observable, the value function is stored
 1880 in a table, and the learning rate is decreased at a suitable rate over time
 1881 [23]. In modern implementations, the value function is approximated by a NN.
 1882 Theoretical convergence proofs no longer hold in this setting. Nevertheless, deep
 1883 methods have demonstrated remarkable empirical success, making them the tool
 1884 of choice for RL tasks.

1885 Critic-only approaches suffer from several limitations. The main limitation
 1886 is scalability to large action spaces. Since the policy must evaluate all possible
 1887 actions at each decision step to select the maximum, the computational cost
 1888 grows with the size of the action set. To address this issue, another family of
 1889 RL algorithms exists: actor-critic methods.

Algorithm 3: Deep Critic-Only Reinforcement Learning**Input:**

MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$
 Architecture $\hat{q}$
 Exploration rate $\varepsilon \in [0, 1]$
 Discount factor $\gamma \in [0, 1]$
 Learning rate $\eta > 0$
 Initial parameters $\mathbf{w}$ (optional)

Output:

Trained policy $\pi_{\mathbf{w}}$

```

1 if  $\mathbf{w}$  is not given then
2   | Initialize  $\mathbf{w}$  randomly;
   Exploration policy as defined in Equation (4.2)
3  $\pi_{\mathbf{w}} \leftarrow \varepsilon$ -greedy policy with respect to  $\hat{q}(\cdot, \cdot, \mathbf{w})$ 
   MSE loss function as defined in Equation (2.2)
4  $\mathcal{L} \leftarrow$  Mean Squared Error (MSE);
5 while stopping criterion not satisfied do
   Data collection see Algorithm 2
6    $\mathcal{D} \leftarrow \text{DataCollection}(\text{MDP}, \pi_{\mathbf{w}}, \gamma)$ 
   observation-action pairs and their returns, from  $\mathcal{D}$ 
7    $\mathcal{D}' \leftarrow \{((o, a), G) \mid (s, o, a, G) \in \mathcal{D}\}$ 
   Model update with SGD see Algorithm 1
8    $\mathbf{w} \leftarrow \text{SGD}(\mathcal{D}', \hat{q}, \mathcal{L}, \eta, \mathbf{w})$ 
   Final policy without exploration as defined in Equation (4.1)
9  $\pi_{\mathbf{w}} \leftarrow$  policy with respect to  $\hat{q}(\cdot, \cdot, \mathbf{w})$ ;
10 Return  $\pi_{\mathbf{w}}$ 

```

4.4.2 Actor-Critic Methods

Two principal modifications distinguish actor-critic methods from critic-only approaches. First, the objective is to approximate the state-value function v_{π} , instead of the state-action value function q_{π} . Second, the policy π is given its own dedicated set of parameters. Two separate components are thus obtained: a value function approximation $\hat{v}(s, o, \mathbf{w})$, called the critic, with parameters $\mathbf{w}$, and a parameterized policy π_{θ} , called the actor, with parameters θ .

Analogously to critic-only methods, $\hat{v}$ is trained using supervised learning on pairs obtained from interaction with the MDP. For each time step t along a trajectory, the input is the state-observation pair (s_t, o_t) and the target is the observed return G_t . The value function is thus trained to correct its predictions based on data collected by the current policy interacting with the environment.

The key difference lies in how the policy π itself is optimized. Since the policy now has its own set of parameters θ , it can be updated directly. The update uses the notion of advantage $A : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, defined as:

$$A(s_t, a_t) = G_t - \hat{v}(s_t, o_t, \mathbf{w}),$$

where G_t is the return obtained by taking action a_t in state s_t and then following π_{θ} for all subsequent steps.

When $A(s_t, a_t) > 0$, the action a_t performed better than expected, and the policy parameters θ are adjusted to increase the probability $\pi_{\theta}(a_t | o_t)$. When $A(s_t, a_t) < 0$, the action a_t performed worse than expected, and the probability is decreased.

Both $\hat{v}$ and π_{θ} are improved simultaneously within the same loop: the critic provides a learned baseline for the actor, and the actor generates the data used to train the critic. This interplay reduces the instability inherent to RL. Algorithm 4 summarizes this whole actor-critic learning loop.

PPO (Proximal Policy Optimization), which is the current standard in RL, belongs to this family of actor-critic methods. It will be used throughout this thesis for all RL experiments.

4.5 Conclusion

This chapter has provided an introduction to the fundamental principles of RL. Only the essential concepts required to understand the RL framework have been presented. Despite this limited scope, the two defining steps of RL can already be observed in its simple learning procedure. The first step consists of estimating the reward that can be obtained from the interaction between the agent and its environment in a given situation. The second step uses this estimate to improve the current policy. Learning therefore results from the repeated alternation of these two basic operations.

This simple iterative structure brings RL conceptually close to DL. In both paradigms, simple operations can lead to complex behaviors when repeated at scale. Their combination is particularly powerful. A wide range of decision-making problems can be formulated as MDPs, allowing RL to be applied to a broad range of tasks. DL, in turn, provides flexible function approximators capable of representing the value functions and policies required to solve these tasks. Together, these two paradigms form the basis of deep RL, which has achieved remarkable success across a wide range of challenging tasks.

From a broader perspective, the evolution of RL has brought the field closer than ever to its initial biological inspiration. In modern implementations, agents progressively improve their behavior through trial and error directly based on experience. Moreover, their decision-making capabilities are represented by neural architectures that resemble their biological counterparts. It is therefore not surprising that RL has become one of the core paradigms for pursuing artificial general intelligence [100].

As the capabilities of RL continue to improve, a new challenge becomes increasingly important. The ability to learn complex behaviors with limited human intervention can make the resulting agent's strategy difficult to understand. To address this challenge, a parallel research direction has emerged, known as Explainable Reinforcement Learning (XRL). This field is the subject of the next chapter.

Algorithm 4: Deep Actor-Critic Reinforcement Learning

Input:

MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$
 Architecture $\hat{v}$ (critic)
 Architecture π (actor)
 Discount factor $\gamma \in [0, 1]$
 Learning rate $\eta > 0$
 Initial parameters $\mathbf{w}$ of the critic (optional)
 Initial parameters $\boldsymbol{\theta}$ of the actor (optional)

Output:

Trained policy $\pi_{\boldsymbol{\theta}}$

```

1 if  $\mathbf{w}$  is not given then
2   | Initialize  $\mathbf{w}$  randomly;
3 if  $\boldsymbol{\theta}$  is not given then
4   | Initialize  $\boldsymbol{\theta}$  randomly;

   MSE loss function as defined in Equation (2.2)
5  $\mathcal{L} \leftarrow \text{MSE}$ ;

6 while stopping criterion not satisfied do
   Data collection see Algorithm 2
7    $\mathcal{D} \leftarrow \text{DataCollection}(\text{MDP}, \pi_{\boldsymbol{\theta}}, \gamma)$ 

   Advantage estimation: how much better/worse each action performed compared
   to the critic's expectation
8   for  $(s_t, o_t, a_t, G_t) \in \mathcal{D}$  do
9     |  $A(s_t, a_t) \leftarrow G_t - \hat{v}(s_t, o_t, \mathbf{w})$ ;

   state-observation pairs and their returns, from  $\mathcal{D}$ 
10   $\mathcal{D}' \leftarrow \{(s, o), G \mid (s, o, a, G) \in \mathcal{D}\}$ 

   Critic update with SGD see Algorithm 1
11   $\mathbf{w} \leftarrow \text{SGD}(\mathcal{D}', \hat{v}, \mathcal{L}, \eta, \mathbf{w})$ 

   Actor update: increase the probability of actions with positive advantage, decrease
   it for those with negative advantage
12   $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \eta \sum_{(s_t, o_t, a_t, G_t) \in \mathcal{D}} A(s_t, a_t) \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t \mid o_t);$ 

13 Return  $\pi_{\boldsymbol{\theta}}$ 

```

Chapter 5

Explainability in Reinforcement Learning

Today, the majority of Artificial Intelligence (AI) agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences [64]. Even more autonomous systems, such as agents based on Large Language Models (LLMs), remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval [85, 93]. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees [107]. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both Reinforcement Learning (RL) and Deep Learning (DL) [40]. As a result, the mechanisms underlying their decisions cannot be understood directly from the learned model. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, the field of Explainable Reinforcement Learning (XRL) has emerged as a subfield of Explainable Artificial Intelligence (XAI). The objective of XRL is to develop explainability methods that make it possible to obtain explanations of RL agents. Due to the relevance of the topic and to foster broader real-world adoption of RL, a growing number of research teams are working on developing new explainability methods. This enthusiasm is reflected in a rapid increase in the number of publications, as illustrated in Figure 5.1.¹

The growing number and diversity of these methods make it increasingly difficult to obtain a unified view of the field. This motivated our work on developing a unified taxonomy of explainability methods for RL, initially presented in [112].

¹The complete search protocol, including the databases consulted and the full set of query terms, has been archived on Zenodo <https://zenodo.org/records/19387785>.

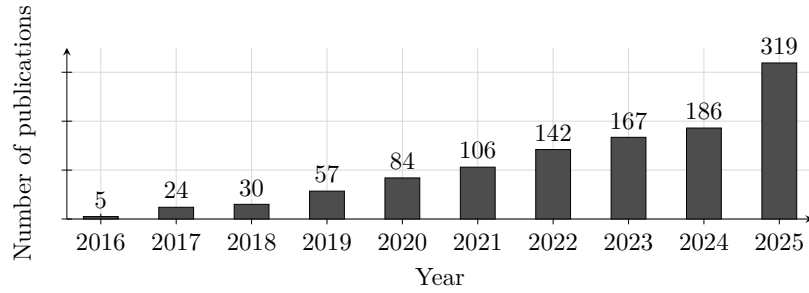


Figure 5.1: Yearly growth of Explainable Reinforcement Learning research with at least one citation on arXiv.

The taxonomy provides a common structure for organizing existing approaches according to the type of information they use and the form of explanation they provide. Building on this work, this chapter provides a detailed presentation of the proposed taxonomy and uses it to organize a broader state-of-the-art review of XRL.

This chapter is organized as follows:

- Section 5.1 begins by clarifying what we mean by an explanation in the context of XAI.
- Section 5.1 then introduces a taxonomy for categorizing existing XRL methods.
- Section 5.3 provides a comprehensive state-of-the-art review structured around this taxonomy.
- Section 5.4 concludes the chapter with a summary and discussion.

5.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in general. We then describe how this concept applies to XAI. Finally, we further detail the properties that characterize an explainability method in the context of XAI.

5.1.1 Explanation

Defining what constitutes an explanation is challenging because the concept is inherently multidisciplinary. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI [73, 92].

For this reason, we propose the following definition of explanation. An explanation involves two entities: the object (Definition 5.1.1) being explained and the receiver of the explanation. The receiver is a human who aims to

understand how the object operates and will be referred to as the observer throughout this thesis (Definition 5.1.2). The observer seeks to understand the object and, in doing so, builds a mental model of its functioning. An explanation is therefore what allows the observer to refine and correct their mental model of how the object operates (Definition 5.1.4). This refinement is obtained through a transformation process that produces an explanation from the object being explained. We refer to this process as an explainability method (Definition 5.1.5), detailed further in Section 5.1.3. For an observer to understand an explanation, the information it contains must be interpretable (Definition 5.1.3). Information is considered interpretable when it can be directly understood by a human without requiring any additional transformation process.

Definition 5.1.1: Object

The object of an explanation is the system the observer wants to understand.

Definition 5.1.2: Observer

The observer is the human who aims to understand an object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

Definition 5.1.3: Interpretable

Information is interpretable when it can be directly understood by an observer without requiring any additional transformation process.

Definition 5.1.4: Explanation

An explanation is any interpretable information that enables an observer to refine or correct their mental model of the object of the explanation.

Definition 5.1.5: Explainability Method

An explainability method is a transformation process that produces an explanation from the object being explained.

Figure 5.2 summarizes this whole explanation process, from the object of the explanation to the mental model refined by the observer, as a chain of three information structures, depicted as boxes and connected by arrows denoting a transformation of information:

- (1) The object of the explanation, that is, the system the observer seeks to understand.
- (2) The explanation, the interpretable information produced from the object. This transformation is performed through the use of an explainability method.
- (3) The mental model held by the observer regarding the object.

A brace spanning boxes (2) and (3) indicates that this second half of the process,

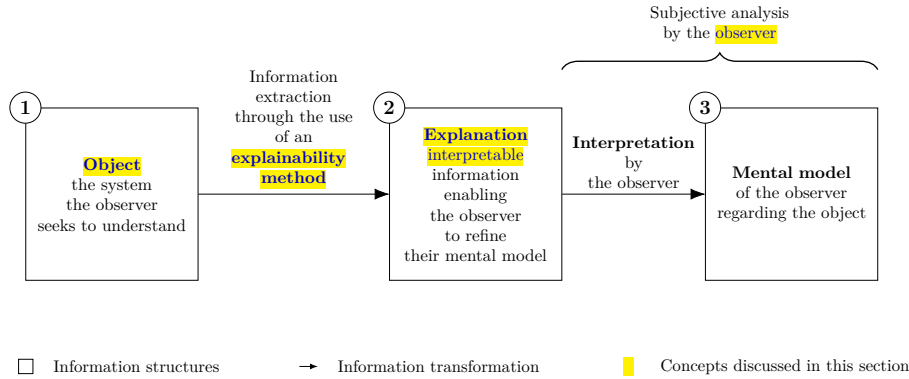


Figure 5.2: Explanation process.

from the explanation to the resulting mental model, constitutes a subjective analysis carried out by the observer.

This definition of explanation establishes the approach to explainability adopted throughout this thesis. Under this definition, explainability is regarded as a continuum that evolves as knowledge develops [81, 118].

This quantitative perspective highlights that not all explanations provide the same level of value [51]. The effectiveness of an explanation depends on its ability to improve the observer's understanding of the system. For example, describing a marginal aspect of a system is different from correcting an observer's misunderstanding of a key mechanism.

5.1.2 Explainable Artificial Intelligence

Now that the general framework of explanation has been established, we can examine how this concept applies in the context of AI. In the context of XAI, the object of explanation is an AI system. More precisely, XAI emerged as a response to the success of DL models [69, 82]. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of Neural Networks (NNs) cannot be directly associated with human concepts. From this starting point, we identify two types of approaches:

- **Substitutive explainability:** In this approach, the assumption is that there is nothing to explain within a NN. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the "black box" view of NNs. Under this assumption, XAI consists of replacing as much of the DL system as possible with alternative AI methods [78].
- **Intrinsic explainability:** In this approach, it is argued that NNs contain mechanisms that can be explained. The limitation is considered to lie not in the networks themselves, but in the absence of adequate tools to analyze their functioning [87].

This thesis considers methods from both approaches in order to provide

a comprehensive overview of the XAI field. Although both approaches have produced convincing results, this thesis primarily adopts the perspective of the *Mechanistic Hypothesis* (Hypothesis 3) [81, 118]. This perspective is more closely aligned with the definition of explainability adopted in this thesis, which considers explainability as a continuum that evolves with the development of knowledge.

Hypothesis 3: Mechanistic Hypothesis

Every AI system results from a sequence of causal computations. Because the rules governing these interactions are mechanical, the behavior of the system can be explained. If an AI system cannot be explained, this reflects a limitation of knowledge rather than an intrinsic property of the system.

Within intrinsic explainability, the *Mechanistic Hypothesis* (Hypothesis 3) underlies the approach that we refer to as Mechanistic Explainability (ME) (Definition 1.1.1), generally called mechanistic interpretability in the literature [87]. The definitions introduced above clarify this terminological choice. According to Definition 5.1.3, the internal mechanisms of a NN are not interpretable, since an observer cannot directly understand them without an additional transformation process. What this approach produces is interpretable information obtained from these mechanisms through an explainability method (Definition 5.1.5), that is, explanations in the sense of Definition 5.1.4. It therefore falls within explainability rather than interpretability.

5.1.3 Explainability Method in XAI

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. In our framework, an explanation in the field of XAI can be characterized by three components:

- **Source:** refers to the origin of the information used by the explainability method. The explainability method transforms the raw information produced by the AI system into an explanation that can be understood by an observer.
- **Form:** refers to the representation through which the explanation is communicated. The form defines how the information is presented to the observer and must allow human interpretation. An explanation can take many different forms, including text, images, computer code, and mathematical equations.
- **Subject:** refers to the aspect of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

These three properties of an explanation are entirely determined by the explainability method (Definition 5.1.5) used to produce it. In this context, an explainability method is a transformation process that maps a source of information into an explanation characterized by a specific form and subject.

2105 Research in XAI focuses on developing new methods capable of performing this
 2106 transformation in order to overcome the lack of interpretability associated with
 2107 DL models.

2108 5.2 Taxonomy of XRL Methods

2109 Now that the concept of XAI has been defined, we turn to its application in RL,
 2110 known as XRL. As in XAI, explanations in XRL are produced by explainability
 2111 methods and can be described in terms of their subject, source, and form. This
 2112 section reviews the possible variations of these three aspects in the specific
 2113 context of RL.

2114 5.2.1 Subject

2115 The first question to address is the subject of explanation in the context of RL.
 2116 To answer this question, it is necessary to identify the different components
 2117 involved in a RL system and determine which of them constitutes the object of
 2118 explanation. A RL system can be described through two main components: the
 2119 environment and the policy.

2120 The environment defines the dynamics in which the agent evolves, including
 2121 the possible states, observations, actions, and transitions. Environments are
 2122 simulations designed by humans, and their dynamics are explicitly defined.
 2123 Therefore, it is assumed that the environment is known to the observer and does
 2124 not require explanation.

2125 The main object of interest is therefore the agent’s policy. The policy
 2126 represents the decision-making mechanism of the agent. It is a function that maps
 2127 observations to actions and is learned through interaction with the environment in
 2128 order to maximize the expected cumulative reward. The policy is the component
 2129 that encodes the learning process and that XRL aims to explain.

2130 Explaining a policy introduces additional challenges compared with classical
 2131 XAI settings. In traditional XAI problems, the objective is to explain an isolated
 2132 decision, such as determining which features of an image led to a classification
 2133 result. In contrast, RL introduces a temporal dimension into the decision-making
 2134 process. An action is not only determined by the current observation but also
 2135 influences future states and rewards. Therefore, understanding the behavior of
 2136 an agent may require considering a sequence of interactions between the agent
 2137 and its environment.

2138 This temporal dimension leads to a distinction between two objectives of
2139 explanation in XRL:

- 2140 • **Decision explanations:** focus on identifying the elements of the current
2141 observation that influenced the action selected by the policy. This type of
2142 explanation is close to classical XAI approaches, as it analyzes the rela-
2143 tionship between inputs and outputs while abstracting away the sequential
2144 nature of the decision process.
- 2145 • **Strategy explanations:** aim to characterize the agent’s behavior over
2146 time. They focus on sequences of decisions and interactions with the
2147 environment in order to understand how the agent achieves its long-term
2148 objective. This type of explanation is specific to RL, as it addresses the
2149 temporal and goal-oriented nature of the learning process.

2150 These two objectives should not be considered as strictly separated categories,
2151 but rather as two extremes of a continuum. Understanding the elements of an
2152 observation that influence a specific decision can help the observer generalize
2153 this information to infer broader aspects of the agent’s behavior. Conversely,
2154 understanding the agent’s overall strategy can provide insights into the reasoning
2155 behind individual decisions.

2156 5.2.2 Source

2157 We identify three sources of explanation in RL, all obtained during or after the
2158 learning process and containing information about the agent:

- 2159 • **Policies:** policies are functions that generate a distribution over the action
2160 space for a given observation. The objective of RL is to learn a policy that
2161 maximizes the expected cumulative future rewards. During the learning
2162 phase, the policy is iteratively improved through interactions with the
2163 environment.
- 2164 • **Trajectories:** trajectories represent sequences of successive interactions
2165 between the agent and the environment. At each step, the agent selects
2166 an action based on its observation, which modifies the environment and
2167 produces a reward as well as a new observation. This process continues
2168 until the end of the trajectory.
- 2169 • **Value functions:** value functions provide predictions about the future
2170 outcome of a trajectory from a given observation. The most commonly
2171 used value function is the critic function, which estimates the expected
2172 sum of future rewards. This prediction guides the learning process, as the
2173 agent updates its policy to maximize the estimated future return. However,
2174 other types of value functions can also be used as sources of explainability.

2175 5.2.3 Form

2176 We identify six forms of explanation in the RL setting:

- 2177 • **Policy Model:** an interpretable model of the agent’s policy function.
- 2178 • **Saliency Map:** a weighting of the input features representing their
2179 influence on the output of the policy function.
- 2180 • **Counterfactual:** a modified observation obtained by perturbing the
2181 original observation in order to produce a different action from the agent.
- 2182 • **Agent Trajectory:** a sequence of successive interactions between the
2183 agent and the environment.
- 2184 • **Trajectory Prediction:** a prediction about the future evolution of the
2185 trajectory produced by a value function.
- 2186 • **Interaction Model:** an interpretable model of the interactions between
2187 the agent and the environment.

2188 Among the three components characterizing an explanation, the form is the
2189 one that varies most significantly across explainability methods. While subjects
2190 and sources tend to be relatively stable in the XRL literature, the diversity in
2191 how explanations are presented is considerably greater. The list above is by no
2192 means definitive and may evolve as new approaches emerge. For this reason, the
2193 following section is organized according to the form of explanations and presents
2194 existing explainability methods based on this criterion.

2195 5.3 State-of-the-Art Review of XRL Methods

2196 The purpose of this section is to detail the functioning of existing explainability
2197 methods (see Table 5.1). They are presented according to the form of explanation
2198 they produce, following the taxonomy introduced in Section 5.2.

2199 A family of methods is defined as a set of methods that share the same
2200 source, form, and subject of explanation, as well as a similar procedure for
2201 generating explanations. Methods belonging to the same family may differ in
2202 their implementation details while relying on the same principle.

2203 The remainder of this review is organized by form of explanation, following
2204 the six forms identified in Section 5.2.3. For each form, we first describe its
2205 principle and its limitations, then present the families of methods that produce
2206 it.

2207 5.3.1 Policy Model

2208 The policy function can be represented by a wide variety of function classes. In
2209 modern RL, NNs are the most commonly used models for representing policies.
2210 Although these models achieve the highest performance, their connectionist
2211 nature makes their internal representations non-interpretable.

2212 For this reason, a substitutive approach to XRL aims to replace these models
2213 with alternative models considered interpretable. Such models can be understood
2214 through direct analysis of their representations. The main classes of interpretable

Explanation Subject	Explanation Form	Explainability Method Family	Source of Explanation		
			Policy	Trajectories	Value functions
Decision	Policy Model	<i>Interpretable policy learning</i>	✓		
		<i>Policy approximation via interpretable model</i>	✓		
	Saliency Map	<i>Observation perturbation</i>	✓		
		<i>Observation gradient analysis</i>	✓		
		<i>Attention-based architecture</i>	✓		
	Counterfactual	<i>Latent space usage for counterfactual generation</i>	✓		
Strategy	Agent Trajectory	<i>Hierarchical decomposition</i>	✓	✓	
		<i>Extraction via critic analysis</i>		✓	✓
	Trajectory Prediction	<i>Critic enrichment</i>			✓
	Interaction Model	<i>Bayesian network learning</i>		✓	
		<i>Latent space usage for MDP generation</i>	✓		

Table 5.1: Taxonomy of explainability methods in reinforcement learning.

models used to represent policy functions include decision trees [101], algorithms [102], formal logic [77], and linear models [84].

Explanations derived from an interpretable model primarily provide insights into the decision-making process, since what is modeled is the mapping between an observation and an action. If the model is sufficiently well structured, the observer can form a mental representation of a sequence of decisions across the environment and thus obtain information about the agent’s strategy.

The main limitation of these methods is that interpretable models rely on symbolic representations. When the policy to be represented is too complex, achieving comparable performance requires a large number of rules or conditions. The resulting model can quickly become difficult to analyze and lose its interpretability.

For example, representing a policy capable of playing Go using a symbolic model would require considering a very large number of possible situations and specific cases. Even if such a representation were possible, the resulting model would likely become intractable for human analysis due to its complexity. This illustrates why NN-based approaches have become dominant in such domains, as they can learn complex decision strategies without requiring an explicit enumeration of rules.

To obtain an interpretable policy model, two families of explainability methods are available: *interpretable policy learning* and *policy approximation via interpretable model*.

Interpretable Policy Learning

This family of methods aims at learning a policy represented by an interpretable model [101, 102]. It is the most direct approach to addressing the challenges posed by XRL. This type of training requires modifying the Markov Decision Process (MDP) in order to adapt it to the search space and to the type of policy

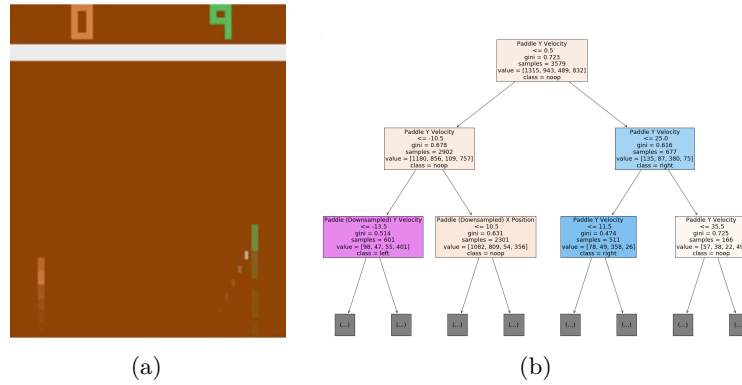


Figure 5.3: Policy approximation via an interpretable model [99].

representation. Moreover, this search space often needs to be reduced to make learning feasible. Such a reduction requires the integration of expert knowledge into the learning process. With this family of methods, the *interpretable policy function* itself is used to provide explanations to the observer.

Policy Approximation via Interpretable Model

An interpretable model can be trained to approximate a policy learned through deep RL [99]. In this setting, the problem is formulated as a supervised learning task, where the initial deep RL policy is used to label each observation with the action it selects, such as the observation from the Atari game **Pong** shown in Figure 5.3a. Using this dataset, the interpretable model is trained to reproduce the behavior of the initial policy, yielding for instance the decision tree illustrated in Figure 5.3b, obtained by approximating a deep RL policy trained on that environment. The resulting interpretable model is then used as an explanation for the observer. If this approximation is sufficiently faithful, it can also be deployed in the environment in place of the initial policy, thereby providing a high degree of control over the agent.

5.3.2 Saliency Map

The influence of each part of the input on the action selected by the policy can be visualized using heatmaps (Figure 5.4). In the RL setting, the input corresponds to the observation received by the agent at a given time step. Heatmaps therefore highlight the regions of the observation that have the greatest influence on the action selection process. By providing this information, they allow the observer to identify which elements of the environment are considered important by the policy when making a decision. In Figure 5.4, both panels correspond to the same observation of the Atari game **Breakout**, with panel (a) obtained by observation perturbation and panel (b) by observation gradient analysis.

A limitation of heatmap-based methods is that the resulting explanations are not always directly meaningful to the observer. As illustrated in Figure 5.4b, interpreting which regions are important often requires human reasoning and involves a degree of subjectivity. This limitation becomes particularly relevant

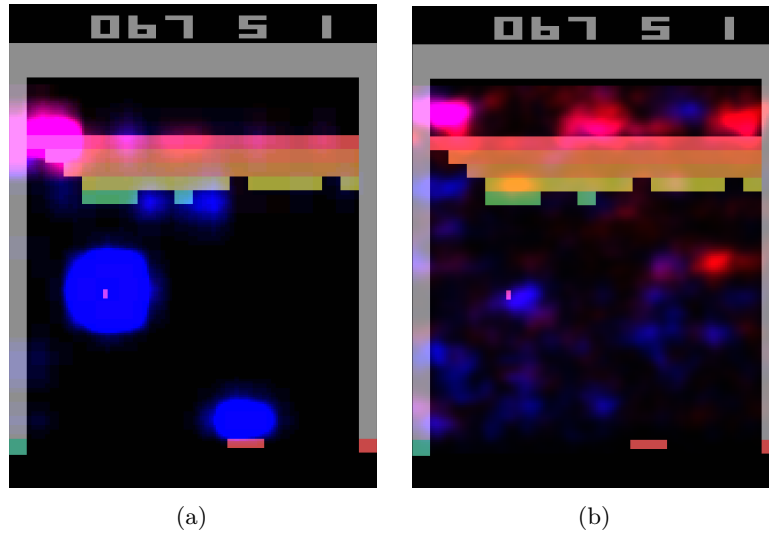


Figure 5.4: Heatmaps for the same observation of the Atari game **Breakout** [58].

in RL, where the objective is not only to explain a single decision but also to understand the factors influencing a sequence of decisions over time.

Three families of explainability methods can be used to obtain this type of explanation: *observation perturbation*, *observation gradient analysis*, and *attention-based architectures*.

Observation Perturbation

Perturbation analysis consists in applying various modifications to the observation in order to study their impact on the output of the policy function [58]. The greater the variation in the output caused by a perturbation applied to a given component of the input vector, the higher the weight assigned to that component. The magnitude of the perturbation is measured by computing the distance between the original output and the output obtained from the perturbed input. Using this set of weights, a *heatmap* is constructed, enabling the observer to visualize which parts of the observation vector are most sensitive in the action-selection process (see Figure 5.4a).

Observation Gradient Analysis

Observation gradient analysis refers to a set of methods that require a differentiable policy function in order to extract information from it [42, 70, 80]. To this end, the gradient of the policy function is computed with respect to all components of the input vector. For each input component, the resulting gradient provides a weighting that represents its importance in the output of the policy function. Based on these weightings, a *heatmap* is constructed to allow the observer to visually identify the parts of the input vector that are the most influential in the policy function's output (see Figure 5.4b).

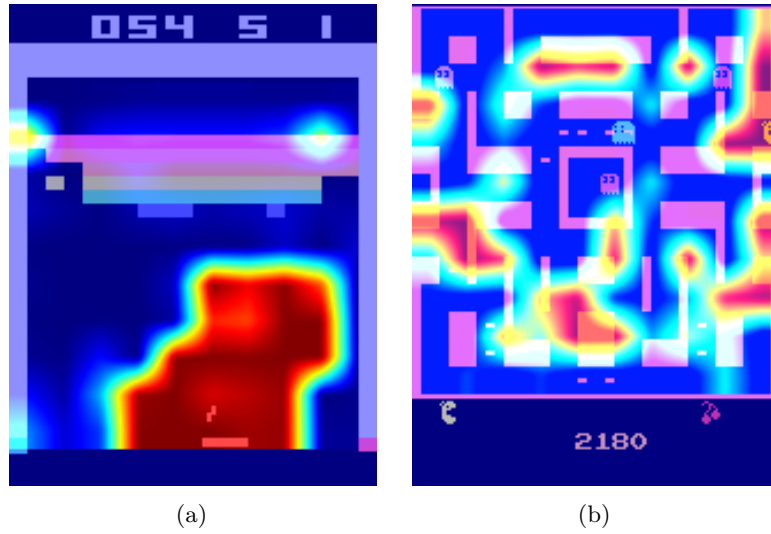


Figure 5.5: Attention-based heatmaps for the Atari games **Breakout** and **Ms. Pac-Man** [74].

2296 *Attention-Based Architecture*

2297 Attention blocks constitute a NN architecture that makes it possible to weight
 2298 the relative importance of one part of a vector with respect to the other parts of
 2299 the same vector [55]. These weights are computed as a function of the vector
 2300 itself and are learned by the NN during training. For a given observation, each
 2301 component of the observation vector is assigned a scalar value. It can therefore
 2302 be assumed that the components associated with the highest weights are the most
 2303 influential in determining the output [74, 94]. This attention-based *heatmap* is
 2304 then provided to the observer as the explanation, as illustrated for the Atari game
 2305 **Breakout** in Figure 5.5a, where the highlighted regions are easily interpretable.
 2306 However, as shown for **Ms. Pac-Man** in Figure 5.5b, the attention weights are
 2307 not always concentrated on a small, easily interpretable set of regions, which
 2308 can make this type of explanation difficult to exploit.

2309 **5.3.3 Counterfactual**

2310 A counterfactual explanation is a new observation, close to the original one, that
 2311 leads the agent to select a different action. Comparing the two observations
 2312 reveals which elements of the environment must change for the agent to behave
 2313 differently, and thus provides insight into its decision-making process. Unlike
 2314 the other forms of explanation, this form is currently associated with a single
 2315 family of methods, based on the *use of the Latent Space (LS) for counterfactual*
 2316 *generation*.

2317 *Latent Space Usage for Counterfactual Generation*

2318 This family of methods uses generative models to build counterfactuals from
 2319 the Latent Representation (LR) of the observation [96, 103]. The observation

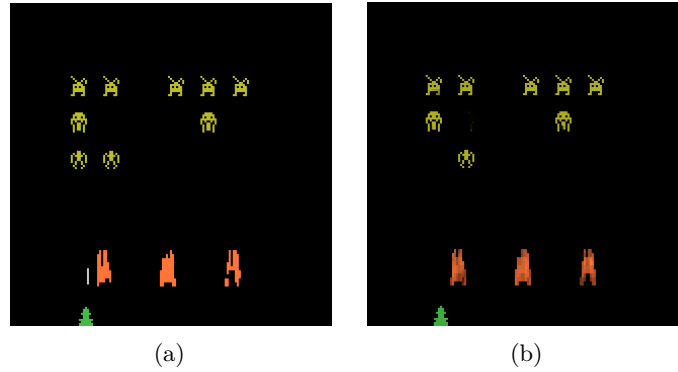


Figure 5.6: Counterfactual explanation of the agent’s behavior for the Atari game *Space Invaders* [96]. Panel a shows the original observation and panel b the counterfactual observation produced by a generative model.

is encoded into the LS, its representation is modified until the agent selects the desired action, and this modified representation is decoded back into a new observation. Working in the LS rather than at the pixel level keeps the counterfactual close to the original state and consistent with the environment. The fidelity of the generated observations to the real game can then be assessed through user studies [96].

Figure 5.6 illustrates this approach on the Atari game *Space Invaders* [96]. In the original observation (Figure 5.6a) the agent selects `LEFT`, whereas in the counterfactual observation (Figure 5.6b) it selects `RIGHT_FIRE`. Here, removing an enemy is enough to change the action, which suggests that the position of that enemy influences the agent’s decision. Such observations remain only indicative. Identifying a reliable pattern requires repeating the analysis across many situations.

5.3.4 Agent Trajectory

Until now, the presented methods have focused on explaining individual decisions made by the agent. The methods considered from this point on aim to provide insight into the agent’s strategy by analyzing its behavior over time.

The most direct approach consists in visualizing agent trajectories, most commonly through complete episodes. By analyzing these trajectories, the observer can infer the dynamics underlying the agent’s behavior and how its decisions evolve through time. This type of visualization is not specific to XRL. In practice, visualizing agent episodes is a standard step in the development and evaluation of RL. It provides a simple way to verify that the agent behaves as expected and to detect potential implementation errors.

This simple approach can be further enhanced using families of methods such as *hierarchical decomposition* and *extraction via critic analysis*.

Extraction via Critic Analysis

In this family of methods, the predictions of the critic are used to determine which trajectories to present to the observer [90]. A set of states is selected from

multiple trajectories. For a given state, all possible actions are examined. The state is assigned a value equal to the difference between the critic’s prediction for the lowest-valued action and the prediction for the highest-valued action. Trajectories containing the states with the highest values according to this procedure are then chosen to be presented to the observer. These trajectories are considered critical moments of the agent and are therefore relevant for analysis. For this family of methods, the *selected trajectories* represent the interpretable information provided to the observer.

Hierarchical Decomposition

Hierarchical agents are composed of a set of sub-policies specialized for specific tasks within subsets of the environment’s states. This family of methods distributes the optimization of the reward function across multiple models, significantly simplifying the learning process. Moreover, this decomposition of the policy function can also provide explainability in the agent’s strategy [67]. If a sub-policy is selected, it indicates that the agent will execute a specialized behavior. This behavioral specialization provides insight into the strategy the agent employs to maximize the reward function. The agent’s *trajectory*, together with the *sub-policy* selected for each action, is used as interpretable information.

5.3.5 Trajectory Prediction

Making predictions about the future of a trajectory allows one to form a representation of what the system expects, based on its training experience. These predictions are produced using value functions. Such functions provide insights into the future evolution of the agent’s trajectory. By their nature, this form of explanation informs the observer about the strategy adopted by the agent. Value functions provide an estimate of the expected cumulative reward from a given state. Consequently, a value function maps an observation to a scalar value in $\mathbb{R}$.

This aggregated prediction provides limited information about the underlying factors that contribute to this value. It does not directly reveal which aspects of the task are being optimized or how the predicted return is composed. Unlike the other forms of explanation, this form is currently associated with a single family of methods, aimed at improving the interpretability of value-based explanations: *critic enrichment*.

Critic Enrichment

In order to obtain interpretable information from the predictions of the critic, it is possible to decompose the critic into multiple sub-functions [71]. The reward function is thus decomposed into multiple sub-reward functions, each representing a sub-objective of the task to be accomplished. Similarly, for each of these sub-reward functions, corresponding sub-critic functions are defined, along with a function that combines their outputs. During training, the agent selects its actions so as to maximize this combination function of the sub-critics. Using this approach, for each observation and action, a prediction is obtained regarding the sub-objectives the agent is attempting to maximize, so that the

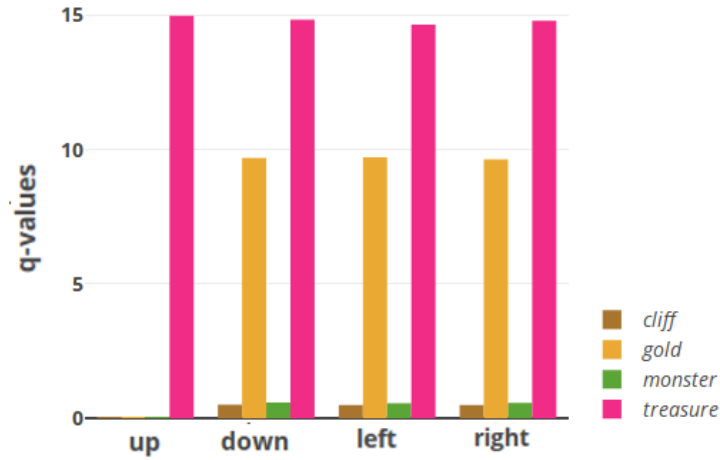


Figure 5.7: Prediction of sub-rewards as a function of the action selected by the agent for a given state [71].

2392 action chosen by the agent reveals which sub-reward it aims to maximize (see
 2393 Figure 5.7). This *prediction* is then used as the explanation for the observer.

2394 5.3.6 Interaction Model

2395 Modeling the interactions between an agent and its environment makes it possible
 2396 to understand how the agent’s decisions are made and what impact they have
 2397 on the environment. By incorporating these interactions into an explanatory
 2398 model, one can better grasp the complex relationships between the agent and its
 2399 environment, thereby identifying the determining factors in the decision-making
 2400 process. By explicitly handling the agent–environment interaction, this type
 2401 of explanation aims to explain the agent’s strategy. The families of methods
 2402 used to obtain explanations in the form of agent–environment interactions are:
 2403 *Bayesian network learning* and *LS usage for MDP generation*.

2404 *Bayesian Network Learning*

2405 The evolution of an agent within an environment can be observed as a set of
 2406 random variables connected through a Bayesian network [86]. Random variables
 2407 are defined to represent both external aspects (the environment) and internal
 2408 aspects of the agent (the actions). By analyzing trajectories, it becomes possible
 2409 to infer causal relationships among these random variables. At the end of this
 2410 process, a weighted causal graph is obtained, which serves as an explanation
 2411 for the observer, as illustrated for the video game *StarCraft II* in Figure 5.8,
 2412 where nodes represent random variables and edges represent causal relationships
 2413 between them. This *Bayesian network*, representing the interactions between
 2414 the agent and the environment, is then used as the explanation.

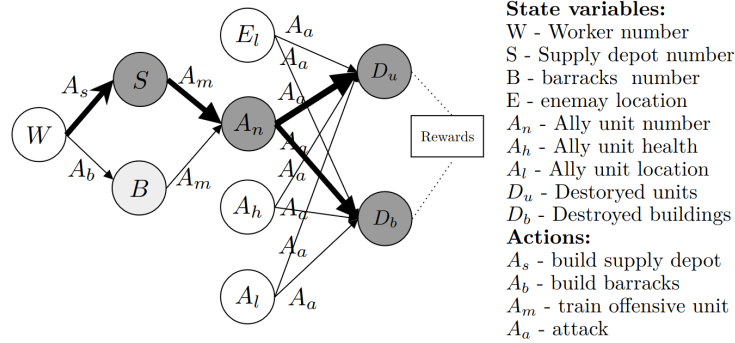


Figure 5.8: Agent-environment interaction model for the video game *StarCraft II* [86].

Latent Space Usage for MDP Generation

Each layer of a NN can be viewed as a projection from one space to another [50]. As information propagates through successive layers of a NN, it becomes increasingly abstract. It can therefore be assumed that two vectors that are close in this projected space are also close in terms of the agent's perception. Based on this observation, it is possible to redefine an MDP by leveraging the abstract representations produced by the final layers of a NN. Once the new MDP has been constructed, a model of the interaction between the agent and its environment is obtained. This model, represented as a *graph*, is then used as the explanation.

Figure 5.9 illustrates this idea using a two-dimensional *t-SNE* projection of the LS of the NN used to represent the policy. Each point is colored according to the value function's prediction for the corresponding state, with warmer colors indicating higher predicted values. An agreement can be observed between the structure of this projected space and the critic's predictions: states that are assigned similar values by the critic also tend to be located close to one another in the projection. This agreement makes it possible to manually identify clusters of states, which likely correspond to states that are perceived as similar by the agent.

5.4 Conclusion

The taxonomy and the state-of-the-art review presented in this chapter make it possible to draw three main observations.

5.4.1 Absence of Explanation of the Learning Process

Regarding the subject of explanation, Table 5.1 shows that none of the reviewed methods addresses the learning process itself. Such an explanation would concern how the policy is progressively updated throughout training, and why it evolves over successive iterations to converge toward a particular solution. Several reasons explain this absence.

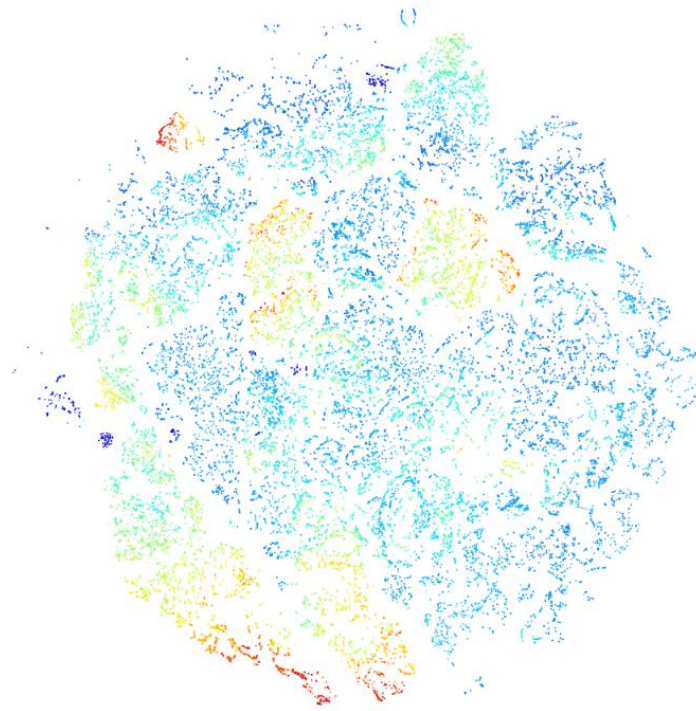


Figure 5.9: *t-SNE* projection of a neural network’s latent space [50].

2443 First, among the different machine learning paradigms, RL is arguably the
 2444 one that would benefit the most from explanations of the learning process.
 2445 Unlike supervised or unsupervised learning, it relies on an explicit exploration–
 2446 exploitation trade-off, making the learning trajectory itself an informative subject
 2447 of explanation. However, most XRL methods are inherited from the XAI
 2448 literature, where explanations focus exclusively on the behavior of already-
 2449 trained models. This limitation has naturally carried over to XRL.

2450 Second, explanations of the learning process have limited value from a
 2451 practical perspective. In most real-world applications, practitioners deploy
 2452 an already-trained agent. Practitioners are therefore primarily interested in
 2453 understanding the behavior of the final policy rather than the optimization
 2454 process that produced it.

2455 Finally, explaining the learning process raises significant methodological
 2456 challenges. Training in RL can be viewed as the iterative improvement of a policy
 2457 through repeated interactions with the environment. In this sense, explaining
 2458 learning would amount to explaining the evolution of an entire sequence of
 2459 policies rather than a single one. Current XRL methods still struggle to provide
 2460 comprehensive explanations of an individual policy, so extending them to a full
 2461 training trajectory appears considerably more difficult.

5.4.2 Lack of Unified Metrics for Explainability

The diversity of explanation forms identified in this review also reveals another limitation shared by nearly all families of XRL methods, which is the absence of reliable metrics for evaluating explainability. In many studies, explanations are not evaluated with human participants. Instead, authors focus on technical properties of the proposed method, such as fidelity, sparsity, or computational efficiency. When human evaluations are conducted, they usually rely on subjective satisfaction questionnaires. Although useful, self-reported satisfaction remains prone to numerous biases and does not necessarily indicate that the explanation has improved the observer’s mental model or understanding of the agent.

This limitation is not specific to XRL, it reflects a broader challenge affecting XAI as a whole. In our view, this limitation comes from how explainability has developed in artificial intelligence. Most of the proposed approaches have been developed by the computer science community. Explainability has consequently often been studied from an algorithmic or mathematical perspective, with less attention given to how humans actually benefit from explanations.

Our definition of explanation explicitly places the observer at the center of the explanatory process. From this perspective, the value of an explanation should not only depend on its technical properties but also on its ability to improve the observer’s understanding. Addressing this challenge will require stronger interdisciplinary collaboration between computer science and disciplines such as cognitive science, psychology, philosophy, and sociology. These fields provide the theoretical and experimental tools needed to study human understanding and to develop evaluation protocols that measure the actual effectiveness of explanations.

5.4.3 Toward Intrinsic Explainability in XRL

Our review also suggests that current XRL research remains largely dominated by substitutive explainability approaches. Replacing a deep model with a simpler one therefore risks losing precisely the complexity that makes deep RL effective. For this reason, we argue that a promising direction for XRL is to develop explanations directly from the NN that forms the agent’s policy. Such approaches aim to analyze the representations learned by the network itself. From this perspective, the NN should not necessarily be regarded as an black box, but as a complex representation whose apparent opacity may instead reflect the limitations of current analysis tools.

Among the methods reviewed, *latent space usage for MDP generation* (Section 5.3.6) appears closest to this intrinsic perspective. However, this approach still relies on a largely manual and qualitative analysis of the LS, which limits its reproducibility and its ability to produce systematic explanations. The next chapter builds upon this idea and proposes a mechanistic XRL approach based on LS analysis.

Chapter 6

Clustering Agent Representations

As discussed in the previous chapter, current explainability methods for Reinforcement Learning (RL) still lack intrinsic approaches capable of explaining the internal mechanisms of deep neural policies. To the best of our knowledge, only [50] have proposed exploiting the Latent Space (LS) learned by neural policies in order to identify the concepts used by the agent. Its main idea is to analyze the Latent Representations (LRs) produced by the Neural Network (NN) and organize them into clusters that correspond to high-level concepts.

While the core idea of this paper is closely aligned with our vision of explainability in RL, this approach suffers from two important limitations. First, the clusters are manually constructed. Such a procedure heavily depends on the user’s expertise and neither guarantees reproducibility or scales. Second, no evaluation protocol is proposed to assess the quality of the obtained clusters. Their relevance is essentially determined through visual inspection.

The Clustering Agent Representation (CAR) method proposed in this chapter addresses these two limitations. First, it automates concept discovery by relying on unsupervised clustering algorithms to identify the Concept Structures (CSs) of the LS. Second, it introduces an objective evaluation metric based on the stability of the extracted representations across independently trained policies.

This chapter is organized as follows:

- Section 6.1 lays out the theoretical assumptions underlying the proposed CAR method, which we term the *Representation Convergence Hypothesis* (Hypothesis 4).
- Section 6.2 then details the CAR method itself, covering both the extraction of LRs using surrogate policies and the automatic detection of CSs through clustering.
- Section 6.3 introduces the metrics we propose for evaluation.
- Section 6.4 situates CAR within the broader landscape of Mechanistic Explainability (ME).

6.1 Representation Convergence Hypothesis

The CAR method relies on a assumption, referred to as the *Representation Convergence Hypothesis* (Hypothesis 4). This hypothesis follows the two assumptions established previously throughout this thesis.

The *Mechanistic Hypothesis* (Hypothesis 3) rejects the notion that deep NNs constitute irreducible black boxes. Adopting a mechanistic perspective, we consider that every process results from a sequence of causal computations. Although these computations may be complex, they remain explainable through the analysis of the internal mechanisms.

The *Concept Hypothesis* (Hypothesis 1) states that NNs achieve generalization by learning concepts. During optimization, the network progressively organizes its LS into representations that capture the information required to solve the task.

Following these two hypotheses, it becomes possible to explain a NN by studying the organization of its LS. This family of approaches corresponds to ME (Definition 1.1.1). Since the CAR method is grounded in these two hypotheses, it falls within this field of research. More specifically, CAR focuses on a specific aspect of ME known as universality. Universality refers to the consistency with which a concept is recovered across different models trained on the same task [119].

To provide a explanation for this phenomenon, we developed the *Representation Convergence Hypothesis* (Hypothesis 4). This hypothesis states that, when NNs are trained to solve the same task, a set of concepts should systematically emerge because these concepts provide a effective way of representing the problem. As a result, independently trained models should develop similar concepts in their learned representations.

Hypothesis 4: Representation Convergence Hypothesis

For a given task, independently trained NNs tend to develop similar sets of concepts because these concepts provide an efficient representation for solving the task.

The CAR method is designed to investigate the *Representation Convergence Hypothesis* (Hypothesis 4) by assessing whether it can consistently recover similar concepts across independently trained models. The hypothesis therefore serves both as the foundation of the method and as the phenomenon it aims to investigate. If CAR consistently recovers the same set of concepts across different training runs, this set is more likely to reflect invariant properties of the task than model-specific artifacts. Conversely, the absence of representation convergence would provide evidence against the hypothesis and would also question the validity of the CAR approach.

It is important to clarify that this hypothesis does not imply convergence at the parameter level. Independently trained models are expected to converge toward different sets of weights. The proposed hypothesis concerns a abstract level of representation. We assume that the conceptual organization induced by their LRs converges toward an equivalent CS, even though the numerical implementation of the networks differ.

The remainder of this chapter develops the methodology used to investigate

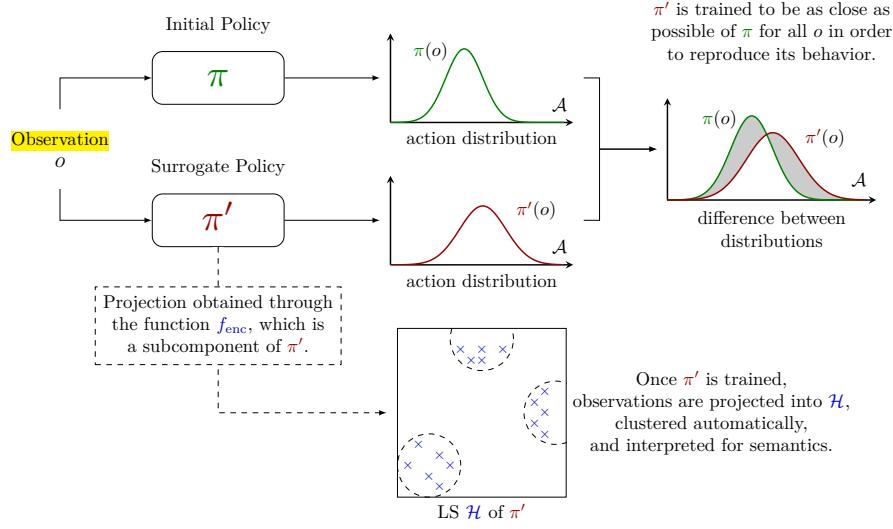


Figure 6.1: Illustration of the Clustering Agent Representation method.

2577 this hypothesis.

2578 6.2 Latent Clustering of Surrogate Policies

2579 Following Zahavy et al. [50], we analyze the LS through clustering techniques,
 2580 placing our approach within the broader field of Deep Clustering. This design
 2581 choice is motivated by our rejection of the *Superposition Hypothesis* (Hypothesis 2)
 2582 and, more generally, of any view that assumes a linearly organized CS, as
 2583 discussed in Section 3.5.

2584 The CAR methodology is composed of two main stages. The first stage trains
 2585 multiple independent NNs, called surrogate policies, which implement the same
 2586 decision policy. The second stage clusters their LRs in order to identify the
 2587 underlying conceptual organization. Once these two stages have been completed,
 2588 the resulting clustering can be analyzed from two complementary perspectives.
 2589 First, the consistency of the resulting clusters across independent training runs
 2590 can be assessed to determine whether the identified clusters are stable. Second,
 2591 the semantic content of the clusters can be examined to determine whether they
 2592 correspond to meaningful concepts. Figure 6.1 provides a visual overview of
 2593 the two stages, from the imitation of the initial policy to the clustering of the
 2594 surrogate representations.

2595 This section is organized into three parts. The first explains why multiple
 2596 surrogate policies are used to study the stability of the representations. The
 2597 second describes how these policies are trained to match the initial policy as
 2598 closely as possible. The third presents how their LRs are clustered to identify
 2599 the CSs.

6.2.1 Multiple surrogate policies

The CAR method was developed to test the *Representation Convergence Hypothesis* (Hypothesis 4). This requires training several NNs independently and comparing their LR. Consistent similarities in the resulting CSs would provide empirical support for this hypothesis.

In supervised learning, this protocol is straightforward since several models can be independently trained on the same target function. RL introduces an additional difficulty. Because of stochastic exploration, independently trained agents are not guaranteed to converge toward the same policy. Even if they achieve comparable performance, they may solve the task using different strategies. This makes it difficult to determine whether differences in the LS result from different representations of the same strategy or from differences in the learned strategies themselves.

To overcome this limitation, we consider an already trained initial policy. This policy is then used as a target for imitation learning [35]. In this process, several surrogate NNs are independently trained in a supervised manner to reproduce the behavior of the same agent. This allows us to investigate whether NNs that learn equivalent policies also develop similar conceptual organizations.

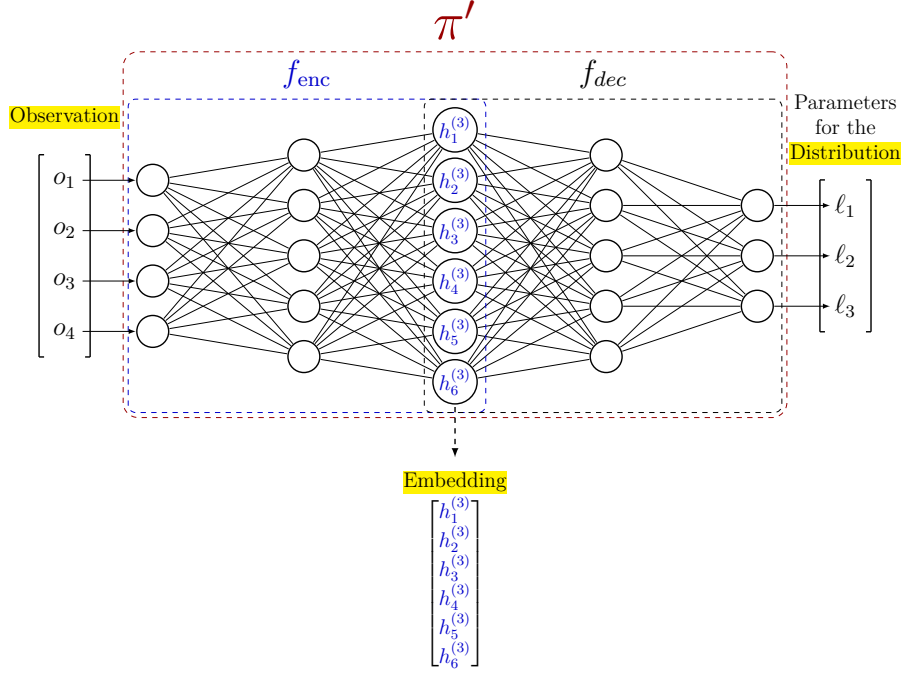
It is important to emphasize that the use of surrogate policies does not make CAR a surrogate explainability method, as defined in Subsection 5.1.2. The surrogate policies remain deep NNs. Their role is not to simplify the decision process but to provide multiple independent realizations of the same policy function, enabling the study of representation convergence.

6.2.2 Training surrogate policies

Throughout the remainder of this thesis, the initial policy we aim to explain is denoted as π . This initial policy is obtained without imposing any constraints on its origin. It may result from classical or deep reinforcement learning [65], alpha-beta algorithm [14], or even from demonstrations provided by humans [32]. The crucial requirement is the availability of a dataset containing a sufficient number of projections from the observation space $\mathcal{O}$ to the action space $\mathcal{A}$ induced by π to train a NN.

The replicated policies are referred to as surrogate policies. To assess the stability of the learned representations, n surrogate policies are trained independently, denoted by $\pi'_1, \dots, \pi'_n$. All surrogate policies share the same architecture. It consists of a NN with two components, an encoder f_{enc} and a decoder f_{dec} , highlighted by the blue and black dashed rectangles, respectively, in Figure 6.2. The encoder maps the observation o to the embedding, and the decoder reconstructs the parameters of the action distribution from this embedding. For a given surrogate policy π'_i , parameterized by θ_i , its encoder and decoder are written $f_{\text{enc},i}$ and $f_{\text{dec},i}$.

To train each surrogate policy π'_i to accurately replicate the action distribution, we define a loss function $\mathcal{L}$ optimized via gradient descent as introduced in Section 2.4. This loss, defined in Equation (6.1), is computed as the Mean Squared Error (MSE) between the parameters of the action distributions, obtained by applying the function ℓ , which maps a distribution to its parameters (the logits). In practice, this function ℓ does not need to be explicitly defined, as neural policies directly output the parameters of the action distribution.

Figure 6.2: Example of a surrogate policy architecture π' .

$$\min_{\theta_i} \sum_{o \in \mathcal{O}} \text{MSE} \left(\ell(\pi(o)), \ell(\pi'_i(o)) \right) \quad (6.1)$$

6.2.3 Cluster extraction

Once the surrogate policies are trained, their encoders are used to generate a point cloud from a set of observations. This point cloud provides a sample of the LS to be analyzed. The proposed CAR method adopts the deep clustering approach introduced in Section 3.5. Under this approach, conceptually similar observations are expected to produce nearby LR. A CS can therefore be viewed as a region of the LS containing related LR. To identify these regions, a clustering algorithm is applied to the point cloud.

The clustering is performed independently for each surrogate policy. For a surrogate policy π'_i , applying its encoder $f_{enc,i}$ to a subset of observations $\mathcal{O}_{\text{data}} \subseteq \mathcal{O}$ yields the set of embeddings $\mathcal{H}_i$. A clustering function f_{clust} then partitions the point cloud $\mathcal{H}_i$ into groups of nearby LR, assigning each observation a cluster label and yielding the partition $\mathcal{C}_i$. The same observation set $\mathcal{O}_{\text{data}}$ is used for every surrogate policy, so that the partitions $\mathcal{C}_1, \dots, \mathcal{C}_n$ are defined over the same observations and can be compared. This comparison is the basis of the evaluation introduced in the following section.

Algorithm 5 states the complete CAR procedure, whose lines map directly onto the stages described above. Lines 1–2 prepare the imitation of the initial policy. They build the training dataset by pairing each observation with the

Algorithm 5: Clustering Agent Representation (CAR)**Input:**Initial policy π Observation dataset $\mathcal{O}_{\text{data}}$ Number of surrogate policies n Surrogate policy architecture π' (encoder f_{enc} , decoder f_{dec})Learning rate η Clustering function f_{clust} **Output:**Trained surrogate policies $\pi'_1, \dots, \pi'_n$ Clusterings $\mathcal{C}_1, \dots, \mathcal{C}_n$ *Behavioral cloning dataset built from the target policy π* **1** $\mathcal{D} \leftarrow \{(o, \ell(\pi(o))) \mid o \in \mathcal{O}_{\text{data}}\};$ *MSE loss between action-distribution parameters, as defined in Equation (6.1)***2** $\mathcal{L} \leftarrow \text{MSE};$ *Independently train n surrogate policies by behavioral cloning***3** **for** $i = 1, \dots, n$ **do****4** Initialize θ_i randomly;*Behavioral cloning of π via SGD, see Algorithm 1***5** $\theta_i \leftarrow \text{SGD}(\mathcal{D}, \pi', \mathcal{L}, \eta, \theta_i);$ *Cluster the latent representations of each surrogate policy***6** **for** $i = 1, \dots, n$ **do***Embeddings produced by the encoder of π'_i* **7** $\mathcal{H}_i \leftarrow \{f_{\text{enc},i}(o) \mid o \in \mathcal{O}_{\text{data}}\};$ *Partition of the embedded observations into clusters***8** $\mathcal{C}_i \leftarrow f_{\text{clust}}(\mathcal{H}_i);$ **9** **Return** $(\pi'_1, \dots, \pi'_n), (\mathcal{C}_1, \dots, \mathcal{C}_n)$

parameters of the action distribution produced by π , and define the MSE loss of Equation (6.1). Lines 3–5 correspond to the first stage, in which each of the n surrogate policies is initialized randomly and trained independently by stochastic gradient descent. Lines 6–8 correspond to the second stage. The encoder of each surrogate policy embeds the shared observation set $\mathcal{O}_{\text{data}}$ (Line 7), and the resulting point cloud is partitioned by f_{clust} (Line 8). Finally, Line 9 returns the surrogate policies together with their partitions, which are the inputs of the evaluation.

Two choices govern this analysis without being part of the clustering algorithm itself. The first is the choice of layer used to extract the representations. Each hidden layer defines its own LS, reflecting the information represented at that stage of the network. CAR can therefore be applied to the LRs extracted from any layer. This allows the conceptual organization of the policy to be examined at different levels of the network.

The second is the number of clusters. CAR does not assume that a task admits a single correct number of concepts. This choice instead determines the

granularity at which the conceptual organization is examined. A small number reveals broad, strategic concepts, while a larger number reveals finer, tactical ones. The influence of both choices is studied in Chapter 7.

Figure 6.3 shows the outcome of this procedure for a surrogate policy trained on **Space Invaders**, using five clusters. The clusters are computed in the original high-dimensional LS. The point cloud is a two-dimensional projection used only for visualization. Each cluster gathers observations that correspond to a distinct, interpretable stage of a game:

- **Cluster 1:** The start of a game, with few or no invaders neutralized.
- **Cluster 2:** The right-hand group of invaders eliminated.
- **Cluster 3:** The pursuit of the bonus saucer.
- **Cluster 4:** The end of a game, with a few fast invaders close to the ground.
- **Cluster 5:** The middle of a game, with the invaders descending but already thinned out.

The position of an observation in the LS thus reflects the stage of the environment it belongs to, suggesting a correspondence between the geometry of the LS and the semantic properties of environment states.

This example illustrates what CAR is expected to produce, but it relies on the visual inspection of a single surrogate policy, which is precisely the type of evidence whose limitations motivated this work. Three aspects must therefore be evaluated before such an analysis can be trusted. First, the surrogate policy must faithfully reproduce the initial policy so that its explanations remain relevant to the target policy. Second, the recovered conceptual organization must be consistent across independently trained surrogate policies. Finally, the clusters must capture genuine abstractions rather than simply reproducing structure already present in the observations or action distributions.

The following section introduces one evaluation metric for each of these aspects and details how it is computed.

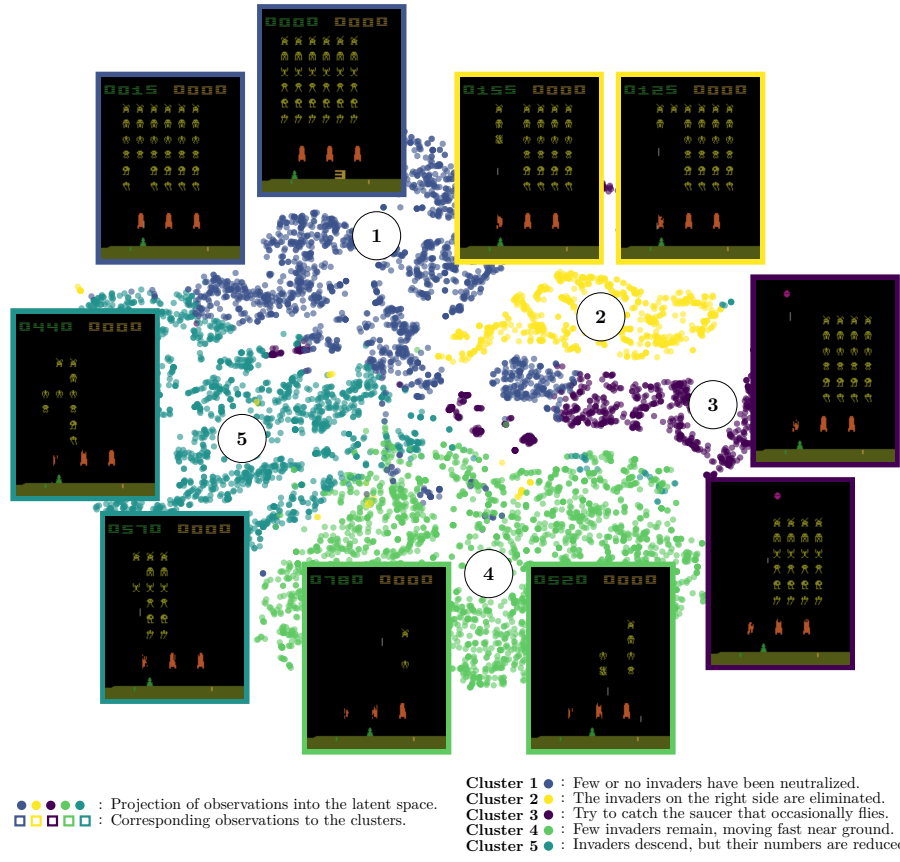


Figure 6.3: Two-dimensional projection of the LS embeddings of a surrogate policy trained on **Space Invaders**, together with the CAR clusters and their corresponding observations.

6.3 Evaluation Metrics

To evaluate the effectiveness of CAR, four evaluation metrics are introduced:

- ***Behavior Reconstruction*** : it measures how well a surrogate policy faithfully reproduces the behavior of the initial policy by comparing their cumulative rewards.
- ***Clustering Universality*** : it evaluates the stability of clusterings by measuring the similarity of clusters obtained from independently trained surrogate policies.
- ***Abstraction Score*** : it quantifies the extent to which LS clusters capture abstractions rather than directly reflecting the structure of observations, of the action distribution, or the network architecture prior to training.
- ***Adjusted Clustering Universality***: it combines the two previous metrics to measure the agreement between surrogate policies beyond what can be explained by the baselines used in *Abstraction Score*.

The remainder of this section details the motivation for each metric and describes how they are calculated. In the first subsection, we develop the *Behavior Reconstruction*. We then introduce the Adjusted Rand Index (ARI) measure, which serves as the foundation for the three other metrics introduced in the subsequent subsections, namely *Clustering Universality*, *Abstraction Score*, and *Adjusted Clustering Universality*.

6.3.1 Behavior Reconstruction

We introduce the *Behavior Reconstruction* metric to assess whether π' faithfully reproduces the behavior of π . This metric compares the returns obtained by the initial policy π and the surrogate policies π' over a set of evaluation episodes. We use the definition of return introduced in Section 4.3.4, where G_1 denotes the discounted sum of rewards collected along a whole trajectory. Here, the return is computed over complete episodes with $\gamma = 1$, so G_1 corresponds to the total reward collected during the episode. Each policy is evaluated over M episodes. We denote by $G_\pi^{(e)}$ the return obtained by the initial policy in episode e , and by $G_{\pi'_j}^{(e)}$ the return obtained by surrogate policy π'_j in episode e . The *Behavior Reconstruction* is then defined as the mean return of the surrogate policies normalized by the mean return of the initial policy:

$$\text{BR} = \frac{1}{n \sum_{e=1}^M G_\pi^{(e)}} \sum_{j=1}^n \sum_{e=1}^M G_{\pi'_j}^{(e)} \quad (6.2)$$

In imitation learning, it is unlikely to expect the surrogate policy π' to outperform the initial policy π it aims to approximate through supervised learning [32]. Consequently this metric ranges in $[0, 1]$. Values approaching 1 indicate minimal loss in reproducing π , while values close to 0 indicate poor reproduction.

We deliberately adopt a reward-based definition rather than alternatives based on actions or action distributions (e.g., training loss, accuracy, KL divergence [41], or optimal transport [88]). Since surrogate policies are trained in a supervised manner, they cannot develop strategies beyond those demonstrated by the initial policy. This means that if the rewards achieved are comparable, the surrogate policy is effectively copying the initial policy’s behavior.

6.3.2 Adjusted Rand Index

Within CAR, concepts are assumed to correspond to clusters in the LS. Combined with the *Representation Convergence Hypothesis* (Hypothesis 4), this suggests that independently trained surrogate policies should develop similar conceptual organizations. Their LRs should therefore produce similar clusterings when analyzed using the same clustering procedure.

When a particular observation is processed by the network, it is expected to activate the same concept regardless of the specific neural implementation of the policy. If independently trained policies converge toward the same conceptual organization, a given observation should activate the same concept across all models. As a consequence, observations grouped together within a cluster for one policy should also be grouped together for another independently trained policy.

In the clustering literature, the most used measures for comparing two partitions is the ARI [17]. This measure quantifies the agreement between cluster assignments independently of the spatial organization of the clusters. Both metrics introduced later in this chapter, *Clustering Universality* and *Abstraction Score*, rely on the ARI. Since the ARI is a complex measure, we review its definition and discuss its main properties.

The ARI is derived from the Rand Index (RI) [13], which measures the agreement between two partitions by considering every possible pair of points in the point cloud. For each pair, four situations are possible:

- **Agreement:** The points belong to the same cluster in both partitions.
- **Agreement:** The points belong to different clusters in both partitions.
- **Disagreement:** The points belong to the same cluster in the first partition but to different clusters in the second.
- **Disagreement:** The points belong to different clusters in the first partition but to the same cluster in the second.

The RI measures the proportion of pairs for which the two partitions agree:

$$\text{RI} = \frac{\text{number of agreeing pairs}}{\text{total number of pairs}}.$$

Although the RI is intuitive, it suffers from an important limitation. In most clustering problems, the majority of pairs of points belong to different clusters. Consequently, even two completely independent partitions will agree on a large number of pairs, simply because both assign them to different clusters. This effect becomes more pronounced as the number of points or clusters increases.

The ARI addresses this issue by accounting for the agreement that would be expected purely by chance. It compares the observed agreement RI to

the expected agreement under random partitions $\mathbb{E}[\text{RI}]$. The score is then normalized by the maximum agreement achievable beyond this random baseline, $\max(\text{RI}) - \mathbb{E}[\text{RI}]$. As a result, an ARI of 0 corresponds to the level of agreement expected from random clusterings, while an ARI of 1 indicates perfect agreement between the two partitions:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}$$

In practice, the ARI is computed directly from the cluster assignments of the two partitions. Let clustering $\mathcal{C}$ be composed of clusters $\{C_i\}$ and clustering $\mathcal{C}'$ of clusters $\{C'_j\}$. Let $a_i = |C_i|$ denote the number of samples in cluster i of clustering $\mathcal{C}$, and $b_j = |C'_j|$ the number of samples in cluster j of clustering $\mathcal{C}'$. Furthermore, let $n_{ij} = |C_i \cap C'_j|$ be the number of samples shared by cluster i of clustering $\mathcal{C}$ and cluster j of clustering $\mathcal{C}'$, and let N be the total number of samples. With these definitions, the ARI can be computed as:

$$\text{ARI} = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{N}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{N}{2}}$$

Because it is bounded by 1, a value such as 0.5 can be misleadingly interpreted as “halfway to a perfect match”. This interpretation is incorrect because the ARI is normalized against the agreement expected from two random partitions. As a result, values close to 1 require the two clusterings to be nearly identical, while a value of 0.5 already indicates substantial agreement [66].

6.3.3 Clustering Universality

Having introduced the ARI, we can now define the *Clustering Universality* metric. While the ARI quantifies the similarity between two partitions, the objective of CAR is to assess the consistency of conceptual organizations across an entire set of independently trained surrogate policies.

To this end, the same set of observations is projected into the LS of each surrogate policy and clustered. Under the *Representation Convergence Hypothesis* (Hypothesis 4), these independently obtained partitions are expected to be similar. *Clustering Universality* quantifies this by computing the mean pairwise ARI:

$$\text{CU} = \frac{1}{\binom{n}{2}} \sum_{i=2}^n \sum_{j=1}^{i-1} \text{ARI}(\mathcal{C}_i, \mathcal{C}_j). \quad (6.3)$$

This metric takes values in $[-1, 1]$. A value close to 1 indicates that surrogate policies produce highly similar partitions, suggesting that the clusters reflect intrinsic properties of the policy. Conversely, a value near 0 implies that the similarity between partitions is no better than that of two randomly generated clusterings, meaning that the clustering fails to capture any meaningful structure in the policy. A negative value, on the other hand, signals that the agreement between partitions is worse than what would be expected by chance, indicating a systematic disagreement between the clusterings.

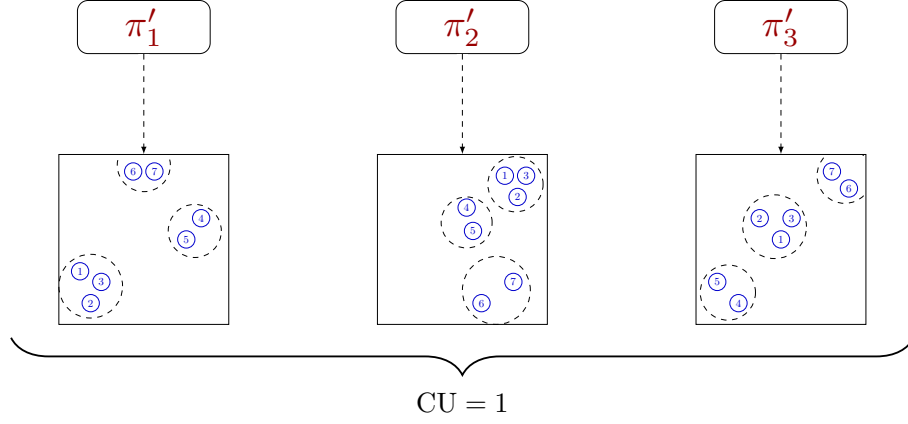


Figure 6.4: Illustration of *Clustering Universality* in the ideal case where projections of the same observations are grouped within the same clusters across all surrogate policies.

Figure 6.4 illustrates *Clustering Universality* in the ideal case. It shows three LSs obtained by projecting the same seven observations, labeled 1 to 7, through three independently trained surrogate policies. Although the embeddings occupy different positions, the clusters group exactly the same elements in all three cases. The partitioning is therefore identical, resulting in an ARI of 1.

6.3.4 Abstraction Score

If *Clustering Universality* is high, it becomes important to determine what underlies this stability. Two possibilities arise:

- (i) the LS provides no abstraction, as defined in Definition 3.1.1, because it either reproduces the observation space or action distribution, or reflects a structure already present in the network architecture before training;
- (ii) the LS captures similarities in decision-making, resulting in behaviorally meaningful clusters.

To distinguish among these cases, we introduce the *Abstraction Score*. This metric contextualizes latent clusterings by examining how their patterns align with three reference baselines: the raw observations, the action distribution of the initial policy, and an untrained network of the same architecture.

We consider $\mathcal{C}_{\text{obs}}$, the clustering of the observations $\mathcal{O}_{\text{data}}$, and $\mathcal{C}_{\text{act}}$, the clustering of the action distribution of the initial policy. Concretely, $\mathcal{C}_{\text{act}}$ is computed on the parameters of this distribution (the logits, before the softmax) produced by the initial policy for the same observations, which are precisely the quantities the surrogate policies are trained to reproduce. We cluster these continuous parameters rather than the sampled discrete action, since a discrete action takes only a few distinct values, which would limit the number of clusters that can be distinguished. Both clusterings use the same algorithm, distance metric, and hyperparameters as those used for $\mathcal{C}_i$ in the LS. This ensures that the three clusterings are obtained under the same conditions.

The third baseline addresses the case where the LS merely reflects an inductive bias of the network architecture rather than something learned. To this end, we introduce U untrained networks sharing the same architecture as the surrogate policies, but with randomly initialized, never-updated weights. Each of these U untrained networks is used to project the same observations into its own latent space, which is then clustered in the same way, yielding partitions $\mathcal{C}_{\text{untr},1}, \dots, \mathcal{C}_{\text{untr},U}$. This baseline isolates what the architecture already imposes on the representation before any learning takes place, so that a high similarity between $\mathcal{C}_i$ and this baseline indicates that the corresponding structure cannot be attributed to training.

For each latent clustering $\mathcal{C}_i$, we define its similarity to these three baselines using the ARI:

$$\begin{aligned}\rho_i^{\text{obs}} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{\text{obs}}) \\ \rho_i^{\text{act}} &= \text{ARI}(\mathcal{C}_i, \mathcal{C}_{\text{act}}) \\ \rho_i^{\text{untr}} &= \frac{1}{U} \sum_{u=1}^U \text{ARI}(\mathcal{C}_i, \mathcal{C}_{\text{untr},u})\end{aligned}$$

The *Abstraction Score* is subsequently calculated as one minus the average of the maximum absolute similarity values across these three reference baselines:

$$\text{AS} = 1 - \frac{1}{n} \sum_{i=1}^n \max(|\rho_i^{\text{obs}}|, |\rho_i^{\text{act}}|, |\rho_i^{\text{untr}}|) \quad (6.4)$$

The *Abstraction Score* ranges from $[0, 1]$. A score around 0 means that the latent clusters are strongly aligned with the observation space, the action distribution, or the untrained network, reflecting no abstraction, case (i). A score close to 1 indicates that the latent clusters are independent of the observation space, the action distribution, and the network’s initialization, capturing abstractions that meaningfully reflect the agent’s behavior, case (ii).

6.3.5 Adjusted Clustering Universality

As discussed above, *Clustering Universality* alone does not indicate whether the recovered structure is meaningful. We therefore introduce the *Adjusted Clustering Universality*, which measures how much of the agreement between the different surrogate policies remains after accounting for the three trivial baselines.

$$\text{ACU} = \text{CU} - \frac{1}{n} \sum_{i=1}^n \max(|\rho_i^{\text{obs}}|, |\rho_i^{\text{act}}|, |\rho_i^{\text{untr}}|) \quad (6.5)$$

This metric ranges in $[-1, 1]$. A value close to 1 indicates that surrogate policies converge toward a partition that is both highly reproducible across seeds and clearly not explained by the observation space, the action distribution, or the network’s initialization, providing the strongest evidence for *Representation Convergence Hypothesis* (Hypothesis 4). A value close to 0, or negative, indicates that whatever agreement exists between independently trained surrogate policies is largely trivial: the surrogate policies converge, but toward a partition that could just as easily emerge from raw pixel similarity, from the action distribution of the initial policy, or from an untrained network sharing the same architecture.

2886 6.4 Conclusion

2887 The CAR method was developed to investigate how concepts are organized
 2888 within the representations of RL policies. It is based on three hypotheses
 2889 introduced throughout this thesis, the *Mechanistic Hypothesis* (Hypothesis 3),
 2890 the *Concept Hypothesis* (Hypothesis 1), and the *Representation Convergence*
 2891 *Hypothesis* (Hypothesis 4).

2892 CAR translates these assumptions into an empirical analysis. Multiple surro-
 2893 gate policies are trained independently to reproduce the same policy. Their LRs
 2894 are then clustered and compared to assess whether similar conceptual structures
 2895 emerge across policies. The *Clustering Universality* and *Abstraction Score* pro-
 2896 vide quantitative measures for evaluating the consistency and abstraction of the
 2897 resulting clusters.

2898 This makes CAR a general procedure for analyzing agent representations.
 2899 The analysis and evaluation metrics are fully automated and require no manual
 2900 intervention or environment-specific adjustments. The method can therefore be
 2901 applied to a wide range of RL settings and provides a basis for systematically
 2902 studying the internal representations of neural policies.

2903 The next chapter applies CAR to several environments to evaluate its ability
 2904 to identify meaningful conceptual structures.

Chapter 7

Experiments

This chapter presents the experimental study of the Clustering Agent Representation (CAR) method introduced in the previous chapter. It focuses on evaluating the method’s ability to identify conceptual structures. The remainder of this chapter is organized as follows:

- Section 7.1 describes the software implementation of the CAR pipeline.
- Section 7.2 details the experimental configuration, covering the environments, the Reinforcement Learning (RL) setup, and the surrogate policy training.
- Section 7.3 presents the training of the initial and surrogate policies and evaluates how faithfully the surrogate policies reproduce the behavior of the initial policy.
- Section 7.4 provides a quantitative analysis of the Latent Representations (LRs) using the *Clustering Universality* (CU), *Abstraction Score* (AS), and *Adjusted Clustering Universality* (ACU) metrics.
- Section 7.5 provides a qualitative analysis of the resulting clusters by manually examining their semantic content.
- Section 7.6 summarizes the findings and discusses their implications.

7.1 Software Implementation

This section details the experimental evaluation of the proposed method. Deep Learning (DL) and RL are strongly empirical fields, where experimental evaluation plays an important role in assessing how methods perform in practice. Particular attention is therefore given to the implementation of CAR. It relies on established libraries that provide efficient implementations of its main components. The code is made publicly available as open-source to support reproducible replication of the reported results.¹

The CAR implementation is organized as a pipeline of five stages. Each stage performs a specific operation and produces the artifacts required by the

¹https://github.com/Maxalaar/Clustering_Agent_Latent_Space

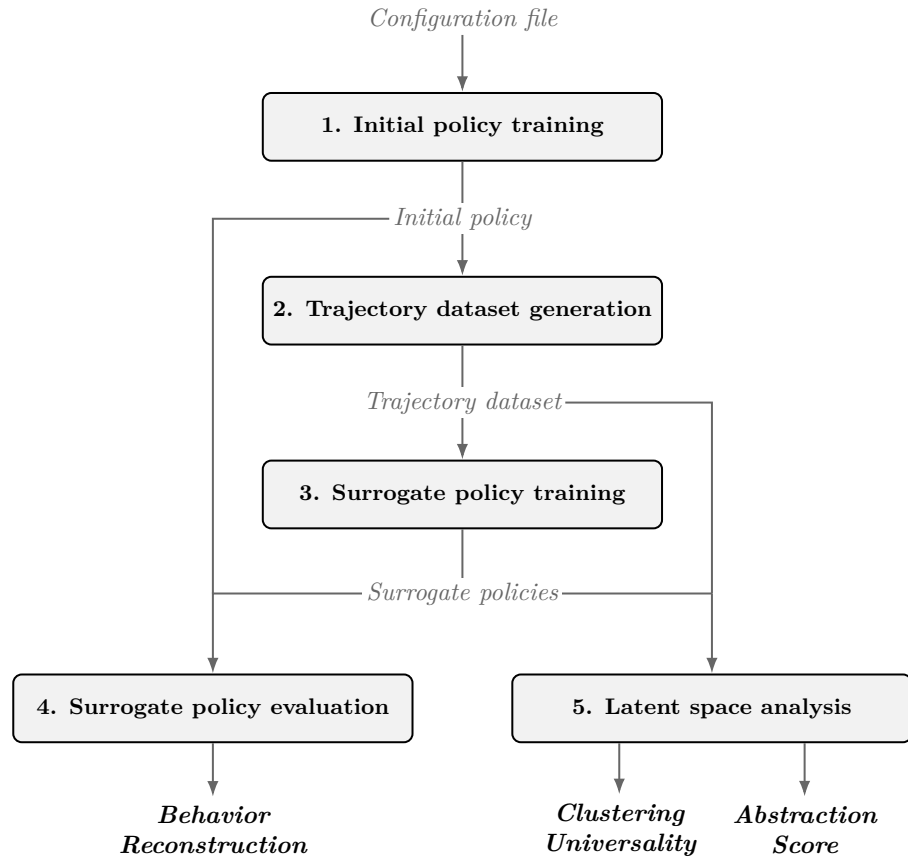


Figure 7.1: The CAR pipeline.

subsequent stages. The stages are implemented as independent executables allowing each stage to be rerun independently. The whole pipeline is driven by a single configuration file, which is the input of the process and gathers the parameters of every stage. The CAR pipeline is illustrated in Figure 7.1, and Table 7.1 summarizes the libraries used throughout the pipeline. The remainder of this section describes the purpose and implementation of each stage.

7.1.1 Initial policy training

The first stage trains the initial policy using RLlib [61, 63]. RLlib provides scalable implementations of RL algorithms and supports distributed training. The use of RLlib also allows CAR to be applied to environments implemented using the Gymnasium interface [124]. By default, the initial policy is trained using Proximal Policy Optimization (PPO) [54]. Other RLlib algorithms, such as Asynchronous Proximal Policy Optimization (APPO) [57], Deep Q-Network (DQN) [40], and DreamerV3 [121], can also be selected through the same configuration.

The policy and critic networks are implemented using PyTorch [76]. PyTorch is the standard DL framework and is widely used for developing and training Neural Networks (NNs). It provides GPU acceleration and supports a wide range of NN architectures. This makes it well suited for implementing the different

Pipeline Stage	Library	Description
Initial policy training	RLlib [61, 63]	Distributed RL library used to train the initial policy.
	PyTorch [76]	Deep learning framework implementing the policy and critic networks.
	Gymnasium [124]	Common interface used to plug in any RL environment.
Trajectory dataset generation	RLlib [61, 63]	Distributed environment runners used to collect trajectories in parallel.
	Apache Arrow [45]	Columnar file format (Parquet) used to store the trajectory datasets.
Surrogate policy training	PyTorch Lightning [68]	Structured training loop used for the behavior cloning of the surrogate policies.
Surrogate policy evaluation	RLlib [61, 63]	Wraps each surrogate policy as an RL policy to evaluate it in the environment.
Latent space analysis	cuML [111]	GPU-accelerated clustering algorithms used to analyze the latent representations.

Table 7.1: Main software libraries used at each stage of the CAR pipeline.

2953 policy and critic architectures used in the experiments.

2954 7.1.2 Trajectory dataset generation

2955 The second stage builds the dataset used to train the surrogate policies. RLlib
 2956 is used to collect trajectories in parallel using multiple environment runners. For
 2957 each step, the dataset stores the observation together with the parameters of
 2958 the action distribution produced by the initial policy.

2959 Trajectories are stored as Parquet files [45]. Its columnar structure allows
 2960 efficient data access and supports concurrent processing. This is useful here
 2961 because the datasets contain around one million observation–action pairs and
 2962 cannot be loaded entirely into memory.

2963 7.1.3 Surrogate policy training

2964 The third stage trains the surrogate policies using Behavioral Cloning (BC)
 2965 on the trajectory dataset. This provides a supervised learning approach for
 2966 replicating the initial policy. The training procedure is implemented using
 2967 PyTorch Lightning [68]. PyTorch Lightning provides a simplified interface for
 2968 training NNs with PyTorch. It handles common training operations such as data
 2969 loading, optimization, checkpointing, and GPU execution.

2970 7.1.4 Surrogate policy evaluation

2971 The fourth stage evaluates how faithfully the surrogate policies reproduce the
 2972 behavior of the initial policy. This stage therefore requires both the surrogate
 2973 policies and the initial policy. Each surrogate is wrapped as an RLlib policy and
 2974 evaluated in the original environment. The initial policy is evaluated under the
 2975 same conditions to provide the reference return. The mean cumulative reward of
 2976 each surrogate over multiple episodes is then compared with that of the initial
 2977 policy. Evaluation episodes are collected using parallel environment runners.

2978 This stage computes the *Behavior Reconstruction*. It therefore verifies that
 2979 the surrogate policies reproduce the behavior of the initial policy sufficiently
 2980 well before their LRs are analyzed. If a surrogate does not meet the required
 2981 level of behavior reconstruction, its training can be repeated.

2982 7.1.5 Latent space analysis

2983 The fifth stage performs the main analysis of CAR. The trajectory dataset and
 2984 trained surrogate policies are used to generate the LRs. These representations
 2985 form the point clouds that are analyzed using clustering algorithms.

2986 The clustering step relies on the cuML library [111], which provides GPU-
 2987 accelerated clustering algorithms. This reduces the computational cost and allows
 2988 point clouds containing tens of thousands of representations to be analyzed.

2989 The resulting partitions are used to compute the *Clustering Universality*
 2990 and the *Abstraction Score*. The stage also produces the material required for
 2991 the qualitative analysis, including the clustered point clouds and the cluster
 2992 assignment of each observation.

2993 7.2 Experimental Configuration

2994 This section describes the configuration of the experiments reported in this
 2995 chapter. It covers the environments, the neural architectures used, and the main
 2996 parameters of the pipeline.

2997 7.2.1 Environments

2998 All experiments presented in this thesis use environments from the Atari bench-
 2999 mark [37], accessed through the Arcade Learning Environment. The use of Atari
 3000 environments offers several advantages:

- 3001 • **Standard benchmark:** Atari is widely used in the RL community, making
 3002 the experimental results easier to compare with existing work.
- 3003 • **Limited environment-specific bias:** The environments were not specif-
 3004 ically designed for RL, reducing the risk of introducing biases related to
 3005 their design.
- 3006 • **Diversity:** The benchmark provides a diverse set of well-known environ-
 3007 ments with different types of tasks and levels of complexity, allowing CAR
 3008 to be evaluated under different conditions.

3009 The configuration of the Atari environments follows common practice in RL
 3010 research [62]. In each environment, the agent receives visual observations in the

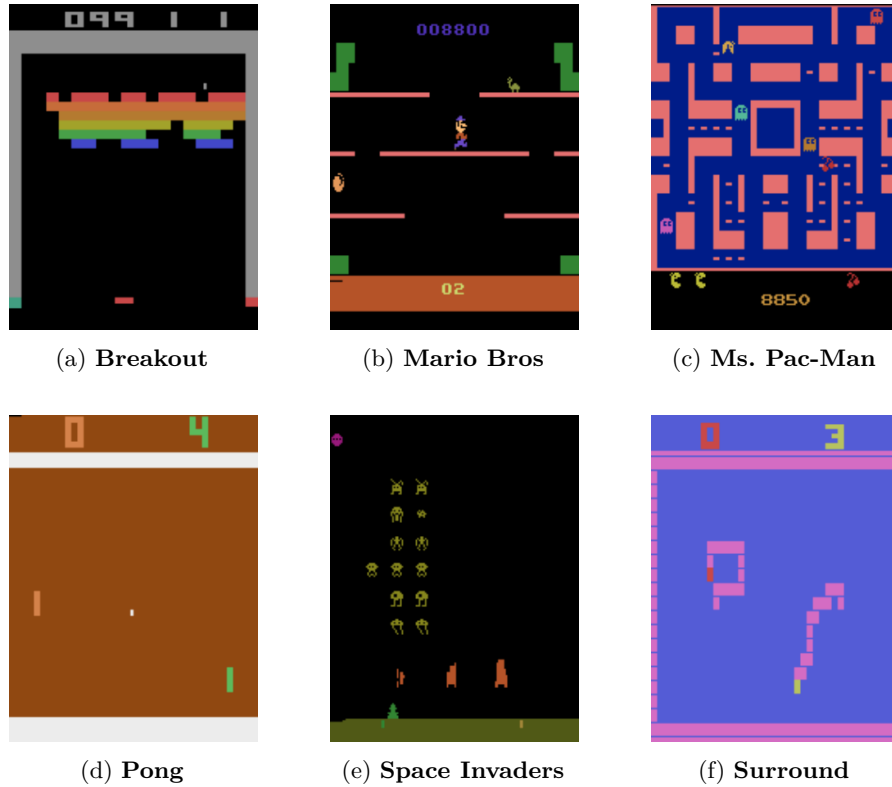


Figure 7.2: The six Atari environments used to evaluate CAR.

form of raw pixels. Each image is resized to 64×64 pixels and converted to grayscale to reduce memory requirements. To provide information about the dynamics of the environment, four consecutive observations are stacked along the channel dimension. The resulting observation has a shape of $64 \times 64 \times 4$. This allows the agent to infer the motion of objects from successive observations.

The actions correspond to the different buttons available on the Atari controller. The resulting action space is therefore discrete. Since Atari games are fully deterministic, sticky actions are introduced to add stochasticity to the trajectories. With a probability of 0.05, the environment repeats the action selected at the previous step instead of applying the new action. This prevents the agent from simply memorizing a fixed sequence of actions during training.

To reduce the duration of each episode, frame skipping is used. The agent selects a new action every five frames and repeats it for the following frames. This reduces the number of decisions required during an episode while maintaining sufficient temporal resolution for effective control.

The experiments use the six environments described below:

- **Breakout:** The agent moves a paddle at the bottom of the screen (Figure 7.2a) to bounce a ball into a wall of bricks. Each broken brick scores one point, and the ball speeds up as the wall is cleared.
- **Mario Bros:** A platform game (Figure 7.2b) in which the player clears each phase by hitting enemies from below to flip them, then kicking them

- 3032 off the platforms before they recover. A central POW block can flip all
 3033 enemies at once but can only be used a few times per phase.
- 3034 • **Ms. Pac-Man:** The player moves through a maze (Figure 7.2c), eating
 3035 pellets while avoiding four ghosts. Power pellets temporarily make the
 3036 ghosts edible.
 - 3037 • **Pong:** A two-player game (Figure 7.2d) in which the agent controls one
 3038 paddle and must send the ball past the opponent.
 - 3039 • **Space Invaders:** The agent moves a cannon along the bottom of the
 3040 screen (Figure 7.2e), firing at descending rows of aliens while avoiding their
 3041 shots and using destructible shields. A bonus saucer occasionally crosses
 3042 the top of the screen.
 - 3043 • **Surround:** A two-player game on a grid (Figure 7.2f) in which each player
 3044 leaves a trail behind a moving cursor. The goal is to force the opponent to
 3045 crash into a trail.

3046 7.2.2 Neural architecture

3047 The architecture of the surrogate policies was deliberately kept simple. The
 3048 objective of this chapter is not to search for the best-performing architecture,
 3049 but to obtain surrogate policies that faithfully reproduce the initial policy while
 3050 remaining easy to analyze. A standard convolutional architecture was therefore
 3051 selected because it is sufficient to reach a high *Behavior Reconstruction* on the
 3052 six environments, as shown in Section 7.3.

3053 The surrogate policies use a Convolutional Neural Network (CNN) to process
 3054 the visual observations. The convolutional part consists of four layers with 16, 32,
 3055 64, and 128 channels, using 4×4 kernels with a stride of 2. Each convolutional
 3056 layer is followed by a LeakyReLU activation. The resulting representation is
 3057 then flattened and passed through a Multilayer Perceptron (MLP) that produces
 3058 the parameters of the action distribution.

3059 The architecture of the MLP varies across environments. For **Pong** and
 3060 **Space Invaders**, it contains a single hidden layer with 128 units. For the other
 3061 four environments, it contains three hidden layers with 512, 256, and 128 units.
 3062 The size of the output layer matches the number of actions available in each
 3063 environment: 4 for **Breakout**, 18 for **Mario Bros**, 9 for **Ms. Pac-Man**, 6 for
 3064 **Pong**, 6 for **Space Invaders**, and 5 for **Surround**.

3065 Figure 7.3 illustrates both configurations, together with the shape of the
 3066 representation produced by each convolutional layer. Throughout the remainder
 3067 of this chapter, the analyzed representations are numbered from 1 to 5. These
 3068 representations correspond to the four convolutional layers and the flattened
 3069 representation, which are common to all six environments.

3070 The representations produced by the MLP are numbered separately. For
 3071 **Pong** and **Space Invaders**, which use a single hidden layer, the representation
 3072 is numbered 6a. For the other four environments, which use three hidden layers,
 3073 the representations are numbered 6b, 7b, and 8b.

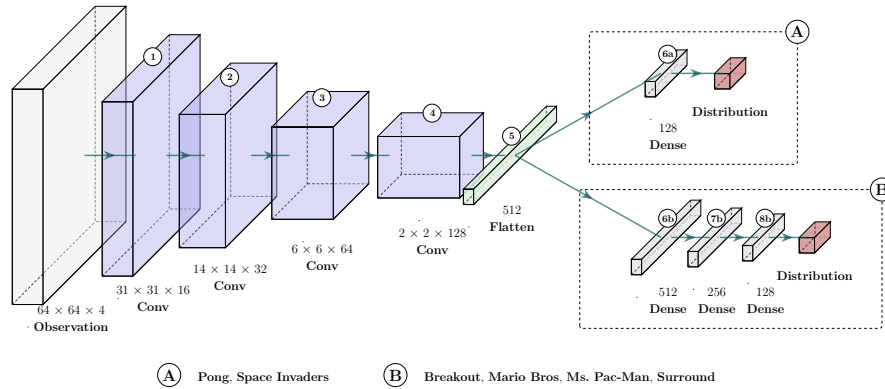


Figure 7.3: Architecture of the surrogate policies.

7.2.3 Other CAR parameters

Several other parameters must be defined for the different stages of the CAR pipeline. These parameters are less critical to the overall methodology than those discussed above, but they still need to be specified to reproduce the experiments. Since each stage introduces its own configuration choices, there are too many parameters to discuss them all in detail. The main parameters used across the experiments are summarized in Table 7.2. All these parameters are provided to the pipeline through the modular configuration file introduced in Section 7.1. Each stage reads its own section of this file, so that a parameter can be changed, or a component replaced, without modifying the code of the pipeline.

These settings are specific to the experiments presented in this chapter and do not restrict the CAR methodology itself. The different stages of the pipeline are modular and can be configured or replaced depending on the application. The evaluation metrics provide an objective basis for comparing these alternatives and assessing their effect on the consistency and meaningfulness of the resulting clusters.

7.3 Policy Performance and Surrogate Fidelity

This section reports the results of training the initial and surrogate policies for CAR, applied to the six environments. The first part verifies that the initial policies reach a competent level of play and evaluates how faithfully the surrogate policies reproduce their behavior using the *Behavior Reconstruction*.

7.3.1 Initial policy training

Figure 7.4a shows how the initial policies improve during PPO training. For each environment, the mean episode return is normalized so that 0% and 100% correspond to the lowest and highest mean return observed during that run. In every environment the return increases and then reaches a plateau, at which point training is stopped. The number of iterations required to reach this plateau varies from a few hundred, for **Pong**, to several thousand, for **Surround** and **Ms. Pac-Man**. For the subsequent analysis, the checkpoint with the best performance is retained rather than the final checkpoint.

Environment configuration	
Frame size	64×64
Frames stacked per observation	4
Frame skip	5
Sticky-action probability	0.05
Initial policy training	
Algorithm	PPO
Optimizer	Adam
Learning rate	1.5×10^{-4}
GAE λ	0.95
Policy clipping ε	0.1
Value-function clipping	10
Entropy coefficient	0.01
KL coefficient	0.5
Epochs per update	10
Gradient clipping (global norm)	100
Surrogate policy training	
Surrogate policies per environment	3
Observation–action pairs per dataset	$\geq 10^6$
Batch size	2048
Optimizer	Adam
Learning rate	1×10^{-4}
Latent space analysis	
Embedding per surrogate policy	20 000
Clustering algorithm	k -means
Distance	Euclidean distance
Number of clusters K	2 to 20

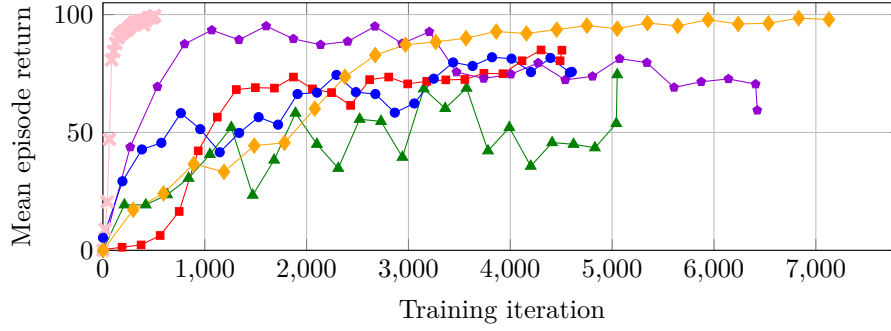
Table 7.2: Main experimental parameters.

3104 To confirm that the trained policies play competently, their scores are com-
3105 pared with the reference scores reported by [62]. Each initial policy reaches a
3106 score of the same order as these references, so it constitutes a reasonable target
3107 for imitation.

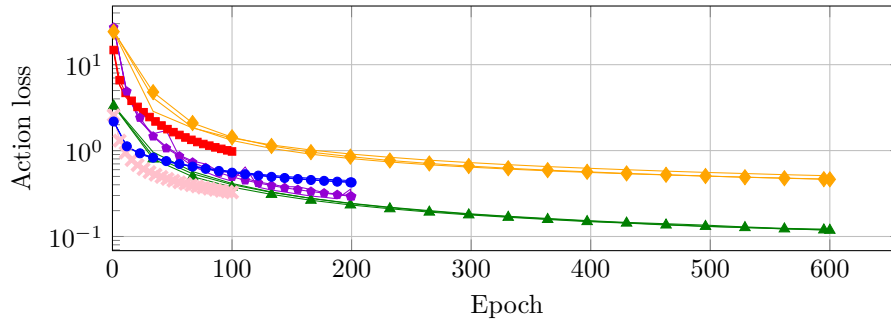
3108 7.3.2 Surrogate policy training

3109 Figure 7.4b shows, for each environment, the training action loss of the three
3110 surrogate policies. In all cases, the loss decreases smoothly, and the three
3111 independently trained policies converge to the same value.

3112 We further evaluate the ability of the surrogate policies to replicate the
3113 behavior of the initial policies using the *Behavior Reconstruction* (Table 7.3).
3114 The surrogate policies reach a *Behavior Reconstruction* of 0.96 on average over
3115 100 evaluation episodes, ranging from 0.83 for **Breakout** to 1.00 for **Ms. Pac-**
3116 **Man** and **Surround**. They therefore reproduce the decisions of the initial
3117 policies closely enough for their LR to be analyzed.



(a) Training of the initial policies with PPO.



(b) Training action loss of the surrogate policies.



Figure 7.4: Training of the initial and surrogate policies.

Environment	<i>Behavior Reconstruction</i>
Breakout	0.83 ± 0.03
Mario Bros	0.99 ± 0.03
Ms. Pac-Man	1.00 ± 0.03
Pong	0.95 ± 0.01
Space Invaders	0.97 ± 0.07
Surround	1.00 ± 0.01

Table 7.3: Evaluation of *Behavior Reconstruction* across the six Atari environments.

7.4 Clustering Quality of the Latent Space

3118

3119 This section analyzes the Latent Space (LS) of the surrogate policies using CU,
 3120 AS, and ACU. To do so, a random subset of 20 000 observations is selected for
 3121 each environment. This subset is then embedded into the LS of each surrogate
 3122 policy at each analyzed layer. This produces a point cloud for each combination
 3123 of environment, surrogate policy, and layer. Each point cloud is partitioned
 3124 using k -means clustering [11] for each number of clusters from 2 to 20.

The three metrics are then computed from these point clouds and their corresponding clusterings. Their interpretation, detailed in Section 6.3, is briefly recalled here:

- **CU** measures the agreement between the partitions obtained from independently trained surrogate policies. A value close to 1 indicates that the surrogate policies group the observations in the same way, while a value close to 0 indicates an agreement no better than chance. Since it relies on the Adjusted Rand Index (ARI), a value of 0.5 already corresponds to a substantial agreement.
- **AS** measures how far the latent partitions are from three trivial baselines: the clustering of the raw observations, of the action distribution of the initial policy, and of untrained networks. A value close to 1 indicates that the clusters capture an abstraction learned by the policy, while a value close to 0 indicates that they merely reproduce one of these baselines.
- **ACU** subtracts the agreement explained by these baselines from CU. A value close to 1 indicates that the surrogate policies converge toward a partition that is both reproducible and non-trivial, providing the strongest support for the *Representation Convergence Hypothesis* (Hypothesis 4). A value close to 0, or negative, indicates that the observed agreement is largely explained by the baselines.

A high CU alone is therefore not sufficient, since it can result from a trivial structure shared by all networks. It is the combination of a high CU and a high AS, summarized by ACU, that supports the existence of a stable conceptual organization.

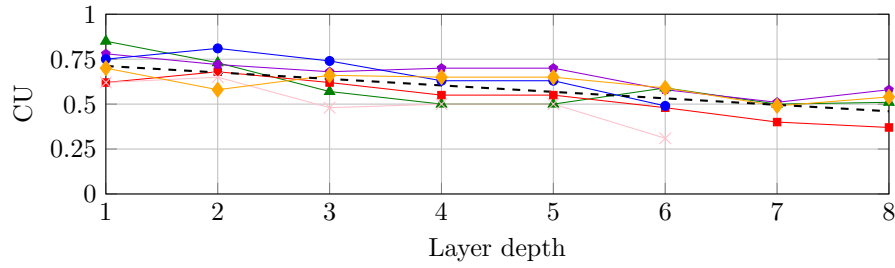
The full results across all environments and analyzed layers are reported in Appendix 1. Two complementary analyses are then conducted on these results. First, the evolution of the three metrics across the different layers is examined to study the effect of representation depth. Second, for each environment, the layer achieving the highest average ACU is selected and analyzed in more detail across the different numbers of clusters.

7.4.1 Analysis across layers

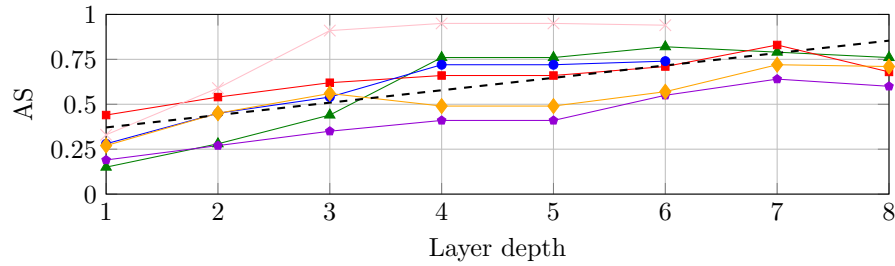
The first analysis examines how the three metrics evolve with the depth of the analyzed layer. For each layer, CU, AS, and ACU are averaged over the tested cluster numbers. Figure 7.5 reports these averages as a function of layer depth.

Across environments, CU tends to be highest in the earliest layers and decreases with depth. AS follows the opposite trend and generally increases as the representation becomes less tied to the raw observation, the action distribution, and the network’s initialization. This increase is however not monotonic in the deepest layers. For **Mario Bros** and **Ms. Pac-Man**, AS decreases at the last layer, from 0.83 to 0.68 and from 0.64 to 0.60 respectively, as the latent clustering becomes closer to the clustering of the action distribution, this layer being directly followed by the output layer. As ACU combines these two effects, its evolution depends on the environment.

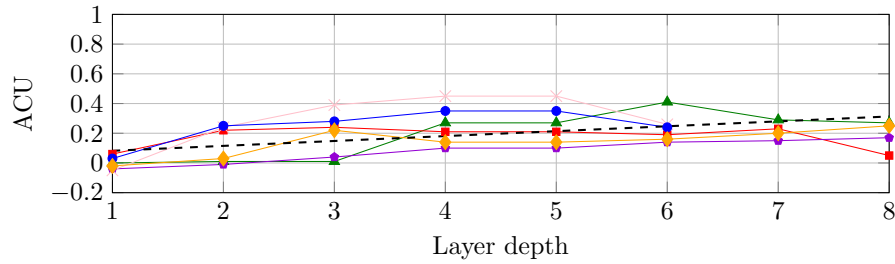
The corresponding linear fits show that CU decreases from 0.72 at the first layer to 0.50 at the last, with a slope of approximately -0.04 per layer, while AS



(a) CU with respect to the layer depth.



(b) AS with respect to the layer depth.



(c) ACU with respect to the layer depth.



Figure 7.5: CU, AS, and ACU averaged over cluster numbers as a function of layer depth.

increases from 0.28 to 0.69, with a slope of approximately $+0.06$ per layer. Over the same range, ACU increases more moderately from 0.00 to 0.19, with a slope of approximately $+0.03$ per layer. This suggests that, despite the environment-specific variations, deeper representations tend on average to provide a better trade-off between CU and AS.

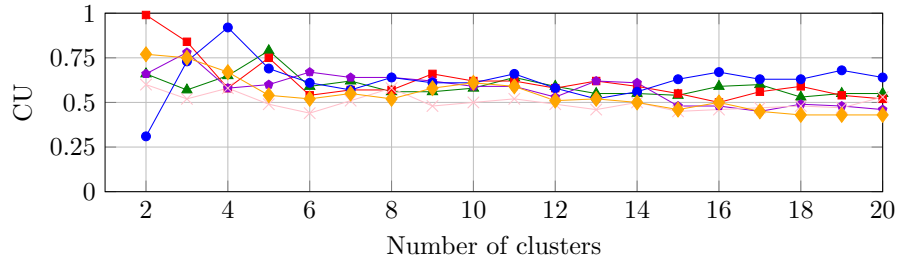
7.4.2 Analysis of the selected layers

The second analysis examines the effect of the number of clusters on the three metrics. For each environment, the layer with the highest average ACU across the tested cluster numbers is selected. Table 7.4 reports the selected layer, together with the corresponding average CU, AS, and ACU.

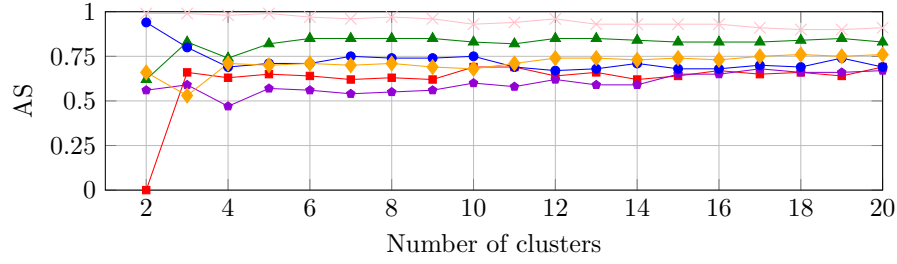
Figure 7.6 reports the evolution of CU, AS, and ACU as the number of clusters

varies. **Pong** achieves the highest average ACU among the six environments, with a value of 0.45 ± 0.06 . This results from a moderate CU combined with an AS that saturates from the third layer onward. In contrast, **Ms. Pac-Man** achieves the lowest AS, with 0.60 ± 0.05 . The resulting ACU, 0.17 ± 0.06 , is the lowest across the six environments.

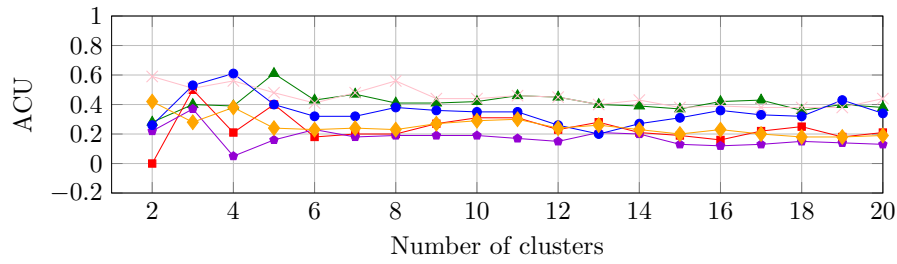
The quantitative analysis provides a measure of the consistency and abstraction of the identified clusters, but does not directly reveal their semantic content. The following section therefore provides a qualitative analysis of the clusters to examine the concepts they represent.



(a) CU with respect to the number of clusters.



(b) AS with respect to the number of clusters.



(c) ACU with respect to the number of clusters.



Figure 7.6: CU, AS, and ACU with respect to the number of clusters, for each of the six environments, for the layer that maximizes the average ACU in that environment.

Environment	Layer	CU	AS	ACU
Breakout	6b	0.59 $\pm$ 0.06	0.82 $\pm$ 0.05	0.41 $\pm$ 0.06
Mario Bros	3	0.62 $\pm$ 0.12	0.62 $\pm$ 0.15	0.24 $\pm$ 0.10
Ms. Pac-Man	8b	0.58 $\pm$ 0.08	0.60 $\pm$ 0.05	0.17 $\pm$ 0.06
Pong	4	0.50 $\pm$ 0.04	0.95 $\pm$ 0.03	0.45 $\pm$ 0.06
Space Invaders	4	0.63 $\pm$ 0.11	0.73 $\pm$ 0.06	0.35 $\pm$ 0.09
Surround	8b	0.54 $\pm$ 0.10	0.71 $\pm$ 0.05	0.25 $\pm$ 0.06

Table 7.4: Layer that maximizes the average ACU across $K \in [2, 20]$ for each environment, together with the corresponding average CU, AS, and ACU over K .

7.5 Semantic Interpretation of the Clusters

The metrics reported in Section 7.4 evaluate the LS clusterings quantitatively but do not indicate what the resulting clusters actually represent. This section complements them with a qualitative, hand-labeled inspection of the clusters obtained for each environment.

For each environment, two clusterings are selected for the qualitative analysis. They are obtained using two different numbers of clusters and may correspond to different layers of the network. The choice of the layer and the number of clusters is based on the following criteria:

- **High ACU:** The selected clustering should achieve a sufficiently high ACU, indicating a good balance between consistency and abstraction.
- **Manual analysis:** The number of clusters must remain small enough to allow the resulting clusters to be examined manually.
- **Limited use of two-cluster partitions:** Clusterings with two clusters often achieve a high ACU, but provide limited information about the organization of the LS. Therefore, at most one of the two selected clusterings uses two clusters, while the other uses a larger number of clusters to provide a more detailed view of the LS.

7.5.1 Breakout

Two clusterings are examined for **Breakout** (Table 7.5), obtained at layer 4 with $K = 2$ and at layer 6b with $K = 5$.

At layer 4 (Figure 7.7), the partition captures the position of the ball relative to the brick wall. The ball remains below the wall in cluster A. The ball has passed behind the wall in cluster B, allowing it to continue clearing bricks without further interaction with the paddle.

At layer 6b (Figure 7.8), the partition again isolates situations in which the ball has passed behind the wall in cluster E. It also identifies the digging of a tunnel through the left side of the wall in cluster C. Finally, it distinguishes different paddle positions at the beginning of the episode in clusters D, F, and G.

The clustering at layer 4 with $K = 2$ is almost perfectly reproducible across surrogate policies, with a CU of 0.99 and an ACU of 0.87. The clustering at layer 6b with $K = 5$ is less consistent, with a CU of 0.79 and an ACU of 0.61, but reveals additional information about the organization of the LS.

Layer 4	$K = 2$	CU 0.99	AS 0.88	ACU 0.87
A	The ball is below the brick wall.			
B	The ball passes behind the brick wall.			
Layer 6b	$K = 5$	CU 0.79	AS 0.82	ACU 0.61
C	Digging a tunnel through the left side of the wall.			
D	Start of the episode, paddle on the right.			
E	The ball is behind the brick wall.			
F	Start of the episode, paddle in the middle.			
G	Start of the episode, paddle on the left.			

Table 7.5: CU, AS, and ACU of the two clusterings analyzed in the **Breakout** environment.

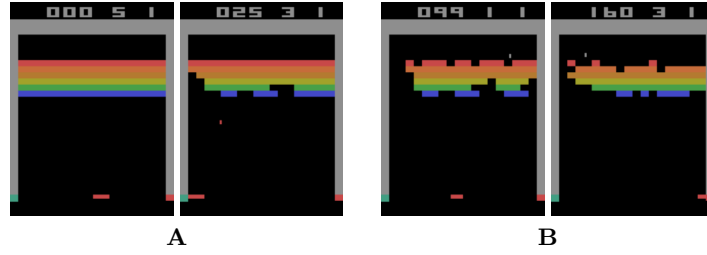


Figure 7.7: Clusters A–B identified by CAR in the **Breakout** environment (layer 4, $K = 2$).

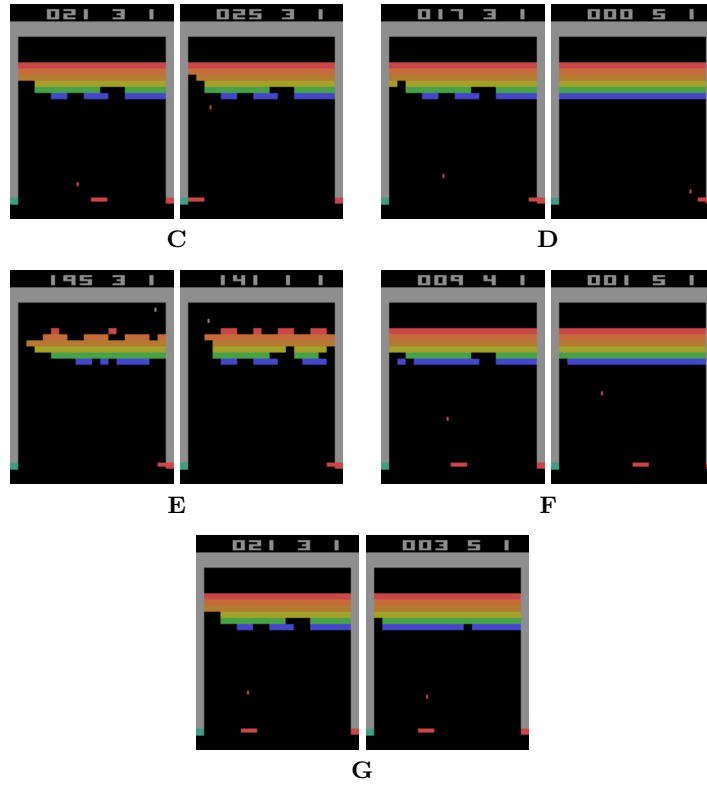


Figure 7.8: Clusters C–G identified by CAR in the **Breakout** environment (layer 6b, $K = 5$).

7.5.2 Mario Bros

Two clusterings are examined for **Mario Bros** (Table 7.6), both obtained at layer 3, with $K = 3$ and $K = 5$.

With $K = 3$ (Figure 7.9), the partition is organized around the location of the action. Mario fights enemies on the left side of the screen in cluster A. He fights enemies on the right side of the screen in cluster C. The POW block is present in cluster B, isolating a fixed and recognizable element of the level.

With $K = 5$ (Figure 7.10), the partition provides a more detailed description of the situations occurring in the level. Mario is on the top-left platform in cluster D. Mario attacks enemies at the top left in cluster F. Mario hits enemies from below in cluster H. Mario simply moves through the level in cluster G. The POW block remains isolated in cluster E.

Increasing K from 3 to 5 lowers CU from 0.84 to 0.75. The AS remains nearly unchanged, with values of 0.66 and 0.65. This results in a decrease in ACU from 0.50 to 0.40.

Layer 3	$K = 3$	CU 0.84	AS 0.66	ACU 0.50
A	Mario fights enemies on the left side of the screen.			
B	The POW block is present.			
C	Mario fights enemies on the right side of the screen.			
Layer 3	$K = 5$	CU 0.75	AS 0.65	ACU 0.40
D	Mario is on the top-left platform.			
E	The POW block is present.			
F	Mario attacks the enemies at the top right.			
G	Mario moves through the level.			
H	Mario hits enemies from below on the left side.			

Table 7.6: CU, AS, and ACU of the two clusterings analyzed in the **Mario Bros** environment.

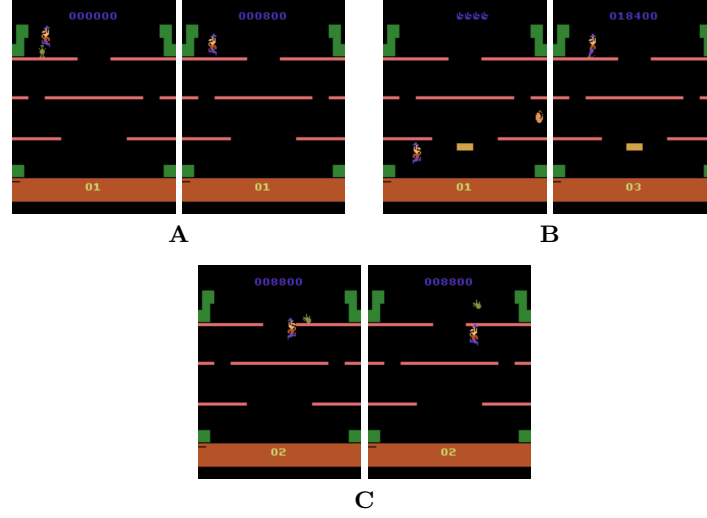


Figure 7.9: Clusters A–C identified by CAR in the **Mario Bros** environment (layer 3, $K = 3$).

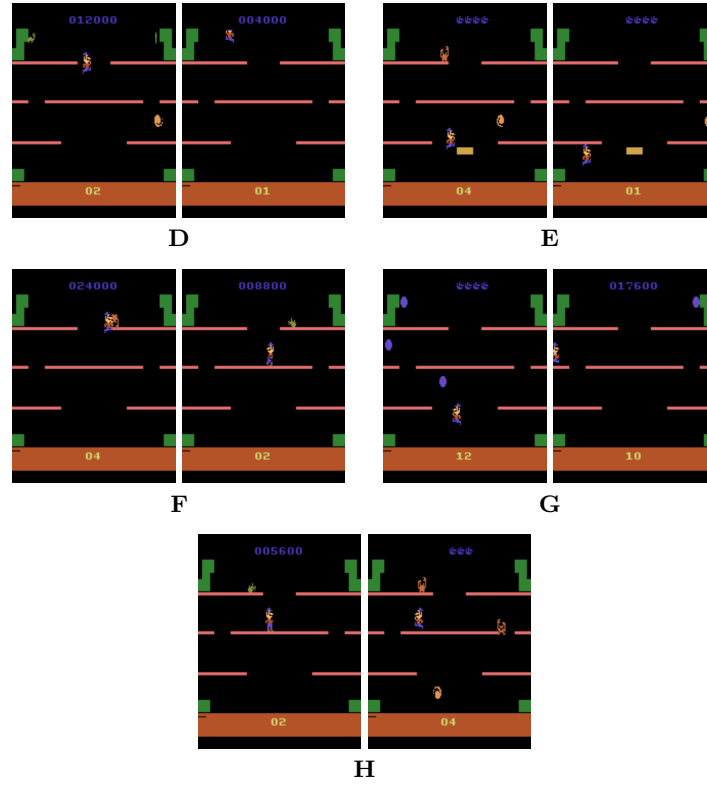


Figure 7.10: Clusters D–H identified by CAR in the **Mario Bros** environment (layer 3, $K = 5$).

7.5.3 Ms. Pac-Man

Two clusterings are examined for **Ms. Pac-Man** (Table 7.7), obtained at layer 6b with $K = 3$ and at layer 4 with $K = 5$.

With $K = 3$ (Figure 7.11), the partition captures the progress and state of the game rather than a specific spatial feature of the observation. The start of the episode is captured in cluster B. The middle of the episode is captured in cluster A. The end of the episode is captured in cluster C.

With $K = 5$ (Figure 7.12), the partition provides a more detailed description of the episode. The initial animation, during which the character cannot move, is isolated in cluster H. The beginning of actual play is captured in cluster F. A situation specific to the game mechanics, in which Pac-Man is in booster mode and waits in front of the ghost spawn to eat the ghosts, is isolated in cluster D. The end of the episode, when at most one power pellet remains, is separated into two similar clusters, E and G.

The two clusterings have the same CU of 0.88. The clustering at layer 4 has a lower AS of 0.47 compared with 0.52 at layer 6b. This results in a lower ACU of 0.35 compared with 0.40.

Layer 6b	$K = 3$	CU 0.88	AS 0.52	ACU 0.40
A	Middle of the episode.			
B	Start of the episode.			
C	End of the episode.			
Layer 4	$K = 5$	CU 0.88	AS 0.47	ACU 0.35
D	Pac-Man is in booster mode and waits in front of the ghost spawn to eat the ghosts.			
E	End of the episode, at most one power pellet remains (similar to cluster G).			
F	Start of the episode.			
G	End of the episode, at most one power pellet remains (similar to cluster E).			
H	Start-of-episode animation, the character cannot move.			

Table 7.7: CU, AS, and ACU of the two clusterings analyzed in the **Ms. Pac-Man** environment.

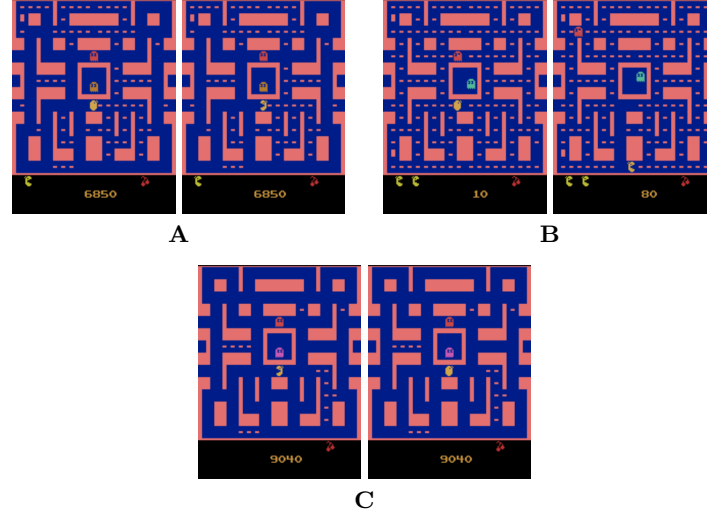


Figure 7.11: Clusters A–C identified by CAR in the **Ms. Pac-Man** environment (layer 6b, $K = 3$).

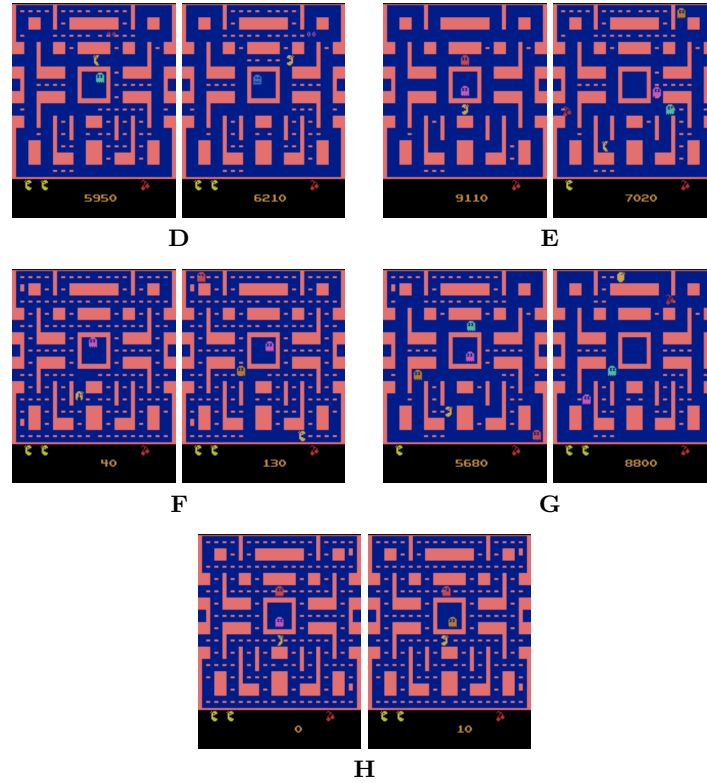


Figure 7.12: Clusters D–H identified by CAR in the **Ms. Pac-Man** environment (layer 4, $K = 5$).

7.5.4 Pong

Two clusterings are examined for **Pong** (Table 7.8), obtained at layer 2 with $K = 2$ and at layer 3 with $K = 3$.

At layer 2 (Figure 7.13), the partition separates the observations according to the number of digits in the player's score. A one-digit score corresponds to cluster A. A two-digit score corresponds to cluster B. This reflects the progress of the game, but only through its visual display at the top of the screen. The partition therefore captures a feature of the observation that provides no information about how the agent plays.

At layer 3 (Figure 7.14), the partition instead follows the vertical position of the ball. The ball is at the top of the field in cluster C. It is in the middle of the field in cluster D. It is at the bottom of the field in cluster E. This feature directly determines where the agent needs to place its paddle.

The clustering at layer 2 is perfectly reproducible and perfectly abstract, with CU, AS, and ACU all equal to 1.00. The clustering at layer 3 is less consistent, with a CU of 0.67, and reaches a lower ACU of 0.65.

Layer 2	$K = 2$	CU 1.00	AS 1.00	ACU 1.00
A	The player's score has two digits.			
B	The player's score has one digit.			
Layer 3	$K = 3$	CU 0.67	AS 0.98	ACU 0.65
C	The ball is at the top.			
D	The ball is in the middle.			
E	The ball is at the bottom.			

Table 7.8: CU, AS, and ACU of the two clusterings analyzed in the **Pong** environment.

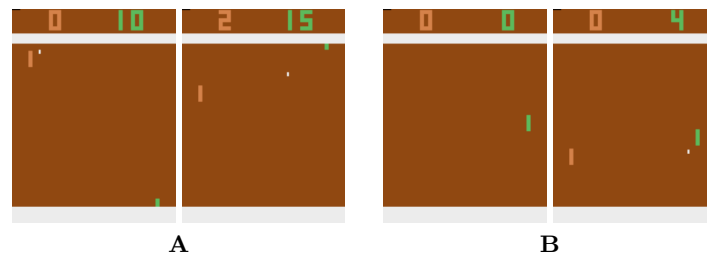


Figure 7.13: Clusters A–B identified by CAR in the **Pong** environment (layer 2, $K = 2$).

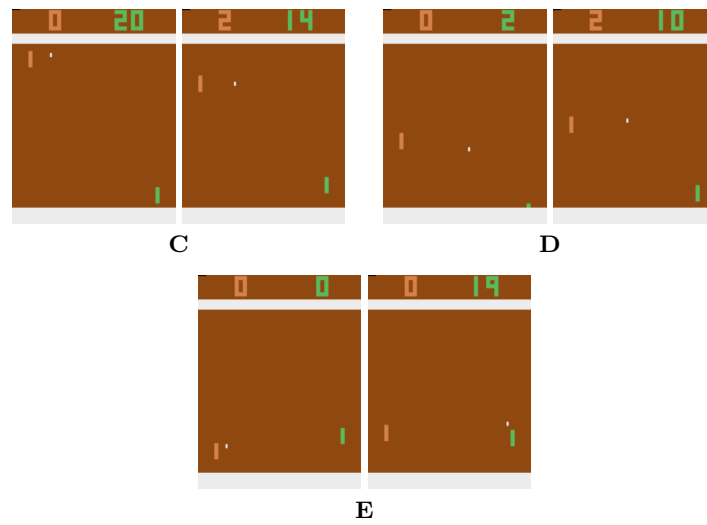


Figure 7.14: Clusters C–E identified by CAR in the **Pong** environment (layer 3, $K = 3$).

7.5.5 Space Invaders

Two clusterings are examined for **Space Invaders** (Table 7.9), obtained at layer 4 with $K = 4$ and at layer 2 with $K = 5$.

At layer 4 (Figure 7.15), the partition tracks the progression of the episode. The start of the episode is captured in cluster D. The middle of the episode is captured in cluster B. The end of the episode is captured in cluster C. A specific behavior in which the agent has cleared the leftmost column and is positioned to catch the flying saucer that occasionally crosses the top of the screen is captured in cluster A.

At layer 2 (Figure 7.16), the partition again tracks the progression of the episode. The start of the episode is captured in cluster G. The middle of the episode is captured in cluster F. The end of the episode is captured in cluster I. Attacking the first two columns of the formation is captured in cluster E. Trying to catch the flying saucer is captured in cluster H.

The clustering at layer 4 has a CU of 0.92 and a AS of 0.69, resulting in a ACU of 0.61. The clustering at layer 2 has the same CU of 0.92, but a lower AS of 0.49, resulting in a lower ACU of 0.41.

Layer 4	$K = 4$	CU 0.92	AS 0.69	ACU 0.61
A	The left column has been cleared; the flying saucer occasionally appears.			
B	Middle of the episode.			
C	End of the episode.			
D	Start of the episode.			
Layer 2	$K = 5$	CU 0.92	AS 0.49	ACU 0.41
E	Attacks the first two columns of the formation.			
F	Middle of the episode.			
G	Start of the episode.			
H	Tries to catch the flying saucer.			
I	End of the episode.			

Table 7.9: CU, AS, and ACU of the two clusterings analyzed in the **Space Invaders** environment.

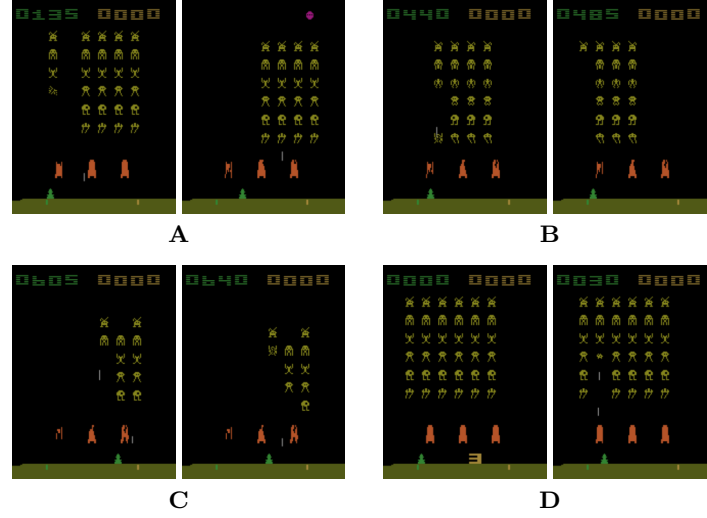


Figure 7.15: Clusters A–D identified by CAR in the **Space Invaders** environment (layer 4, $K = 4$).

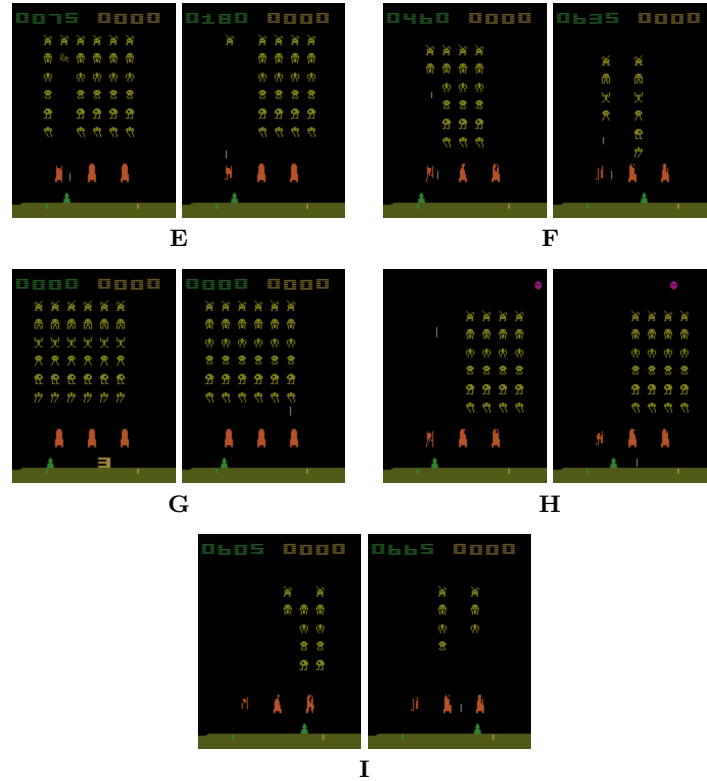


Figure 7.16: Clusters E–I identified by CAR in the **Space Invaders** environment (layer 2, $K = 5$).

7.5.6 Surround

Two clusterings are examined for **Surround** (Table 7.10), obtained at layer 8b with $K = 4$ and at layer 7b with $K = 6$.

At layer 8b (Figure 7.17), the partition captures the start of the episode in cluster B. The outcome of the episode, when the opponent loses, is captured in cluster D. The player's trail is confined to the bottom-left area in cluster A. The player reaches the top of the level from the left and then moves along the top in cluster C.

At layer 7b (Figure 7.18), the partition again captures the start of the episode in cluster J. The outcome of the episode, when the opponent loses, is captured in cluster F. The player moves in the upper-left corner in cluster E. The player moves in the lower-left corner in cluster G. The end of the game is played in the lower-left corner in cluster H. The player crosses the level from the top left to the bottom right in cluster I.

The clustering at layer 8b has a CU of 0.67, a AS of 0.71, and a ACU of 0.38. The clustering at layer 7b reaches the same values, with a CU of 0.67, a AS of 0.71, and a ACU of 0.38.

Layer 8b	$K = 4$	CU 0.67	AS 0.71	ACU 0.38
A	The agent's trail remains confined to the bottom-left area.			
B	Start of the episode.			
C	The agent moves along the top of the level, arriving from the left.			
D	The opponent loses by running into a trail.			
Layer 7b	$K = 6$	CU 0.67	AS 0.71	ACU 0.38
E	The player moves in the upper-left corner.			
F	The opponent loses the confrontation.			
G	The player moves in the lower-left corner.			
H	The end of the game is played in the lower-left corner.			
I	The player starts at the top left and ends at the bottom right.			
J	Start of the episode.			

Table 7.10: CU, AS, and ACU of the two clusterings analyzed in the **Surround** environment.

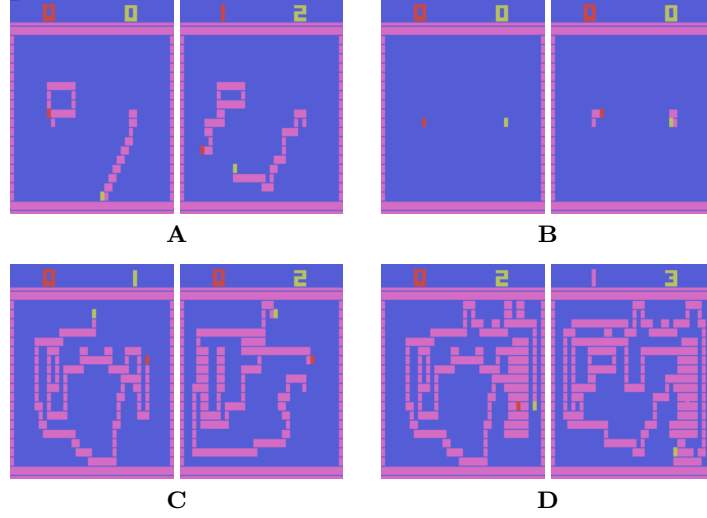


Figure 7.17: Clusters A–D identified by CAR in the **Surround** environment (layer 8b, $K = 4$).

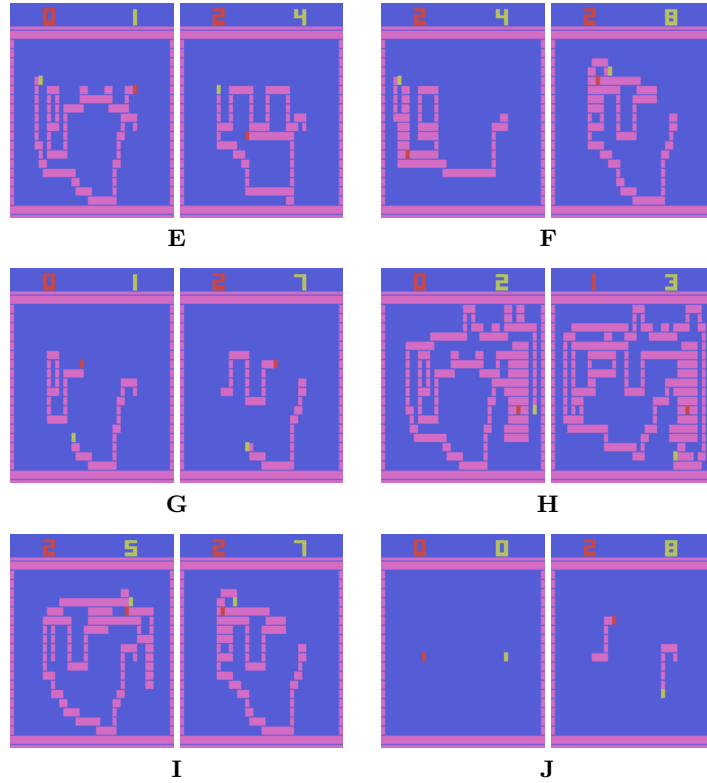


Figure 7.18: Clusters E–J identified by CAR in the **Surround** environment (layer 7b, $K = 6$).

3306 7.6 Conclusion

3307 The quantitative results show that CAR consistently recovers similar clustering
 3308 structures across independently trained surrogate policies. Across the six envi-
 3309 ronments, the CU frequently exceeds 0.5, indicating that independently trained
 3310 policies can develop similarly structured LRs [66]. The CU also tends to decrease
 3311 as the network gets deeper, with the highest values generally observed in the
 3312 earliest layers. Considering consistency alone would therefore favor the earliest
 3313 layers of the network.

3314 To complement this analysis, the level of abstraction of these representations
 3315 must also be considered. For this purpose, AS is analyzed and shows that the
 3316 earliest layers tend to exhibit lower levels of abstraction. Their representations
 3317 remain strongly related to the observations and the initial structure of the
 3318 network. In the deepest layers, the abstraction can decrease again in some
 3319 environments, as the latent clustering becomes closer to the clustering of the
 3320 action distribution.

3321 This highlights the importance of considering both consistency and abstrac-
 3322 tion when identifying meaningful LRs. The ACU addresses this limitation by
 3323 combining these two properties into a single measure, providing a practical crite-
 3324 rion for identifying layers that achieve a suitable balance between representational
 3325 consistency and abstraction.

3326 These quantitative analyses are complemented by the qualitative analysis,
 3327 which shows that the clusters identified by CAR correspond to interpretable
 3328 properties of the environment and the agent’s behavior. Across the different
 3329 environments, the clusters capture recurring aspects such as episode progression,
 3330 specific game situations, and behavioral patterns. These observations provide
 3331 qualitative evidence that the structures recovered by CAR can reflect semantically
 3332 aspects of the underlying task.

3333 The qualitative analysis also reveals a trade-off associated with the number of
 3334 clusters. Increasing the number of clusters allows more specific situations to be
 3335 distinguished, but tends to reduce the stability of the resulting partitions across
 3336 surrogate policies. Finer clusters have less clear semantic boundaries, reducing
 3337 agreement across independently trained policies. The choice of the number of
 3338 clusters therefore involves a trade-off between recovering stable, shared concepts
 3339 and capturing finer distinctions between situations.

3340 Overall, these results provide empirical support for the assumptions un-
 3341 derlying CAR. Independently trained surrogate policies exhibit a degree of
 3342 convergence in the organization of their LRs that cannot be explained solely
 3343 by the considered baselines. The qualitative analysis also shows that these
 3344 clusters can correspond to meaningful properties of the environment and the
 3345 agent’s behavior. Taken together, these findings provide empirical support for
 3346 the *Representation Convergence Hypothesis* (Hypothesis 4).

Chapter 8

Conclusion

This final chapter reviews the work presented in this thesis. It begins by revisiting each research question based on the contributions presented. It then discusses the limitations of these contributions. Finally, it presents the directions for future research that these limitations open.

8.1 Summary of Contributions

This section revisits the four research questions introduced in Section 1.2. For each of them, it summarizes the answer provided by this thesis.

Research Question 1 concerned the approaches proposed in the literature to analyze the internal workings of Neural Networks (NNs). To answer it, we first made explicit the working hypothesis shared by most of this literature, the *Concept Hypothesis* (Hypothesis 1). According to this hypothesis, NNs generalize by organizing their Latent Spaces (LSs) around concepts. We then organized existing methods into four families: probing, interpretable directions, Sparse Autoencoders (SAEs), and clustering.

Research Question 2 concerned the organization of existing Explainable Reinforcement Learning (XRL) methods [112, 125]. To answer this question, we first clarified what an explanation is. Building on this framework, we identified the concepts and sources of explanation. Regarding the subject of explanation, we distinguish two levels: a single decision and the agent’s strategy. For the source of explanation, we identify three possibilities: policies, trajectories, and value functions. From these elements, we proposed a taxonomy that accounts for the specificities of Reinforcement Learning (RL) and used it to review the state of the art.

Research Question 3 concerned the use of Latent Representations (LRs) to identify the mechanisms underlying the decisions of RL agents. To answer it, we introduced Clustering Agent Representation (CAR), an XRL method based on the principles of Mechanistic Explainability (ME) [126]. CAR trains several surrogate policies independently to reproduce the same agent. It then clusters their LRs and compares the resulting partitions. This procedure relies on the *Representation Convergence Hypothesis* (Hypothesis 4), according to which independently trained networks develop similar concepts for a given task. Applied to six Atari environments, CAR recovers similar clustering structures

across surrogate policies. The qualitative analysis shows that these clusters capture interpretable aspects of the task, such as episode progression, specific game situations, and behavioral patterns.

Research Question 4 concerned the objective quantification of explanation quality. To answer it, we introduced three metrics. The *Clustering Universality* (CU) measures the consistency of the clusters across independently trained surrogate policies. The *Abstraction Score* (AS) measures how far these clusters depart from the structure of the observations, of the actions, and of an untrained network. The experiments show that these two properties evolve in opposite directions across the network. The earliest layers are the most consistent, but also the least abstract. The *Adjusted Clustering Universality* (ACU) combines both properties into a single measure and provides a practical criterion for selecting the layer to analyze.

Taken together, these contributions form a coherent chain. The *Mechanistic Hypothesis* (Hypothesis 3) states that NNs can be explained through their internal mechanisms. The *Concept Hypothesis* (Hypothesis 1) states that these mechanisms are organized around concepts. The *Representation Convergence Hypothesis* (Hypothesis 4) states that these concepts reflect the task. CAR translates these hypotheses into an empirical procedure, and the experiments provide support for them.

8.2 Limitations

Although the research presented in this thesis has produced encouraging results, it remains subject to limitations. This section discusses the four main limitations identified at this stage.

First, CAR describes how the LS of an agent is organized, but not what each cluster means. The metrics quantify the consistency and abstraction of the clusters, but they say nothing about their content. In this thesis, the meaning of each cluster was assigned through manual inspection of the corresponding observations. This step is time-consuming, subjective, and difficult to scale to a large number of clusters or environments.

Second, the clusters identified by CAR are not yet directly related to the decisions of the agent. We observe that the agent distinguishes certain situations, but not how these distinctions lead to a given action. As a result, CAR cannot yet explain a specific failure or reveal a bias in the behavior of the agent.

Third, the experiments were conducted on six Atari environments. These environments are diverse, but they remain limited in size and complexity. Whether CAR scales to more complex environments and larger networks has not yet been established. In addition, only one clustering method was evaluated, based on k -means with Euclidean distance. Other distances and other clustering methods could reveal different structures in the LS. Since the pipeline of CAR is modular, these alternatives can be tested and compared using the proposed metrics.

Finally, the proposed metrics have only been applied to clustering. The other families of methods reviewed in Chapter 3 have not yet been evaluated within this framework. This is particularly the case for methods based on the *Superposition Hypothesis* (Hypothesis 2). These methods are widely used, but their assumptions remain debated. Our metrics could be used to test these assumptions quantitatively, and to compare these methods with CAR on the

3428 same tasks.

3429 8.3 Perspectives

3430 The limitations discussed above open several directions for future research. This
3431 section presents three of them.

3432 A first direction is to automate the semantic interpretation of the clusters.
3433 One approach is to rely on vision-language models. Given observations from
3434 the same cluster, such models could generate a natural language description of
3435 what distinguishes it from the others. This approach would provide a textual
3436 description for each cluster, helping identify its specific semantics and fully
3437 automate the CAR method.

3438 A second direction is to use the identified clusters to explain the strategy of
3439 the agent. This thesis has shown that the LS can be used to identify meaningful
3440 concepts. The next step is to use these concepts to describe how the agent
3441 behaves over time. Since each cluster groups states that are similar from the
3442 agent’s perspective, it can be treated as an abstract state. The transitions
3443 observed between these clusters during episodes would then form a simplified
3444 model of the environment, as perceived by the agent.

3445 A third direction concerns the evaluation of explanations. The metrics pro-
3446 posed in this thesis measure structural properties of the LRs. They are necessary,
3447 but they are not sufficient. According to our definition, an explanation is valuable
3448 only if it improves the observer’s understanding of the agent. Measuring this
3449 improvement requires experimental protocols from disciplines such as cognitive
3450 science and psychology. Such protocols could assess whether an explanation
3451 actually improves the observer’s mental model of the agent.

3452 Now that the main directions for future research have been presented, we
3453 can step back and consider this work as a whole. In XRL, a common approach
3454 is to replace NNs with simpler models that can be directly interpreted. The
3455 approach followed in this thesis is different. Rather than viewing the NN as
3456 an obstacle, we propose explainability methods that build on its specificities.
3457 From this perspective, explainability is not achieved despite the use of NNs, but
3458 through them. The results presented in this thesis show that this approach is
3459 feasible. It therefore offers a way to understand our creations and to ensure that
3460 they remain aligned with our intentions.

Appendix 1

Additional Results

This appendix reports, for each of the six Atari environments studied in Chapter 7, the full evolution of CU, AS, and ACU with respect to the number of clusters $K \in [2, 20]$, for every analyzed layer, together with the three baseline similarities entering the AS, namely the similarity to the observation clustering (ρ^{obs}), to the untrained-network latent clustering (ρ^{untr}), and to the clustering of the action distribution, i.e., of the policy logits before the softmax (ρ^{act}), underlying the analysis of Section 7.4. All these quantities are computed with the adjusted Rand index, consistent with the rest of this thesis.

For each environment, one table is reported per metric and per baseline, with the analyzed layers as rows and $K \in [2, 20]$ as columns. In each table, a cell's background shade is proportional to its value (the closer to 1, the darker); a value in **bold** is the highest value of its column, i.e., the best layer for that number of clusters; a value that is underlined is the highest value of its row, i.e., the best number of clusters for that layer; and a value both **bold and underlined** is simultaneously the best of its row and of its column. The last column and the last row report, respectively, the mean $\pm$ standard deviation across K for each layer, and the mean $\pm$ standard deviation across layers for each K , with the best mean of each in bold; the bottom-right cell reports the mean $\pm$ standard deviation over the whole table. The same convention is applied to the baseline tables, where a high value (dark cell) therefore indicates a strong alignment of the latent clustering with the baseline, i.e., a *lower* contribution to the AS.

3484

1.1 Breakout

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Table 1.1: CU — **Breakout**.

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Table 1.2: AS — **Breakout**.

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Table 1.3: ACU — **Breakout**.

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Table 1.4: Observation baseline (ρ^{obs}) — **Breakout**.

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Table 1.5: Untrained-network (latent) baseline (ρ^{untr}) — **Breakout**.

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Table 1.6: Action baseline (ρ^{act}) — **Breakout**.

3485

1.2 Mario Bros

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Table 1.7: CU — **Mario Bros.**

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Table 1.8: AS — **Mario Bros.**

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Table 1.9: ACU — **Mario Bros.**

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Table 1.10: Observation baseline (ρ^{obs}) — **Mario Bros.**

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Table 1.11: Untrained-network (latent) baseline (ρ^{untr}) — **Mario Bros.**

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Table 1.12: Action baseline (ρ^{act}) — **Mario Bros.**

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1.3 Ms. Pac-Man

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Table 1.13: CU — Ms. Pac-Man.

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Table 1.14: AS — Ms. Pac-Man.

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Table 1.15: ACU — **Ms. Pac-Man**.

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Table 1.16: Observation baseline (ρ^{obs}) — **Ms. Pac-Man**.

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Table 1.17: Untrained-network (latent) baseline (ρ^{untr}) — **Ms. Pac-Man**.

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Table 1.18: Action baseline (ρ^{act}) — **Ms. Pac-Man**.

3487

1.4 Pong

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Table 1.19: CU — **Pong**.

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Table 1.20: AS — **Pong**.

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Table 1.21: ACU — **Pong**.

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Table 1.22: Observation baseline (ρ^{obs}) — **Pong**.

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Table 1.23: Untrained-network (latent) baseline (ρ^{untr}) — **Pong**.

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Table 1.24: Action baseline (ρ^{act}) — **Pong**.

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1.5 Space Invaders

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Table 1.25: CU — **Space Invaders**.

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Table 1.26: AS — **Space Invaders**.

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Table 1.27: ACU — **Space Invaders**.

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Table 1.28: Observation baseline (ρ^{obs}) — **Space Invaders**.

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Table 1.29: Untrained-network (latent) baseline (ρ^{untr}) — **Space Invaders**.

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Table 1.30: Action baseline (ρ^{act}) — **Space Invaders**.

3489

1.6 Surround

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Table 1.31: CU — **Surround**.

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Table 1.32: AS — **Surround**.

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Table 1.33: ACU — **Surround**.

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Table 1.34: Observation baseline (ρ^{obs}) — **Surround**.

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Table 1.35: Untrained-network (latent) baseline (ρ^{untr}) — **Surround**.

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Table 1.36: Action baseline (ρ^{act}) — **Surround**.

Acronyms

ACU	<i>Adjusted Clustering Universality</i> The Clustering Universality minus the similarity of the latent clusterings to the three trivial baselines of the Abstraction Score (see Section 6.3.5).
AI	Artificial Intelligence The field of computer science concerned with building systems that perform tasks usually associated with human intelligence, such as perception, reasoning, learning and decision-making.
APPO	Asynchronous Proximal Policy Optimization A variant of Proximal Policy Optimization in which data collection by the workers and policy updates run asynchronously, to scale training.
ARI	Adjusted Rand Index The Rand Index corrected for chance, equal to 0 for random partitions and to 1 for identical partitions (see Section 6.3.2).
AS	<i>Abstraction Score</i> A metric measuring how far latent clusterings differ from those of the observations, of the action distribution and of untrained networks (see Section 6.3.4).
BC	Behavioral Cloning The training of a policy by supervised learning to reproduce the actions of another policy on a dataset of its interactions.
CAR	Clustering Agent Representation The method proposed in this thesis, which trains several surrogate policies of an agent and clusters their latent representations to detect concept structures (see Chapter 6).
CNN	Convolutional Neural Network A neural network whose layers apply learned convolution filters, suited to data with a spatial structure such as images.

CS	Concept Structure The specific form in which a concept is instantiated within the latent space, for instance a direction or a cluster (see Definition 3.1.5).
CU	Clustering Universality The mean pairwise Adjusted Rand Index between the latent clusterings of independently trained surrogate policies (see Section 6.3.3).
DDPG	Deep Deterministic Policy Gradient An actor–critic reinforcement learning algorithm for continuous action spaces, which learns a deterministic policy.
DL	Deep Learning The family of machine learning methods based on neural networks with many successive layers, trained from data by gradient descent (see Chapter 2).
DQN	Deep Q-Network A reinforcement learning algorithm that approximates the action-value function with a deep neural network.
GPU	Graphics Processing Unit A processor specialized in massively parallel computation, used to accelerate the training of neural networks.
LLM	Large Language Model A neural network trained on large text corpora to predict the next token of a sequence, used for language understanding and generation.
LR	Latent Representation The internal encoding of an input given by an intermediate layer of a trained neural network (see Definition 3.1.2).
LS	Latent Space The set of all latent representations produced by a layer over a dataset (see Definition 3.1.3).
LSTM	Long Short-Term Memory A recurrent neural network architecture whose gating mechanism allows information to be retained over long sequences.
MDP	Markov Decision Process The mathematical framework modeling sequential decision-making as an agent interacting with an environment through states, actions, rewards and transitions (see Section 4.3.1).

ME	Mechanistic Explainability The explainability approach that seeks to explain the behavior of a neural network by analyzing its internal mechanisms (see Definition 1.1.1).
MLP	Multilayer Perceptron A neural network made of a sequence of fully connected layers.
MSE	Mean Squared Error A loss function equal to the average of the squared differences between predictions and targets.
NN	Neural Network A parameterized function made of successive layers of artificial neurons, whose parameters are learned from data.
PPO	Proximal Policy Optimization A policy-gradient reinforcement learning algorithm that stabilizes training by limiting the size of each policy update through a clipped objective.
RI	Rand Index A measure of the agreement between two partitions of the same set of points, computed over all pairs of points (see Section 6.3.2).
RL	Reinforcement Learning The learning paradigm in which an agent learns a policy by interacting with an environment so as to maximize the rewards it collects (see Chapter 4).
SAE	Sparse Autoencoder An autoencoder trained with a sparsity constraint, used to decompose latent representations into a larger set of more interpretable directions (see Section 3.4).
SGD	Stochastic Gradient Descent An optimization algorithm that updates the parameters of a model in the direction opposite to the gradient of the loss, estimated on a random subset of the training data (see Section 2.4).
XAI	Explainable Artificial Intelligence The subfield of artificial intelligence that develops methods to make the decisions of a model understandable to a human observer (see Section 5.1.2).
XRL	Explainable Reinforcement Learning The branch of explainability dedicated to reinforcement learning agents, whose decisions are sequential and depend on the interaction with an environment (see Chapter 5).

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LLM Use Policy

In the course of this thesis, Large Language Models (LLMs) were used as assistance for the following tasks:

- **Writing:** translation, wording improvement, and proofreading.
- **Software development:** code implementation and review.
- **Scientific work:** identification of relevant documents and discussion of approaches. They were not used as autonomous scientific sources.

All generated outputs were systematically reviewed by the author. The author takes full responsibility for the content presented in this thesis.

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Mechanistic Explainability in Reinforcement Learning

Maxime Alaarabiou

Deep reinforcement learning agents achieve strong performance on complex sequential decision-making tasks, but the neural networks that implement their policies remain difficult to interpret. This thesis addresses the explainability of such agents from a mechanistic perspective. It seeks to explain their decisions by analyzing the internal representations learned by their neural networks.

The thesis first reviews methods for analyzing the latent space and makes explicit the hypothesis they share. We refer to this hypothesis as the Concept Hypothesis, which states that neural networks generalize by organizing their latent space around concepts.

It then clarifies what constitutes an explanation and proposes a unified taxonomy of explainable reinforcement learning methods. This taxonomy is organized according to the subject of the explanation, which can be a single decision or the agent’s overall strategy, and according to its source, which can be policies, trajectories, or value functions.

Building on these foundations, the thesis introduces Clustering Agent Representation, or CAR, as a mechanistic explainability method. CAR independently trains several surrogate policies to reproduce the behavior of the same agent. It then clusters their latent representations and compares the resulting partitions.

The method relies on the Representation Convergence Hypothesis, which states that independently trained networks develop similar concepts for a given task.

To objectively evaluate the recovered structures, three metrics are proposed. Clustering Universality measures their consistency across surrogate policies. Abstraction Score measures their departure from the structure of the observations, the actions, and an untrained network. Adjusted Clustering Universality combines these two measures.

Experiments conducted on six Atari environments show that independently trained surrogate policies converge toward similar clustering structures that cannot be explained by the considered baselines alone. Consistency and abstraction evolve in opposite directions across network layers. Adjusted Clustering Universality provides a practical criterion for selecting the layer to analyze.

Finally, a qualitative analysis shows that the recovered clusters correspond to interpretable aspects of the task, including episode progression, specific game situations, and behavioral patterns.

Keywords: Explainable Artificial Intelligence, Deep Learning, Reinforcement Learning, explainability, Mechanistic Interpretability, Latent Space, Clustering.